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(54) **LEPIDOPTERAN-ACTIVE Cry1Da1 AMINO ACID SEQUENCE VARIANT PROTEINS**

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(58) **Field of Classification Search**

None
See application file for complete search history.

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(57) **ABSTRACT**

Engineered Cry1Da amino acid sequences are provided that exhibit improved Lepidopteran insecticidal activity and an enhanced Lepidopteran spectrum compared to the naturally occurring Cry1Da protein toxin. Polynucleotide sequences intended for use in expression of the improved proteins in plants are also provided. Particular embodiments provide compositions containing insect inhibitory amounts of the engineered proteins, as well as recombinant plants, plant parts, and seeds containing polynucleotide constructs encoding one or more of the improved engineered proteins.

21 Claims, 1 Drawing Sheet

Specification includes a Sequence Listing.

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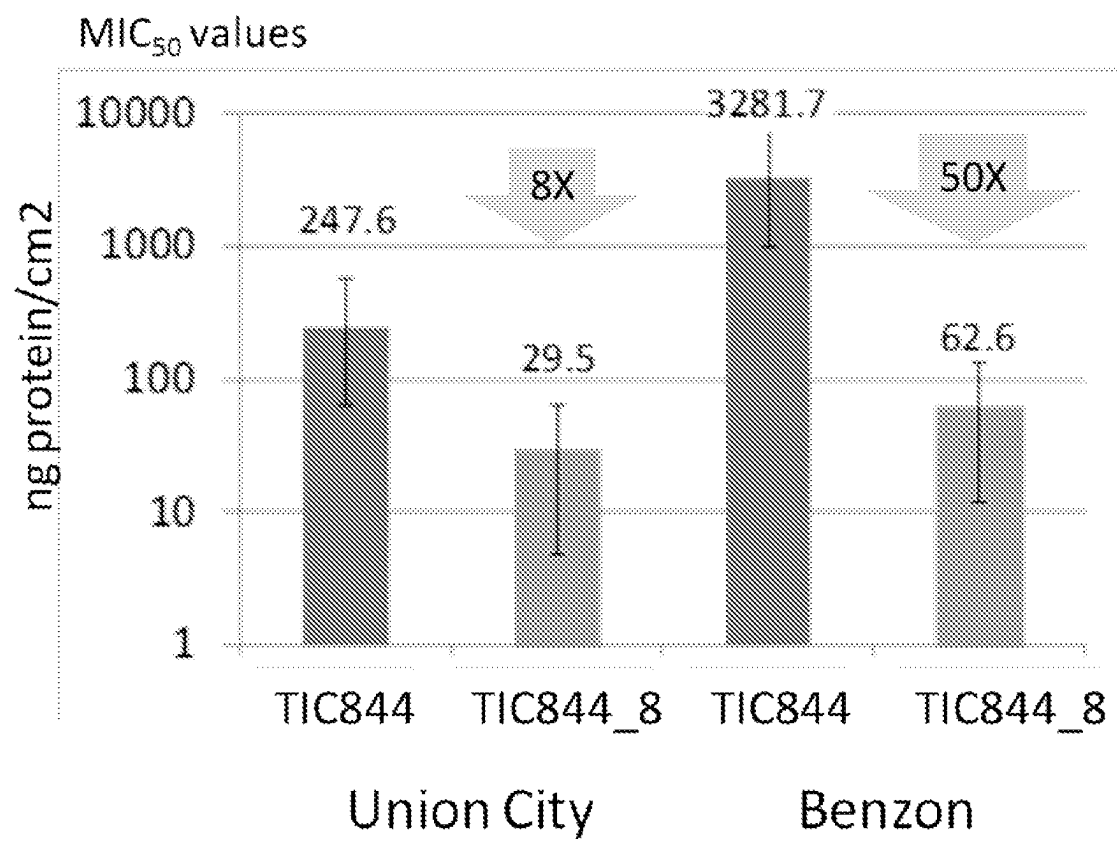
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LEPIDOPTERAN-ACTIVE Cry1Da1 AMINO ACID SEQUENCE VARIANT PROTEINS

REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/815,122, filed Jul. 26, 2022, which is a continuation of U.S. patent application Ser. No. 17/028,518, filed Sep. 22, 2020, now U.S. Pat. No. 11,396,661, which is a continuation of U.S. patent application Ser. No. 16/573,538, filed Sep. 17, 2019, now U.S. Pat. No. 10,801,037, which is a continuation of U.S. patent application Ser. No. 16/047,187, filed Jul. 27, 2018, now U.S. Pat. No. 10,457,958, which is a continuation of U.S. patent application Ser. No. 14/884,432, filed Oct. 15, 2015, now U.S. Pat. No. 10,059,959, which claims the benefit of U.S. Provisional Application No. 62/065,017, filed Oct. 17, 2014, and U.S. Provisional Application No. 62/064,994, filed Oct. 16, 2014, all of which are incorporated by reference in their entireties herein.

INCORPORATION OF SEQUENCE LISTING

A computer readable form of the Sequence Listing is filed herewith by electronic submission. The Sequence Listing is incorporated by reference in its entirety, is contained in the file created on Dec. 22, 2023, having the file name MONS517USC5_ST26.xml, and which is 161,016 bytes in size (as measured in the MS-Windows® operating system).

FIELD OF THE INVENTION

The invention generally relates to the field of insect inhibitory proteins. A novel class of engineered proteins exhibiting insect inhibitory activity against agriculturally-relevant pests of crop plants and seeds is disclosed. In particular, the disclosed class of engineered inhibitory proteins has insecticidal activity against the Lepidopteran order of insect pests. Plants, plant parts, and seeds containing a polynucleotide construct encoding one or more of the disclosed engineered inhibitory proteins are provided.

BACKGROUND OF THE INVENTION

Helicoverpa zea is a significant Lepidopteran pest of major agricultural crops, including corn, cotton, and soy. Known as the corn earworm (CEW), cotton bollworm (CBW), and soy podworm (SPW), this polyphagous insect species is particularly difficult to control with insecticidal proteins from *Bacillus thuringiensis* (Bt) or other bacterial species. *H. zea* is considered at risk for resistance development to current insect control traits, given its ability to feed on many different crops and the current absence of a high-dose control strategy. Accordingly, new modes of action (MoA) are required to ensure the durability of transgenic plants protected from *H. zea* feeding damage.

The Cry1Da1 protein is a Lepidopteran-active protein that was first described by Hofte, et al. "Nucleotide sequence and deduced amino acid sequence of a new Lepidoptera-specific crystal protein gene from *Bacillus thuringiensis*." *Nucleic Acids Res.* 18(18) (1990): 5545. This protein exhibits excellent insecticidal activity towards *Spodoptera* species including *Spodoptera frugiperda* (fall armyworm, FAW), a pest of several row crops, including corn, cotton and soybean. However, Cry1Da1 exhibits low-to-moderate activity towards a variety of other major Lepidopteran pests, including bollworms (e.g., *Helicoverpa armigera* and *H. zea*),

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borers (e.g., *Ostrinia nubilalis* and *Diatraea grandiosella*) and soybean looper (*Pseudoplusia includens*). Because of its narrow insecticidal spectrum and its inability to provide commercial-level protection against a range of important Lepidopteran agricultural pests such as CEW, the Cry1Da1 insecticidal protein has limited value as a transgenic plant insect control trait. As a result, no current commercial varieties of insect-protected crops utilize Cry1Da1 as a plant-incorporated protectant.

Despite its narrow insecticidal spectrum, Cry1Da1 is an interesting insecticidal protein because it appears that the Cry1Da1 protein uses an alternative MoA for controlling certain Lepidopteran pests. Evidence for this comes from studies with multiple resistant insect colonies. For example, field-derived colonies of *Plutella xylostella* (diamondback moth) and *Pectinophora gossypiella* (pink bollworm) that are resistant to Cry1Ac intoxication retain full sensitivity to the Cry1Da1 protein (Tabashnik, et al. "Cross-Resistance of Pink Bollworm (*Pectinophora gossypiella*) to *Bacillus thuringiensis* toxins." *Appl. Environ. Microbiol.* 66 (2000): 4582-4584; Tabashnik, et al. "Cross-Resistance to *Bacillus thuringiensis* Toxin Cry1Ja in a Strain of Diamondback Moth Adapted to Artificial Diet." *J. Invert. Pathol.* 76: (2000): 81-83). These lines of evidence indicate that Cry1Da1 recognizes Lepidopteran midgut receptors distinct from those recognized by Lepidopteran-active proteins currently deployed in transgenic crops, including Cry1Ac, Cry1Ab, Cry1A.105, Cry1Fa, Cry2Ae, and Cry2Ab2. In view of this apparent novel MoA, optimization of Cry1Da1-like proteins for improved activity against a broader spectrum of *Helicoverpa* species while maintaining or increasing their insecticidal activity towards *Spodoptera* would create a high-value plant-incorporated protectant for insect resistance management.

SUMMARY OF THE INVENTION

In the present invention, several amino acid sequence variants of the TIC844 and Cry1 Da scaffold proteins have been identified that exhibit markedly improved activity (compared to the Cry1Da1 native toxin) towards *H. zea* while retaining excellent activity towards *S. frugiperda*. The improved variants of TIC844 and Cry1 Da have been engineered to be expressed in crop plants (e.g., corn, soybean, cotton, sugarcane), and provide novel options for in-plant resistance management and Lepidopteran insect pest control in view of the apparent unique mode-of-action of Cry1 Da coupled with the engineered improvement in activity against *H. zea*.

The engineered Lepidopteran toxic proteins described herein (referred to as "engineered toxin proteins", "engineered toxic proteins", or "engineered insecticidal proteins") are derivatives of the naturally occurring *Bacillus thuringiensis* insecticidal toxin Cry1Da1 (SEQ ID NO:2) or the chimeric homolog of Cry1Da1, TIC844 (SEQ ID NO:14), which comprises the Cry1Da1 core toxin but substitutes the Cry1Ab3 protoxin for the native Cry1Da1 protoxin domain. The engineered insecticidal proteins of the present invention each contain at least one amino acid substitution, one amino acid addition, or one amino acid deletion compared to the scaffold proteins set forth in any of SEQ ID NO:2 or SEQ ID NO:14. The engineered insecticidal proteins of the present invention are particularly toxic to insects of the *Helicoverpa zea* (corn earworm, soy podworm, cotton bollworm) and *Spodoptera frugiperda* (fall armyworm) species. While the scaffold proteins TIC844 (SEQ ID NO:14) and Cry1Da1 (SEQ ID NO:2) display low toxicity to *H. zea*, the

engineered insecticidal proteins of the present invention exhibit surprising and unexpectedly improved insecticidal activity and an enhanced insecticidal spectrum against Lepidopteran insect pests including *H. zea*.

In certain embodiments, an engineered insecticidal protein comprising an amino acid sequence as set forth in any of SEQ ID NO:44, SEQ ID NO: 40, SEQ ID NO: 12, SEQ ID NO:26, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38 or SEQ ID NO:42, or an insect inhibitory fragment thereof is disclosed. In certain embodiments, the engineered insecticidal protein exhibits inhibitory activity against an insect species of the order Lepidoptera. The target Lepidopteran pest species inhibited by the Lepidopteran toxic proteins of the present invention include at least fall armyworm (*Spodoptera frugiperda*), beet armyworm (*Spodoptera exigua*), bertha armyworm (*Mamestra configurata*), black cutworm (*Agrotis ipsilon*), cabbage looper (*Trichoplusia ni*), soybean looper (*Chrysodeixis includens*), velvetbean caterpillar (*Anticarsia gemmatilis*), green cloverworm (*Hypena scabra*), tobacco budworm (*Heliothis virescens*), granulate cutworm (*Agrotis subterranea*), armyworm (*Pseudaletia unipuncta*), western cutworm (*Agrotis orthogonia*), European corn borer (*Ostrinia nubilalis*), navel orangeworm (*Amyelois transitella*), corn root webworm (*Crambus caliginosellus*), sod webworm (*Herpetogramma licarsisalis*), sunflower moth (*Homoeosoma electellum*), lesser cornstalk borer (*Elasmopalpus lignosellus*), codling moth (*Cydia pomonella*), grape berry moth (*Endopiza viteana*), oriental fruit moth (*Grapholita molesta*), sunflower bud moth (*Suleima helianthana*), diamondback moth (*Plutella xylostella*), pink bollworm (*Pectinophora gossypiella*), pink stem borer (*Sesamia inferens*), gypsy moth (*Lymantria dispar*), cotton leaf worm (*Alabama argillacea*), fruit tree leaf roller (*Archips argyrospila*), European leafroller (*Archips rosana*), Asiatic rice borer, or rice stem borer (*Chilo suppressalis*), rice leaf roller (*Cnaphalocrocis medinalis*), corn root webworm (*Crambus caliginosellus*), bluegrass webworm (*Crambus teterrellus*), southwestern corn borer (*Diatraea grandiosella*), sugarcane borer (*Diatraea saccharalis*), spiny bollworm (*Earias insulana*), spotted bollworm (*Earias vittella*), Old World cotton bollworm (*Helicoverpa armigera*), corn earworm, soy podworm or cotton bollworm (*Helicoverpa zea*), sod webworm (*Herpetogramma licarsisalis*), European grape vine moth (*Lobesia botrana*), citrus leafminer (*Phyllocnistis citrella*), large white butterfly (*Pieris brassicae*), imported cabbageworm, or small white butterfly (*Pieris rapae*), tobacco cutworm, or cluster caterpillar (*Spodoptera litura*), and tomato leafminer (*Tuta absoluta*).

Also disclosed herein is a polynucleotide encoding an engineered insecticidal protein or pesticidal fragment thereof, wherein the polynucleotide is operably linked to a heterologous promoter and the engineered insecticidal protein comprises the amino acid sequence as set forth in any of SEQ ID NO:44, SEQ ID NO: 40, SEQ ID NO: 12, SEQ ID NO:26, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38 or SEQ ID NO:42.

In another embodiment, disclosed herein is a polynucleotide encoding an engineered insecticidal protein, wherein the polynucleotide comprises a nucleotide sequence that optionally hybridizes under stringent conditions to the reverse complement of the polynucleotide sequence as set

forth in any of SEQ ID NO: 43, SEQ ID NO: 39, SEQ ID NO: 11, SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO:31, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37 or SEQ ID NO: 41; or encodes the engineered insecticidal protein comprising an amino acid sequence as set forth in any of SEQ ID NO:44, SEQ ID NO: 40, SEQ ID NO: 12, SEQ ID NO:26, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38 or SEQ ID NO:42.

Also provided herein is a host cell comprising the polynucleotide set forth in any of SEQ ID NO: 43, SEQ ID NO: 39, SEQ ID NO: 11, SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO:31, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37 or SEQ ID NO: 41, wherein the host cell is selected from the group consisting of a bacterial host cell or a plant host cell. Contemplated bacterial host cells include bacterial host cells selected from the group consisting of *Agrobacterium*, *Rhizobium*, *Bacillus*, *Brevibacillus*, *Escherichia*, *Pseudomonas*, *Klebsiella*, and *Erwinia*, wherein the *Bacillus* species is a *Bacillus cereus* or a *Bacillus thuringiensis*, said *Brevibacillus* is a *Brevibacillus laterosporus*, and said *Escherichia* is an *Escherichia coli*. Further, contemplated plant host cells include monocots or dicots.

In yet another embodiment, provided herein is an insect inhibitory composition comprising an engineered insecticidal protein comprising an amino acid sequence as set forth in any of SEQ ID NO:44, SEQ ID NO: 40, SEQ ID NO: 12, SEQ ID NO:26, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38 or SEQ ID NO:42 or an insect inhibitory fragment thereof. It is contemplated that this insect inhibitory composition can further comprise at least one insect inhibitory agent different from the engineered insecticidal protein. Contemplated insect inhibitory agents include an insect inhibitory protein, an insect inhibitory dsRNA molecule, and an insect inhibitory chemistry. It is contemplated that the at least one other pesticidal agent can exhibit activity against one or more pest species of the orders Lepidoptera, Coleoptera, Hemiptera, Homoptera, or Thysanoptera.

Also disclosed herein are is a seed comprising an insect inhibitory effective amount of an engineered insecticidal protein comprising the amino acid sequence as set forth in any of SEQ ID NO:44, SEQ ID NO: 40, SEQ ID NO: 12, SEQ ID NO:26, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38 or SEQ ID NO:42; or a polynucleotide set forth in any of SEQ ID NO: 43, SEQ ID NO: 39, SEQ ID NO: 11, SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO: 31, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37 or SEQ ID NO: 41.

A method for controlling a Lepidopteran pest, the method comprising contacting the Lepidopteran pest with an inhibitory amount of an engineered insecticidal protein is also disclosed herein in another embodiment.

In yet another embodiment, disclosed herein is a transgenic plant cell, plant or plant part comprising an engineered insecticidal protein. Methods are provided for controlling a Lepidopteran pest, comprising exposing the pest to the transgenic plant cell, plant or plant part comprising an engineered insecticidal protein. Commodity products derived from the plant cell, plant or plant part comprising an engineered insecticidal protein wherein the product comprises a detectable amount of the engineered insecticidal protein are also contemplated. Contemplated commodity products include plant biomass, oil, meal, animal feed, flour, flakes, bran, lint, hulls, and processed seed.

Another method disclosed herein is a method of producing a seed comprising the engineered insecticidal protein, the method comprising: planting at least one seed comprising the engineered insecticidal protein; growing plants from said seed; and harvesting seed from the plants, wherein said harvested seed comprises the engineered insecticidal protein.

Yet another method disclosed in this application is a method of inhibiting Lepidopteran pests from feeding on a crop plant comprising modifying one or more amino acid residue(s) of SEQ ID NO: 2 or SEQ ID NO:14 through substitution of the one or more amino acid residue(s) to produce a modified SEQ ID NO:2 or SEQ ID NO:14; and making available a Lepidopteran-inhibiting amount of the modified SEQ ID NO: 2 or SEQ ID NO:14 within, on the surface, or in the vicinity of tissues of said crop plant; wherein the SEQ ID NO:2 or SEQ ID NO:14 modified amino acid residue is selected from the group consisting of serine at position 282 replaced by lysine or valine, tyrosine at position 316 replaced by serine, isoleucine at position 368 replaced by proline or arginine, serine at 374 replaced by arginine, asparagine at position 375 replaced by histidine, and isoleucine at position 432 replaced by leucine.

Recombinant polynucleotide molecules that encode the engineered insecticidal proteins of the present invention are also provided. Contemplated recombinant polynucleotide molecules comprise a polynucleotide sequence selected from the group consisting of SEQ ID NO: 43, SEQ ID NO: 39, SEQ ID NO: 11, SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 15, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO:31, SEQ ID NO: 33, SEQ ID NO: 35, SEQ ID NO: 37 or SEQ ID NO: 41; and optionally a polynucleotide sequence encoding an insect inhibitory agent different from the engineered insecticidal protein.

Another method disclosed in this application is method for increasing the Lepidopteran activity and enhancing the Lepidopteran inhibitory spectrum of a scaffold protein, the method comprising modifying one or more amino acid residue(s) of SEQ ID NO: 2 or SEQ ID NO: 14 through substitution of the amino acid residue(s) to produce an engineered insecticidal protein, wherein the SEQ ID NO:2 or SEQ ID NO:14 modified amino acid residue is selected from the group consisting of serine at position 282 replaced by lysine or valine, tyrosine at position 316 replaced by serine, isoleucine at position 368 replaced by proline or arginine, serine at 374 replaced by arginine, asparagine at position 375 replaced by histidine, and isoleucine at position 432 replaced by leucine. In certain embodiments of this method, the engineered insecticidal protein has at least an eight-fold increase in *Helicoverpa zea* lethality relative to the scaffold protein.

Other embodiments, features, and advantages of the invention will be apparent from the following detailed description, the examples, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates the MIC₅₀ values of the scaffold protein TIC844 (SEQ ID NO: 14) compared to the engineered insecticidal protein TIC844_8 (SEQ ID NO: 26) for two different *Helicoverpa zea* (CEW) colonies, Union City and Benzon.

BRIEF DESCRIPTION OF THE SEQUENCES

SEQ ID NO:1 is a nucleotide sequence encoding a Cry1Da1 protein.
 SEQ ID NO:2 is an amino acid sequence of a Cry1Da1 protein toxin.
 SEQ ID NO:3 is a nucleotide sequence encoding a Cry1Da1_3 protein.
 SEQ ID NO:4 is an amino acid sequence of a Cry1Da1_3 protein toxin.
 SEQ ID NO:5 is a nucleotide sequence encoding a Cry1Da1_4 protein.
 SEQ ID NO:6 is an amino acid sequence of a Cry1Da1_4 protein toxin.
 SEQ ID NO:7 is a nucleotide sequence encoding a Cry1Da1_5 protein.
 SEQ ID NO:8 is an amino acid sequence of a Cry1Da1_5 protein toxin.
 SEQ ID NO:9 is a nucleotide sequence encoding a Cry1Da1_6 protein.
 SEQ ID NO:10 is an amino acid sequence of a Cry1Da1_6 protein toxin.
 SEQ ID NO:11 is a nucleotide sequence encoding a Cry1Da1_7 protein.
 SEQ ID NO:12 is an amino acid sequence of a Cry1Da1_7 protein toxin.
 SEQ ID NO:13 is a nucleotide sequence encoding a TIC844 protein.
 SEQ ID NO:14 is an amino acid sequence of a TIC844 protein toxin.
 SEQ ID NO:15 is a nucleotide sequence encoding a TIC844_2 protein.
 SEQ ID NO:16 is an amino acid sequence of a TIC844_2 protein toxin.
 SEQ ID NO:17 is a nucleotide sequence encoding a TIC844_4 protein.
 SEQ ID NO:18 is an amino acid sequence of a TIC844_4 protein toxin.
 SEQ ID NO:19 is a nucleotide sequence encoding a TIC844_5 protein.
 SEQ ID NO:20 is an amino acid sequence of a TIC844_5 protein toxin.
 SEQ ID NO:21 is a nucleotide sequence encoding a TIC844_6 protein.
 SEQ ID NO:22 is an amino acid sequence of a TIC844_6 protein toxin.
 SEQ ID NO:23 is a nucleotide sequence encoding a TIC844_7 protein.
 SEQ ID NO:24 is an amino acid sequence of a TIC844_7 protein toxin.
 SEQ ID NO:25 is a nucleotide sequence encoding a TIC844_8 protein.
 SEQ ID NO:26 is an amino acid sequence of a TIC844_8 protein toxin.

SEQ ID NO:27 is a polynucleotide sequence designed for use in expressing a Cry1Da1 protein in plants.

SEQ ID NO:28 is an amino acid sequence of a Cry1Da1 protein toxin.

SEQ ID NO:29 is a polynucleotide sequence designed for use in expressing a Cry1Da1_2.nno protein in plants.

SEQ ID NO:30 is an amino acid sequence of a Cry1Da1_2.nno protein toxin.

SEQ ID NO:31 is a polynucleotide sequence designed for use in expressing a Cry1Da1_3.nno protein in plants.

SEQ ID NO:32 is an amino acid sequence of a Cry1Da1_3.nno protein toxin.

SEQ ID NO:33 is a polynucleotide sequence designed for use in expressing a Cry1Da1_4.nno protein in plants.

SEQ ID NO:34 is an amino acid sequence of a Cry1Da1_4.nno protein toxin.

SEQ ID NO:35 is a polynucleotide sequence designed for use in expressing a Cry1Da1_5.nno protein in plants.

SEQ ID NO:36 is an amino acid sequence of a Cry1Da1_5.nno protein toxin.

SEQ ID NO:37 is a polynucleotide sequence designed for use in expressing a Cry1Da1_6.nno protein in plants.

SEQ ID NO:38 is an amino acid sequence of a Cry1Da1_6.nno protein toxin.

SEQ ID NO:39 is a polynucleotide sequence designed for use in expressing a Cry1Da1_7.nno protein in plants.

SEQ ID NO:40 is an amino acid sequence of a Cry1Da1_7.nno protein toxin.

SEQ ID NO:41 is a polynucleotide sequence designed for use in expressing a TIC844_9.nno protein in plants.

SEQ ID NO:42 is an amino acid sequence of a TIC844_9.nno protein toxin.

SEQ ID NO:43 is a polynucleotide sequence designed for use in expressing a TIC844_11.nno protein in plants.

SEQ ID NO:44 is an amino acid sequence of a TIC844_11.nno protein toxin.

DETAILED DESCRIPTION OF THE INVENTION

Engineered insecticidal proteins that exhibit surprisingly higher levels of toxic activity against Lepidopteran species and a broader insecticidal spectrum compared to other previously known Lepidopteran insecticidal proteins are provided herein. These engineered insecticidal proteins are derived from insecticidal scaffold proteins, which serve as templates for various amino acid modifications. Examples of such insecticidal scaffold proteins include but are not limited to Cry1Da1 and TIC844 (a homolog of Cry1Da1). TIC844 comprises the Cry1Da1 core toxin (i.e., domains I, II and III) but utilizes the Cry1Ab3 protoxin domain to ensure good expression in *Bacillus thuringiensis* (Bt). Expression of Cry1Da1 in Bt is poor when using the native Cry1Da1 protoxin domain. However, as demonstrated in this application, the expression of Cry1Da1 core toxin is remarkably improved in a crystalliferous strains of Bt when the native protoxin domain is removed and the Cry1Da1 core toxin coding segment is fused in frame with a segment encoding the Cry1Ab3 protoxin domain. Notably, the scaffold proteins TIC844 (SEQ ID NO:14) and Cry1Da1 (SEQ ID NO:2) do not exhibit the commercially useful Lepidopteran inhibitory spectrum and improved Lepidopteran inhibitory activity observed in the engineered insecticidal proteins.

The engineered insecticidal proteins disclosed herein are related by amino acid modifications such that the modified proteins exhibit enhanced Lepidopteran inhibitory spectrum and/or improved Lepidopteran inhibitory activity compared

to the parent scaffold protein, TIC844 or Cry1Da1. The phrases “more active”, “improved activity”, “enhanced specificity”, “increased toxic potency”, “increased toxicity”, “improved Lepidopteran inhibitory activity”, “greater Lepidopteran inhibitory activity”, and “enhanced Lepidopteran inhibitory spectrum” refer to a comparison of the activity of an engineered insecticidal protein to the activity of a scaffold protein (TIC844 or Cry1Da1) against a Lepidopteran insect, wherein the activity attributed to the engineered insecticidal protein is greater than the activity attributed to the scaffold protein. In certain embodiments, the engineered insecticidal proteins provided herein exhibit an enhanced Lepidopteran inhibitory spectrum and/or improved or greater Lepidopteran inhibitory activity when compared to the activities of the scaffold TIC844 or Cry1Da1 protein where the Lepidopteran pest species include, but are not limited to, *Helicoverpa zea* and *Spodoptera frugiperda*.

As used herein, the terms and phrases “active” or “activity”; “pesticidal activity” or “pesticidal”; or “insecticidal activity”, “insect inhibitory”, “insecticidal”, or “an insect inhibitory amount”, refer to efficacy of a toxic agent, such as an insecticidal protein, in inhibiting (inhibiting growth, feeding, fecundity, or viability), suppressing (suppressing growth, feeding, fecundity, or viability), controlling (controlling the pest infestation, controlling the pest feeding activities on a particular crop containing an effective amount of a disclosed engineered insecticidal protein) or killing (causing the morbidity, mortality, or reduced fecundity of) a pest. Similarly, a “Lepidopteran inhibitory amount” refers to an amount of a toxic agent, such as an insecticidal protein, that results in any measurable inhibition of Lepidopteran viability, Lepidopteran growth, Lepidopteran development, Lepidopteran reproduction, Lepidopteran feeding behavior, Lepidopteran mating behavior and/or any measurable decrease in the adverse effects caused to a plant by Lepidopteran feeding. These terms are intended to include the result of providing a pesticidally effective amount of a toxic agent to a pest where the exposure of the pest to the toxic agent results in morbidity, mortality, reduced fecundity, or stunting. These terms also include repulsion of the pest from the plant, a tissue of the plant, a plant part, seed, plant cells, or from the particular geographic location where the plant may be growing, as a result of providing a pesticidally effective amount of the toxic agent in or on the plant. In general, pesticidal activity refers to the ability of a toxic agent to be effective in inhibiting the growth, development, viability, feeding behavior, mating behavior, fecundity, or any measurable decrease in the adverse effects caused by an insect feeding on this protein, protein fragment, protein segment or polynucleotide of a particular target pest, including but not limited to insects of the order Lepidoptera. The toxic agent can be produced by the plant or can be applied to the plant or to the environment within the location where the plant is located.

A pesticidally effective amount of a toxic agent, when provided in the diet of a target pest, exhibits pesticidal activity when the toxic is ingested by the pest. A toxic agent can be a pesticidal protein or one or more chemical agents known in the art. Pesticidal or insecticidal chemical agents and pesticidal or insecticidal protein agents can be used alone or in combinations with each other. Chemical agents include but are not limited to dsRNA molecules targeting specific genes for suppression in a target pest, organochlorides, organophosphates, carbamates, pyrethroids, neonicotinoids, and ryanoids. Pesticidal or insecticidal protein agents include the engineered insecticidal proteins set forth in this application, as well as other proteinaceous toxic

agents including those that target Lepidopteran pest species, as well as protein toxins that are used to control other plant pests such as Cry proteins available in the art for use in controlling Coleopteran, Hemipteran and Homopteran species.

The term “segment” or “fragment” is used herein to describe consecutive amino acid or nucleic acid sequences that are shorter than the complete amino acid or nucleic acid sequence describing the engineered insecticidal proteins.

It is intended that reference to a pest, particularly a pest of a crop plant, means insect pests of crop plants, particularly those Lepidopteran insect pests that are controlled by the disclosed engineered insecticidal proteins. However, reference to a pest can also include Coleopteran, Hemipteran and Homopteran insect pests of plants, as well as nematodes and fungi, when toxic agents targeting these pests are co-localized or present together with the disclosed engineered insecticidal proteins.

The disclosed engineered insecticidal proteins exhibit insecticidal activity towards insect pests from the Lepidopteran insect species, including adults, pupae, larvae, and neonates. The insects of the order Lepidoptera include, but are not limited to, armyworms, cutworms, loopers, and heliothines in the Family Noctuidae, e.g., fall armyworm (*Spodoptera frugiperda*), beet armyworm (*Spodoptera exigua*), bertha armyworm (*Mamestra configurata*), black cutworm (*Agrotis ipsilon*), cabbage looper (*Trichoplusia ni*), soybean looper (*Pseudoplusia includens*), velvetbean caterpillar (*Anticarsia gemmatilis*), green cloverworm (*Hypena scabra*), tobacco budworm (*Heliothis virescens*), granulate cutworm (*Agrotis subterranea*), armyworm (*Pseudaletia unipuncta*), western cutworm (*Agrotis orthogonia*); borers, casebearers, webworms, coneworms, cabbageworms and skeletonizers from the Family Pyralidae, e.g., European corn borer (*Ostrinia nubilalis*), navel orangeworm (*Amyelois transitella*), corn root webworm (*Crambus caliginosellus*), sod webworm (*Herpetogramma licarsialis*), sunflower moth (*Homoeosoma electellum*), lesser cornstalk borer (*Elasmopalpus lignosellus*); leafrollers, budworms, seed worms, and fruit worms in the Family Tortricidae, e.g., codling moth (*Cydia pomonella*), grape berry moth (*Endopiza viteana*), oriental fruit moth (*Grapholita molesta*), sunflower bud moth (*Suleima helianthana*); and many other economically important Lepidoptera, e.g., diamondback moth (*Plutella xylostella*), pink bollworm (*Pectinophora gossypiella*) and gypsy moth (*Lymantria dispar*). Other insect pests of order Lepidoptera include, e.g., *Alabama argillacea* (cotton leaf worm), *Archips argyrospila* (fruit tree leaf roller), *Archips rosana* (European leafroller) and other *Archips* species, *Chilo suppressalis* (Asiatic rice borer, or rice stem borer), *Cnaphalocrocis medinalis* (rice leaf roller), *Crambus caliginosellus* (corn root webworm), *Crambus teterrellus* (bluegrass webworm), *Diatraea grandiosella* (southwestern corn borer), *Diatraea saccharalis* (surgarcane borer), *Earias insulana* (spiny bollworm), *Earias vittella* (spotted bollworm), *Helicoverpa armigera* (American bollworm), *Helicoverpa zea* (corn earworm or cotton bollworm), *Heliothis virescens* (tobacco budworm), *Herpetogramma licarsialis* (sod webworm), *Lobesia botrana* (European grape vine moth), *Phyllocnistis citrella* (citrus leafminer), *Pieris brassicae* (large white butterfly), *Pieris rapae* (imported cabbageworm, or small white butterfly), *Plutella xylostella* (diamondback moth), *Spodoptera exigua* (beet armyworm), *Spodoptera litura* (tobacco cutworm, cluster caterpillar), and *Tuta absoluta* (tomato leafminer).

Reference in this application to an “isolated DNA molecule”, or an equivalent term or phrase, is intended to mean

that the DNA molecule is one that is present alone or in combination with other compositions, but not within its natural environment. For example, nucleic acid elements such as a coding sequence, intron sequence, untranslated leader sequence, promoter sequence, transcriptional termination sequence, and the like, that are naturally found within the DNA of the genome of an organism are not considered to be “isolated” so long as the element is within the genome of the organism and at the location within the genome in which it is naturally found. However, each of these elements, and subparts of these elements, would be “isolated” within the scope of this disclosure so long as the element is not within the genome of the organism and at the location within the genome in which it is naturally found. Similarly, a nucleotide sequence encoding an insecticidal protein or any naturally occurring insecticidal variant of that protein would be an isolated nucleotide sequence so long as the nucleotide sequence was not within the DNA of the bacterium from which the sequence encoding the protein is naturally found. A synthetic nucleotide sequence encoding the amino acid sequence of a naturally occurring insecticidal protein would be considered to be isolated for the purposes of this disclosure. For the purposes of this disclosure, any transgenic nucleotide sequence, i.e., the nucleotide sequence of the DNA inserted into the genome of the cells of a plant or bacterium, or present in an extrachromosomal vector, would be considered to be an isolated nucleotide sequence whether it is present within the plasmid or similar structure used to transform the cells, within the genome of the plant or bacterium, or present in detectable amounts in tissues, progeny, biological samples or commodity products derived from the plant or bacterium.

As described further in the Examples, repetitive rounds of engineering, testing and selecting of over two thousand (2000) amino acid sequence variants of TIC844 and Cry1Da1 resulted in the identification of certain amino acid residues that may be substituted, inserted or deleted from the given scaffold protein to produce engineered insecticidal proteins that exhibit an expanded Lepidopteran inhibitory spectrum and/or improved Lepidopteran inhibitory activity (i.e., more toxic; less insecticidal protein required to obtain same level of mortality) when compared to the spectrum and activity of the baseline scaffold proteins, TIC844 or Cry1Da1. These repetitive rounds of engineering, testing and selecting also resulted in the identification of neutral amino acid residue substitutions, insertions or deletions in the TIC844 and Cry1Da1 scaffold proteins that do not change the proteins' insect inhibitory spectrum or activity. The specific amino acid residues in the TIC844 and Cry1Da1 scaffold that can be modified to yield an enhanced Lepidopteran inhibitory spectrum and/or improved Lepidopteran inhibitory activity relative to TIC844 or Cry1Da1 are identified herein. In certain embodiments, the engineered insecticidal protein provided herein can exhibit about an eight fold or greater Lepidopteran inhibitory activity against a Lepidopteran pest species than a scaffold protein of SEQ ID NO:14 (TIC844) or SEQ ID NO:2 (Cry1Da1).

The “engineering” in these repetitive rounds included identifying relevant residues in the scaffold protein to modify to create a modified test protein, and cloning and expressing the resultant modified test proteins. The atomic structure of the scaffold proteins was used to guide and complement semi-random approaches of selecting amino acid residues to modify for engineering. The “testing” in these repetitive rounds included comparing the Lepidopteran species activities of a modified test protein to its parent scaffold protein. The “selecting” in these repetitive

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rounds included identifying modified test proteins with improved activity (improved variants) and the relevant residues which were engineered to create these improved variants (these improved variants are referred to herein as “engineered insecticidal proteins”).

Examples of methods for testing and selecting engineered insecticidal proteins include administering identical amounts of a modified test protein and of a scaffold protein (TIC844 or Cry1Da1) to an insect pest under controlled assay conditions and measuring and comparing the potency of the modified test and scaffold proteins. Another method for testing and selecting engineered insecticidal proteins includes determining the protein doses (e.g., protein concentration in diet) of a modified test protein and of a scaffold protein (TIC844 or Cry1Da1) which elicit equivalent insect population responses under controlled assay conditions (i.e., obtaining a dose response curve). A statistically robust dose response value used for comparison would be the median lethal concentration (LC_{50}) required to kill 50% of a test population or the molting inhibition concentration (“ MIC_{50} ”), the median concentration required to inhibit molting by 50%).

In certain embodiments, the engineered insecticidal proteins described herein include at least one amino acid

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modification of the following relative positions of TIC844 (SEQ ID NO:14) or Cry1Da1 (SEQ ID NO:2): serine at position 282 replaced by lysine or valine, tyrosine at position 316 replaced by serine, isoleucine at position 368 replaced by proline or arginine, serine at 374 replaced by arginine, asparagine at position 375 replaced by histidine, and isoleucine at position 432 replaced by leucine. The engineered insecticidal proteins can also include at least two, three, four, or more of these amino acid substitutions or deletions within the same engineered insecticidal protein sequence.

The engineered insecticidal proteins that include these amino acid modifications include the proteins set forth as SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:12, SEQ ID NO:16, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:26, SEQ ID NO:32, SEQ ID NO:34, SEQ ID NO:36, SEQ ID NO:38, SEQ ID NO:40, and SEQ ID NO:44, and insect inhibitory fragments thereof. Each of these engineered insecticidal proteins has a measured mass of about 132 k-Daltons. Individual characteristics of the insecticidal scaffold proteins TIC844 and Cry1Da1 and the engineered insecticidal proteins derived therefrom are reported in Table 1.

TABLE 1

Characteristics of TIC844, Cry1Da1 and the Engineered Insecticidal Proteins.								
Protein (Name/SEQ ID NO.)	Molecular Weight (in Daltons)	Amino Acid Length	Isoelectric Point	Charge at PH 7.0	No. of Strongly Basic (—) Amino Acids	No. of Strongly Acidic Amino Acids	No. of Hydrophobic Amino Acids	No. of Polar Amino Acids
Cry1Da1/ NO: 2	132481.87	1165	5.087	−39.319	113	156	388	347
Cry1Da1_3/ NO: 4	132405.77	1165	5.087	−39.318	113	156	388	347
Cry1Da1_4/ NO: 6	132550.98	1165	5.112	−38.319	114	156	388	346
Cry1Da1_5/ NO: 8	132448.80	1165	5.112	−38.318	114	156	387	347
Cry1Da1_6/ NO: 10	132430.82	1165	5.112	−38.319	114	156	387	346
Cry1Da1_7/ NO: 12	132401.78	1165	5.087	−39.318	113	156	388	346
TIC844/ NO: 14	129182.91	1139	5.026	−39.540	110	153	382	340
TIC844_2/ NO: 16	129129.85	1139	5.048	−39.373	110	153	382	339
TIC844_4/ NO: 18	129106.81	1139	5.026	−39.539	110	153	382	340
TIC844_5/ NO: 20	1291118.08	1069	5.325	−27.535	105	136	363	321
TIC844_6/ NO: 22	129252.02	1139	5.050	−38.540	111	153	382	339
TIC844_7/ NO: 24	129149.84	1139	5.050	−38.539	111	153	381	340
TIC844_8/ NO: 26	129102.82	1139	5.026	−39.539	110	153	382	339
Cry1Da1/ NO: 28	132481.87	1165	5.087	−39.319	113	156	388	347
Cry1Da1_2.nno/ NO: 30	132552.95	1166	5.087	−39.319	113	156	389	347
Cry1Da1_3.nno/ NO: 32	132476.85	1166	5.087	−39.318	113	156	389	347
Cry1Da1_4.nno/ NO: 34	132622.06	1166	5.112	−38.319	114	156	389	346
Cry1Da1_5.nno/ NO: 36	132519.88	1166	5.112	−38.318	114	156	388	347
Cry1Da1_6.nno/ NO: 38	132501.90	1166	5.112	−39.319	114	156	388	346
Cry1Da1_7.nno/ NO: 40	132472.86	1166	5.087	−39.318	113	156	389	346

TABLE 1-continued

Characteristics of TIC844, Cry1Da1 and the Engineered Insecticidal Proteins.								
Protein (Name/SEQ ID NO.)	Molecular Weight (in Daltons)	Amino Acid Length	Isoelectric Point	Charge at PH 7.0	No. of Strongly Basic (—) Amino Acids	No. of Strongly Acidic Amino Acids	No. of Hydrophobic Amino Acids	No. of Polar Amino Acids
TIC844_9.nno/ NO: 42	129253.99	1140	5.026	-39.540	110	153	383	340
TIC844_11.nno/ NO: 44	129173.90	1140	5.026	-39.539	110	153	383	339

Fragments of the engineered insecticidal proteins described herein can be truncated forms wherein one or more amino acids are deleted from the N-terminal end, C-terminal end, the middle of the protein, or combinations thereof without a loss of insect inhibitory activity. These fragments should retain the insect inhibitory activity of the parent engineered insecticidal protein.

Proteins that resemble the engineered insecticidal proteins can be identified by comparison to each other using various computer based algorithms known in the art. For example, amino acid sequence identities of proteins related to the engineered insecticidal proteins can be analyzed using a Clustal W alignment using these default parameters: Weight matrix: blosum, Gap opening penalty: 10.0, Gap extension penalty: 0.05, Hydrophilic gaps: On, Hydrophilic residues: GPSNDQERK, Residue-specific gap penalties: On (Thompson, et al (1994) Nucleic Acids Research, 22:4673-4680). Percent amino acid identity is further calculated by the product of 100% multiplied by (amino acid identities/length of subject protein). Other alignment algorithms are also available in the art and provide results similar to those obtained using a Clustal W alignment.

As described further in the Examples of this application, synthetic or artificial sequences encoding the scaffold proteins and the engineered insecticidal proteins were designed for use in plants. Exemplary synthetic nucleotide sequences that were designed for use in plants are set forth in SEQ ID NO:27 (Cry1Da1.nno), SEQ ID NO:29 (Cry1Da1_2.nno), SEQ ID NO:31 (Cry1Da1_3.nno), SEQ ID NO:33 (Cry1Da1_4.nno), SEQ ID NO:35 (Cry1Da1_5.nno), SEQ ID NO:37 (Cry1Da1_6.nno), SEQ ID NO:39 (Cry1Da1_7.nno), SEQ ID NO:41 (TIC844_9.nno) and SEQ ID NO:43 (TIC844_11.nno).

Expression cassettes and vectors containing these synthetic or artificial nucleotide sequences were constructed and introduced into corn, cotton and soybean plant cells in accordance with transformation methods and techniques known in the art. Transformed cells were regenerated into transformed plants that were observed to be expressing the engineered insecticidal protein or the scaffold protein. To test pesticidal activity, bioassays were performed in the presence of Lepidopteran pest larvae using plant leaf disks obtained from the transformed plants.

Recombinant nucleic acid molecule compositions that encode the engineered insecticidal proteins are contemplated. For example, an engineered insecticidal protein can be expressed with recombinant DNA constructs in which a polynucleotide molecule with an ORF encoding the engineered insecticidal protein is operably linked to genetic expression elements such as a promoter and any other regulatory element necessary for expression in the system for which the construct is intended. Non-limiting examples include a plant-functional promoter operably linked to the

synthetic engineered insecticidal protein encoding sequences for expression of the engineered insecticidal protein in plants or a Bt-functional promoter operably linked to an engineered insecticidal protein encoding sequence for expression of the protein in a Bt bacterium or other *Bacillus* species. Other elements can be operably linked to the engineered insecticidal protein encoding sequences including, but not limited to, enhancers, introns, untranslated leaders, encoded protein immobilization tags (HIS-tag), translocation peptides (i.e., plastid transit peptides, signal peptides), polypeptide sequences for post-translational modifying enzymes, ribosomal binding sites, and RNAi target sites. Exemplary recombinant polynucleotide molecules provided herein include, but are not limited to, a heterologous promoter operably linked to a polynucleotide such as SEQ ID NO:3, SEQ ID NO:5, SEQ ID NO:7, SEQ ID NO:9, SEQ ID NO:11, SEQ ID NO:15, SEQ ID NO:17, SEQ ID NO:19, SEQ ID NO:21, SEQ ID NO:23, SEQ ID NO:25, SEQ ID NO:31, SEQ ID NO:33, SEQ ID NO:35, SEQ ID NO:37, SEQ ID NO:39 and SEQ ID NO:43, that encodes the polypeptide or protein having the amino acid sequence as set forth in SEQ ID NO:4 (Cry1Da1_3), SEQ ID NO:6 (Cry1Da1_4), SEQ ID NO:8 (Cry1Da1_5), SEQ ID NO:10 (Cry1Da1_6), SEQ ID NO:12 (Cry1Da1_7), SEQ ID NO:16 (TIC844_2), SEQ ID NO:18 (TIC844_4), SEQ ID NO:20 (TIC844_5), SEQ ID NO:22 (TIC844_6), SEQ ID NO:24 (TIC844_7), SEQ ID NO:26 (TIC844_8), SEQ ID NO:32 (Cry1Da1_3.nno), SEQ ID NO:34 (Cry1Da1_4.nno), SEQ ID NO:36 (Cry1Da1_5.nno), SEQ ID NO:38 (Cry1Da1_6.nno), SEQ ID NO:40 (Cry1Da1_7.nno) and SEQ ID NO:44 (TIC844_11.nno). A heterologous promoter can also be operably linked to synthetic DNA coding sequences encoding a plastid targeted engineered insecticidal protein and untargeted engineered insecticidal protein. It is contemplated that the codons of a recombinant nucleic acid molecule encoding for an engineered insecticidal protein disclosed herein can be substituted by synonymous codons (known in the art as a silent substitution).

A recombinant DNA molecule or construct comprising an engineered insecticidal protein encoding sequence can further comprise a region of DNA that encodes for one or more toxic agents which can be configured to concomitantly express or co-express with a DNA sequence encoding an engineered insecticidal protein, a protein different from an engineered insecticidal protein, an insect inhibitory dsRNA molecule, or an ancillary protein. Ancillary proteins include, but are not limited to, co-factors, enzymes, binding-partners, or other agents that function to aid in the effectiveness of an insect inhibitory agent, for example, by aiding its expression, influencing its stability in plants, optimizing free energy for oligomerization, augmenting its toxicity, and increasing its spectrum of activity. An ancillary protein may

facilitate the uptake of one or more insect inhibitory agents, for example, or potentiate the toxic effects of the toxic agent.

A recombinant DNA molecule or construct can be assembled so that all proteins or dsRNA molecules are expressed from one promoter or each protein or dsRNA molecule is under separate promoter control or some combination thereof. The proteins of this invention can be expressed from a multi-gene expression system in which an engineered insecticidal protein is expressed from a common nucleotide segment which also contains other open reading frames and promoters, depending on the type of expression system selected. For example, a bacterial multi-gene expression system can utilize a single promoter to drive expression of multiply-linked/tandem open reading frames from within a single operon (i.e., polycistronic expression). In another example, a plant multi-gene expression system can utilize multiply-unlinked expression cassettes, each expressing a different protein or other toxic agent such as one or more dsRNA molecules.

Recombinant nucleic acid molecules or recombinant DNA constructs comprising an engineered insecticidal protein encoding sequence can be delivered to host cells by vectors, e.g., a plasmid, baculovirus, synthetic chromosome, virion, cosmid, phagemid, phage, or viral vector. Such vectors can be used to achieve stable or transient expression of an engineered insecticidal protein encoding sequence in a host cell, or subsequent expression of the encoded polypeptide. An exogenous recombinant polynucleotide or recombinant DNA construct that comprises an engineered insecticidal protein sequence encoding sequence and that is introduced into a host cell is referred herein as a "transgene".

Transgenic bacteria, transgenic plant cells, transgenic plants, and transgenic plant parts that contain a polynucleotide that encodes any one or more of the engineered insecticidal proteins are provided herein. The term "bacterial cell" or "bacterium" can include, but is not limited to, an *Agrobacterium*, a *Bacillus*, an *Escherichia*, a *Salmonella*, a *Pseudomonas*, or a *Rhizobium* cell. The term "plant cell" or "plant" can include but is not limited to a dicotyledonous cell or a monocotyledonous cell. Contemplated plants and plant cells include, but are not limited to, alfalfa, banana, barley, bean, broccoli, cabbage, brassica, carrot, cassava, castor, cauliflower, celery, chickpea, Chinese cabbage, citrus, coconut, coffee, corn, clover, cotton, a cucurbit, cucumber, Douglas fir, eggplant, eucalyptus, flax, garlic, grape, hops, leek, lettuce, Loblolly pine, millets, melons, nut, oat, olive, onion, ornamental, palm, pasture grass, pea, peanut, pepper, pigeonpea, pine, potato, poplar, pumpkin, Radiata pine, radish, rapeseed, rice, rootstocks, rye, safflower, shrub, sorghum, Southern pine, soybean, spinach, squash, strawberry, sugar beet, sugarcane, sunflower, sweet corn, sweet gum, sweet potato, switchgrass, tea, tobacco, tomato, triticale, turf grass, watermelon, and wheat plant cell or plant. In certain embodiments, transgenic plants and transgenic plant parts regenerated from a transgenic plant cell are provided. In certain embodiments, the transgenic plants can be obtained from a transgenic seed, by cutting, snapping, grinding or otherwise disassociating the part from the plant. In certain embodiments, the plant part can be a seed, a boll, a leaf, a flower, a stem, a root, or any portion thereof, or a non-regenerable portion of a transgenic plant part. As used in this context, a "non-regenerable" portion of a transgenic plant part is a portion that cannot be induced to form a whole plant or that cannot be induced to form a whole plant that is capable of sexual and/or asexual reproduction. In certain embodiments, a non-regenerable portion of a plant part is a portion of a transgenic seed, boll, leaf, flower, stem, or root.

Methods of making transgenic plants that comprise Lepidoptera-inhibitory amounts of an engineered insecticidal proteins are provided. Such plants can be made by introducing a polynucleotide that encodes the engineered insecticidal proteins provided in this application into a plant cell, and selecting a plant derived from said plant cell that expresses an insect or Lepidoptera-inhibitory amount of the engineered insecticidal protein. Plants can be derived from the plant cells by regeneration, seed, pollen, or meristem transformation techniques. Methods for transforming plants are known in the art.

Plants expressing the engineered insecticidal proteins can be crossed by breeding with transgenic events expressing other insecticidal proteins and/or expressing other transgenic traits such as other insect control traits, herbicide tolerance genes, genes conferring yield or stress tolerance traits, and the like, or such traits can be combined in a single vector so that the traits are all linked.

Processed plant products, wherein the processed product comprises a detectable amount of an engineered insecticidal protein, an insect inhibitory segment or fragment thereof, or any distinguishing portion thereof, are also disclosed in this application. In certain embodiments, the processed product is selected from the group consisting of plant parts, plant biomass, oil, meal, sugar, animal feed, flour, flakes, bran, lint, hulls, processed seed, and seed. In certain embodiments, the processed product is non-regenerable. The plant product can comprise commodity or other products of commerce derived from a transgenic plant or transgenic plant part, where the commodity or other products can be tracked through commerce by detecting nucleotide segments or expressed RNA or proteins that encode or comprise distinguishing portions of an engineered insecticidal protein.

Methods of controlling insects, in particular Lepidoptera infestations of crop plants, with the engineered insecticidal proteins are also disclosed in this application. Such methods can comprise growing a plant comprising an insect- or Lepidoptera-inhibitory amount of the engineered insecticidal protein. In certain embodiments, such methods can further comprise any one or more of: (i) applying any composition comprising or encoding an engineered insecticidal protein to a plant or a seed that gives rise to a plant; and (ii) transforming a plant or a plant cell that gives rise to a plant with a polynucleotide encoding an engineered insecticidal protein. In general, it is contemplated that engineered insecticidal protein can be provided in a composition, provided in a microorganism, or provided in a transgenic plant to confer insect inhibitory activity against Lepidopteran insects.

In certain embodiments, the engineered insecticidal protein is the insecticidally active ingredient of an insect inhibitory composition prepared by culturing recombinant *Bacillus* or any other recombinant bacterial cell transformed to express an engineered insecticidal protein under conditions suitable for expression. Such a composition can be prepared by desiccation, lyophilization, homogenization, extraction, filtration, centrifugation, sedimentation, or concentration of a culture of such recombinant cells expressing/producing the engineered insecticidal protein. Such a process can result in a *Bacillus* or other entomopathogenic bacterial cell extract, cell suspension, cell homogenate, cell lysate, cell supernatant, cell filtrate, or cell pellet. By obtaining the engineered insecticidal protein so produced, a composition that includes the engineered insecticidal protein can include bacterial cells, bacterial spores, and parasporal inclusion bodies and can be formulated for various uses,

including as agricultural insect inhibitory spray products or as insect inhibitory formulations in diet bioassays.

In an embodiment, in order to reduce the likelihood of resistance development, an insect inhibitory composition or transgenic plant comprising an engineered insecticidal protein can further comprise at least one additional toxic agent that exhibits insect inhibitory activity against the same Lepidopteran insect species, but which is different from the engineered insecticidal protein. Possible additional toxic agents for such a composition include an insect inhibitory protein and an insect inhibitory dsRNA molecule. One example for the use of such ribonucleotide sequences to control insect pests is described in Baum, et al. (U.S. Patent Publication 2006/0021087 A1). Such additional polypeptide(s) for the control of Lepidopteran pests may be selected from the group consisting of an insect inhibitory protein, such as, but not limited to, Cry1A (U.S. Pat. No. 5,880,275), Cry1Ab, Cry1Ac, Cry1A.105, Cry1Ae, Cry1B (U.S. patent Publication Ser. No. 10/525,318), Cry1C (U.S. Pat. No. 6,033,874), Cry1D, Cry1E, Cry1F, and Cry1A/F chimeras (U.S. Pat. Nos. 7,070,982; 6,962,705; and 6,713,063), Cry1G, Cry1H, Cry1I, Cry1J, Cry1K, Cry1L, Cry2A, Cry2Ab (U.S. Pat. No. 7,064,249), Cry2Ae, Cry4B, Cry6, Cry7, Cry8, Cry9, Cry15, Cry43A, Cry43B, Cry51Aa1, ET66, TIC400, TIC800, TIC834, TIC1415, Vip3A, VIP3Ab, VIP3B, AXMI-001, AXMI-002, AXMI-030, AXMI-035, AND AXMI-045 (U.S. Patent Publication 2013-0117884 A1), AXMI-52, AXMI-58, AXMI-88, AXMI-97, AXMI-102, AXMI-112, AXMI-117, AXMI-100 (U.S. Patent Publication 2013-0310543 A1), AXMI-115, AXMI-113, AXMI-005 (U.S. Patent Publication 2013-0104259 A1), AXMI-134 (U.S. Patent Publication 2013-0167264 A1), AXMI-150 (U.S. Patent Publication 2010-0160231 A1), AXMI-184 (U.S. Patent Publication 2010-0004176 A1), AXMI-196, AXMI-204, AXMI-207, AXMI-209 (U.S. Patent Publication 2011-0030096 A1), AXMI-218, AXMI-220 (U.S. Patent Publication 2014-0245491 A1), AXMI-221z, AXMI-222z, AXMI-223z, AXMI-224z, AXMI-225z (U.S. Patent Publication 2014-0196175 A1), AXMI-238 (U.S. Patent Publication 2014-0033363 A1), AXMI-270 (U.S. Patent Publication 2014-0223598 A1), AXMI-345 (U.S. Patent Publication 2014-0373195 A1), DIG-3 (U.S. Patent Publication 2013-0219570 A1), DIG-5 (U.S. Patent Publication 2010-0317569 A1), DIG-11 (U.S. Patent Publication 2010-0319093 A1), AfIP-1A and derivatives thereof (U.S. Patent Publication 2014-0033361 A1), AfIP-1B and derivatives thereof (U.S. Patent Publication 2014-0033361 A1), PIP-1A/PIB (U.S. Patent Publication 2014-0007292 A1), PSEEN3174 (U.S. Patent Publication 2014-0007292 A1), AECFG-592740 (U.S. Patent Publication 2014-0007292 A1), Pput_1063 (U.S. Patent Publication 2014-0007292 A1), Pput_1064 (U.S. Patent Publication 2014-0007292 A1), GS-135 and derivatives thereof (U.S. Patent Publication 2012-0233726 A1), GS153 and derivatives thereof (U.S. Patent Publication 2012-0192310 A1), GS154 and derivatives thereof (U.S. Patent Publication 2012-0192310 A1), GS155 and derivatives thereof (U.S. Patent Publication 2012-0192310 A1), SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2012-0167259 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2012-0047606 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2011-0154536 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2011-0112013 A1, SEQ ID NO:2 and 4 and derivatives thereof as described in U.S. Patent Publication 2010-0192256 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication

2010-0077507 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2010-0077508 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Patent Publication 2009-0313721 A1, SEQ ID NO:2 or 4 and derivatives thereof as described in U.S. Patent Publication 2010-0269221 A1, SEQ ID NO:2 and derivatives thereof as described in U.S. Pat. No. 7,772,465 (B2), CF161_0085 and derivatives thereof as described in WO2014/008054 A2, Lepidopteran toxic proteins and their derivatives as described in US Patent Publications US2008-0172762 A1, US2011-0055968 A1, and US2012-0117690 A1; SEQ ID NO:2 and derivatives thereof as described in U.S. Pat. No. 7,510,878(B2), SEQ ID NO:2 and derivatives thereof as described in U.S. Pat. No. 7,812,129(B1); and the like.

In other embodiments, an insect inhibitory composition or transgenic plant can further comprise at least one additional toxic agent that exhibits insect inhibitory activity to an insect pest that is not inhibited by the engineered insecticidal proteins of the present invention (such as Coleopteran, Hemipteran and Homopteran pests), in order to expand the spectrum of insect inhibition obtained.

Such additional toxic agent for the control of Coleopteran pests may be selected from the group consisting of an insect inhibitory protein, such as, but not limited to, Cry3Bb (U.S. Pat. No. 6,501,009), Cry1C variants, Cry3A variants, Cry3, Cry3B, Cry34/35, 5307, AXMI134 (U.S. Patent Publication 2013-0167264 A1) AXMI-184 (U.S. Patent Publication 2010-0004176 A1), AXMI-205 (U.S. Patent Publication 2014-0298538 A1), axmi207 (U.S. Patent Publication 2013-0303440 A1), AXMI-218, AXMI-220 (U.S. Patent Publication 20140245491A1), AXMI-221z, AXMI-223z (U.S. Patent Publication 2014-0196175 A1), AXMI-279 (U.S. Patent Publication 2014-0223599 A1), AXMI-R1 and variants thereof (U.S. Patent Publication 2010-0197592 A1, TIC407, TIC417, TIC431, TIC807, TIC853, TIC901, TIC1201, TIC3131, DIG-10 (U.S. Patent Publication 2010-0319092 A1), eHIPs (U.S. Patent Application Publication No. 2010/0017914), IP3 and variants thereof (U.S. Patent Publication 2012-0210462 A1), and ω -Hexatoxin-Hv1a (U.S. Patent Application Publication US2014-0366227 A1).

Such additional toxic agent for the control of Hemipteran pests may be selected from the group consisting of Hemipteran-active proteins such as, but not limited to, TIC1415 (US Patent Publication 2013-0097735 A1), TIC807 (U.S. Pat. No. 8,609,936), TIC834 (U.S. Patent Publication 2013-0269060 A1), AXMI-036 (U.S. Patent Publication 2010-0137216 A1), and AXMI-171 (U.S. Patent Publication 2013-0055469 A1). Additional polypeptides for the control of Coleopteran, Lepidopteran, and Hemipteran insect pests can be found on the *Bacillus thuringiensis* toxin nomenclature website maintained by Neil Crickmore (on the world wide web at bt.nomenclature.info).

Engineered insecticidal protein-encoding sequences and sequences having a substantial percentage identity to the engineered insecticidal proteins can be identified using methods known to those of ordinary skill in the art such as polymerase chain reaction (PCR), thermal amplification and hybridization. For example, the engineered insecticidal proteins can be used to produce antibodies that bind specifically to related proteins, and can be used to screen for and to find other proteins that are closely related.

Furthermore, nucleotide sequences encoding the engineered insecticidal proteins can be used as probes and primers for screening to identify other members of the class using thermal-cycle or isothermal amplification and hybridization methods. For example, oligonucleotides derived from sequences as set forth in SEQ ID NO: 3 can be used to

determine the presence or absence of an engineered insecticidal transgene in a deoxyribonucleic acid sample derived from a commodity product. Given the sensitivity of certain nucleic acid detection methods that employ oligonucleotides, it is anticipated that oligonucleotides derived from sequences as set forth in any of SEQ ID NO: 3 can be used to detect the respective engineered insecticidal protein in commodity products derived from pooled sources where only a fraction of the commodity product is derived from a transgenic plant containing any of SEQ ID NO: 3.

Other features and advantages of the invention will be apparent from the following Examples and claims.

EXAMPLES

In view of the foregoing, those of skill in the art should appreciate that changes can be made in the specific aspects which are disclosed and still obtain a like or similar result without departing from the spirit and scope of the invention. Thus, specific structural and functional details disclosed herein are not to be interpreted as limiting. It should be understood that the entire disclosure of each reference cited herein is incorporated within the disclosure of this application.

Example 1

Design of Modified Test Proteins and Sample Preparation for Insect Bioassay Testing

This Example illustrates the methods undertaken to identify relevant amino acid residues in the scaffold proteins to modify to create modified test proteins, and the cloning and expressing of the resultant modified test proteins.

Several molecular engineering techniques were employed in a tiered approach to construct improved variants of Cry1Da1 having an enhanced Lepidopteran inhibitory spectrum and/or improved Lepidopteran inhibitory activity compared to the scaffold proteins of Cry1Da1 and TIC844, a homolog of Cry1Da1. The first tier, or initial round of design, was primarily hypothesis driven. The second and third tiers were statistically-driven rounds of design. For example, in the second tier of design, statistically non-deleterious mutations were combined with putative beneficial mutations to produce double mutations which satisfied defined statistical criteria. In the third tier of design, all the data from the previous tests was analyzed using multiple statistical methods. Only mutations showing statistically significant improvement in more than one statistical method were selected to the final pool of mutations. The variants designed in this tier contained one or two more positive mutations from variants previously confirmed positive. Thus, the third tier design significantly enriched the active variants compared to the first and second tier. As demonstrated in the subsequent Examples, the use of the three-tiered design strategy identified both single and synergistic mutations that provided significant improvement in activity against CEW for certain improved variants relative to the TIC844 and Cry1Da1 scaffolds.

The methods which were utilized to create the modified test proteins included, but were not limited to, semi-random modifications, directed modifications of variances in alignment of TIC844/Cry1Da1 with other native *Bacillus thuringiensis* (Bt) proteins, and structure/function assisted design. Examples of utilized molecular engineering techniques include the following.

Receptor binding. Susceptibility of Lepidopteran pests, specifically Corn Earworm (CEW, *Heliothis zea*) to Cry1Da1/TIC844 improved variants may be attributable to different targeted gut receptors. Designs which were utilized to improve binding to receptors in the gut, thus increasing toxicity, included: (1) mutating every position in the apical loops of domain II to all amino-acid types; and (2) swapping all possible combinations of the apical loops of domain II with those from other Cry1Da1 homologs (e.g., Cry1Db1, Cry1Dc1) and CEW-active three-domain toxins (e.g., Cry1Bb1, Cry1Ja1 and Cry2Ab2).

Alignment based approaches. Alignment of Cry1Da1 with other homologs (e.g., Cry1Db1 and Cry1Dc1) was used to identify regions of variability. As a result of the alignment, one hundred fifty (150) positions and two hundred ninety five (295) unique single mutations were identified. These positions were located throughout the three domains. Positions within four (4) amino acids from one another were grouped together. Only mutations from the same parental sequences were nominated for every group of positions, rendering one hundred thirty two (132) unique variants.

Surface mutagenesis approaches. The polynucleotides encoding the surface positions in domains II and III of the scaffold proteins were mutagenized by a scan. Amino acid residues were modified to alanine where an alanine was not already present in the scaffold protein. At surface positions where the native residues were lysine, arginine mutations were introduced in addition to the alanine mutations. The rationale for the lysine to arginine mutations was based on the observation that Lepidopteran-active toxins tend to have very few lysine and many arginine and, therefore, it was hypothesized that changing the surface lysine positions in domains II and III to arginine would increase the Lepidopteran activity of the modified test protein.

Alteration of proteolytic events. The proteolytic process was hypothesized to be an important aspect of the activity of three-domain toxins in the Lepidopteran insect guts. In order to test this, several sets of mutations were made to potentially alter any proteolytic cleavage. Potential cleavage sites are located at the N-terminus and between domain III and the protoxin. The mutational positions included predicted loop regions from the N-terminus to the beginning of helix 4 and from the C-terminus of domain III to ~40 amino acids into the protoxin. Generally, glycine residues were hypothesized to promote proteolysis either through proteolytic site recognition or by increasing the protein flexibility, thereby rendering it more susceptible to proteolytic cleavage. Further, trypsin and chymotrypsin are two proteases that are widely accepted as viable proteases in Lepidopteran midguts. Lysine residues provide recognition sites for trypsin and tyrosine residues provide recognition sites for chymotrypsin. Thus, selected mutational positions in the potential cleavage sites were mutated to either glycine, lysine or tyrosine.

Potential hot-spot mutations from other CEW-active toxins. Activity and absence of activity data against CEW for a large set of proteins (including chimeras, fragments and native sequences) was analyzed. Information gained from a statistical analysis of this data was utilized to identify potential specific mutations or positions for mutation that would be likely to increase CEW activity in the resultant modified test proteins.

The modified test proteins which resulted from the molecular engineering methodologies described above were cloned using methods known in the art into a recombinant Bt plasmid expression vector downstream of a sporulation specific expression promoter and transformed into an acrytalliferous Bt host cell.

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Example 2

Testing of Modified Test Proteins in Diet Bioassays
Against Lepidopteran Pests

This Example illustrates the testing of the modified test proteins created from the engineering efforts described in Example 1.

From the engineering efforts described in Example 1, about two thousand five hundred (2,500) recombinant Bt strains were produced which expressed more than two thousand three hundred (2,300) different modified test proteins. These modified test proteins were expressed in Bt and assayed for toxicity to various species of Lepidoptera. Feeding assays were conducted with neonate larvae (<24 hour post hatch) of various Lepidopteran species, including corn earworm (CEW, *Helicoverpa zea*) and fall armyworm (FAW, *Spodoptera frugiperda*). Insect eggs for the CEW testing were obtained from two different laboratory colonies: Benson Research, Carlisle, PA and Monsanto Company, Union City, TN. All of the expressed modified test proteins were tested on CEW and some of those modified test proteins demonstrating improved activity against CEW compared to their parent scaffold proteins were tested on FAW, in addition to performing additional bioassays to confirm CEW activity.

Various protocols for bioassays and scoring insects for mortality and stunting are known in the art. Variations of methods, such as those described in PCT Patent Application Publication No. WO 2012/139004 and in U.S. Pat. No. 7,927,598, were used.

Example 3

Modified Test Proteins Exhibiting Improved CEW
Activity

This Example illustrates the discovery of an enhanced Lepidopteran inhibitory spectrum and/or improved or greater Lepidopteran inhibitory activity for some of the modified test proteins when compared to the activities of the scaffold TIC844 or Cry1Da1 proteins in multiple testing rounds.

The modified test proteins created from the engineering efforts described in Example 1 and tested in insect bioassay as described in Example 2 were tested in repetitive rounds in which the Lepidopteran species activities of the modified test proteins were compared to their respective parent scaffold proteins (i.e., TIC844 or Cry1Da1). In a first round, three hundred and seventy (370) different modified test proteins demonstrated increased toxicity against CEW relative to TIC844 or Cry1Da1 in diet bioassays. In each of these diet bioassays, identical amounts of the protein (either modified test protein or scaffold protein) was provided to CEW under controlled single-dose assay conditions. The potency of the modified test proteins and scaffold proteins was determined by measuring and comparing the observed mortality and stunting of each of the modified test protein bioassays to the observed mortality and stunting of the parent scaffold protein bioassays.

Of the three hundred and seventy (370) modified test proteins which demonstrated increased toxicity against CEW when compared to the scaffold proteins in single-dose assay screens, about one hundred eighty (180) of them were further tested in FAW bioassays to determine whether these modified test proteins maintained or exhibited increased FAW activity compared to their scaffold protein parents.

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About forty (40) to fifty (50) of these modified test proteins exhibited similar or better FAW activity than their parent scaffold proteins. These further-screened modified test proteins were also tested in additional CEW bioassays to confirm CEW activity. These rounds of selecting and testing modified test proteins which demonstrated improved CEW activity while maintaining or improving FAW activity resulted in a final list of improved variants (referred to herein as the “engineered insecticidal proteins”). Table 2 identifies these engineered insecticidal proteins and the amino acid mutations in each engineered insecticidal protein. Table 2 also demonstrates the activity of the scaffold and the engineered insecticidal proteins against CEW and FAW (insecticidal activity is demonstrated in LC₅₀ value (the toxin concentration required to kill 50% of an insect population during a fixed exposure duration. The lower the LC₅₀ value, the greater the toxicity) and the MIC₅₀ value (the concentration required to inhibit molting to a specific instar of 50% of the larvae during a fixed exposure duration). This Table demonstrates that the engineered insecticidal proteins have improved CEW-activity, while maintaining or improving FAW activity.

TABLE 2

Amino Acid Mutations and Activity Data for Scaffold Proteins and Engineered Insecticidal Proteins.				
Protein (Name/SEQ ID NO.)	Amino Acid Mutations*	LC ₅₀ (µg/cm ²) against CEW Benzon colony with spore-crystal bioassay prep	MIC ₅₀ (µg/cm ²) against CEW Benzon colony with spore-crystal bioassay prep	
Cry1Da1/ NO: 2, 28	None (scaffold protein)	NA**	~3.0	
Cry1Da1_3/ NO: 4	Cry1Da1 + Y316S	NA**	NA**	
Cry1Da1_4/ NO: 6	Cry1Da1 + S374R	NA**	NA**	
Cry1Da1_5/ NO: 8	Cry1Da1 + Y316S_I368R	NA**	NA**	
Cry1Da1_6/ NO: 10	Cry1Da1 + S282K_Y316S_I368P	NA**	NA**	
Cry1Da1_7/ NO: 12	Cry1Da1 + S282V_Y316S_I368P	NA**	NA**	
TIC844/ NO: 14	None (scaffold protein)	41.90	3.73	
TIC844_2/ NO: 16	TIC844 + Y316S_N375H_I432L	0.81	0.65	
TIC844_4/ NO: 18	TIC844 + Y316S	0.98	0.57	
TIC844_5/ NO: 20	TIC844 + S282K_Y316S_I368P	0.32	0.33	
TIC844_6/ NO: 22	TIC844 + S374R	4.09	1.39	
TIC844_7/ NO: 24	TIC844 + Y316S_I368R	0.93	0.61	
TIC844_8/ NO: 26	TIC844 + S282V_Y316S_I368P	0.221	.064	

*The amino acid mutations are identified using the standard IUPAC amino acid code. See IUPAC-IUB Joint Commission on Biochemical Nomenclature, Nomenclature and Symbolism for Amino Acids and Peptides, Eur. J. Biochem. 138:9-37(1984). The first amino acid sequence abbreviation indicates the original amino acid in the given scaffold protein, the number represents the position of the amino acid, and the second amino acid sequence abbreviation indicates the amino acid placed in that position in the improved variant protein.

**The core toxin of Cry1Da1 is identical to the core toxin of TIC844.

Further demonstrating the enhanced Lepidopteran inhibitory spectrum and improved Lepidopteran inhibitory activity of the engineered insecticidal proteins, the lethality of engineered insecticidal protein TIC844_8 relative to its parent scaffold protein is demonstrated in FIG. 1. The bar chart of FIG. 1 demonstrates the MIC₅₀ values of TIC844_8 compared to the scaffold protein TIC844 for two different

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CEW colonies, Union City and Benzon. The bioassay results depicted in FIG. 1 were calculated from sucrose gradient-purified bioassay preparations. The reason these secondary bioassays were run with sucrose gradient-purified preparation of the proteins opposed to spore-crystal preparations of the proteins was to ensure that the improved activity of TIC844_8 persisted with more extensive purification. Further, the Union City colony was tested to confirm the improved activity observed on the Benzon colony. As demonstrated in FIG. 1, the mutations in three residues for TIC844_8 (S282V_Y316S_I368P), imparted an 8-fold improvement in CEW lethality, relative to TIC844, for the Union City colony and a 50-fold improvement in CEW lethality, relative to TIC844, for the Benzon colony.

Even further demonstrating enhanced Lepidopteran inhibitory spectrum and improved Lepidopteran inhibitory activity of the engineered insecticidal proteins, the insect

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activity profiles for TIC844 and TIC844_8 from diet bioassay studies, conducted against a broad spectrum of Lepidopteran insect species, are shown in Table 3. The insects tested against in the bioassay studies in Table 3 include black cutworm (BCW, *Agrotis ipsilon*), corn earworm (CEW, *Helicoverpa zea*), fall armyworm (FAW, *Spodoptera frugiperda*), southern armyworm (SAW, *Spodoptera eridania*), cabbage looper (CLW, *Trichoplusia ni*), European corn borer (ECB, *Ostrinia nubilalis*), southwestern corn borer (SWC, *Diatraea grandiosella*), tobacco budworm (TBW, *Heliothis virescens*), velvetbean caterpillars (VBC, *Anticarsia gemmatilis*), soybean looper (SBL, *Chrysodeixis includes*), and sugarcane borer (SCB, *Diatraea saccharalis*). This Table 3 demonstrates the enhanced Lepidopteran inhibitory spectrum of TIC844_8 compared to the parent scaffold protein TIC844, specifically with improved activity against CEW and VBC.

TABLE 3

Insect activity spectrum for TIC844 and TIC844_8.												
SEQ ID NO.	Protein	BCW	CEW	FAW	SAW	CLW	ECB	SWC	TBW	VBC	SBL	SCB
26	TIC844_8		*	*	*	*				*	*	*
14	TIC844			*	*	*					*	*

* Active against the indicated insect species.

The enhanced Lepidopteran inhibitory spectrum of the engineered insecticidal proteins is further demonstrated in Table 4 which depicts the insect activity profile for certain engineered insecticidal proteins from diet bioassay studies. The insects tested against in the bioassay studies in Table 4 include Old World cotton bollworm (CBW, *Helicoverpa armigera*), tobacco cutworm (TCW, *Spodoptera litura*), beet armyworm (BAW, *Spodoptera exigua*), pink bollworm (PBW, *Pectinophora gossypiella*), pink stem borer (PSB, *Sesamia inferens*) and spotted bollworm (SBW, *Earias vitella*). The results depicted in Table 4 demonstrates the enhanced Lepidopteran inhibitory spectrum of the listed engineered insecticidal proteins compared to the scaffold protein Cry1Da1, specifically with improved activity against CBW, PBW (Cry1Ac resistant), PBW (field collected) and SBW.

TABLE 4

Insect Activity Profile Comparison for Cry1Da1 and Engineered Insecticidal Proteins.								
SEQ ID NO.	Protein	CBW	TCW	BAW	PBW (Lab raised)	PBW (Cry1Ac resistant)	PBW (Field collected)	PSB SBW
2	Cry1Da1		+	+	+	+	+	+
12	Cry1Da1_7	+	+	+	+	+	+	+
18	TIC844_4	+	+	+	+	+		+
20	TIC844_5	+	+	+	+	+		+
24	TIC844_7	+	+	+	+	+		

+ Active against the indicated insect species.

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Example 4

Synthesis of Genes Encoding Engineered Insecticidal Proteins and Scaffold Proteins for Expression in Plants

This Example illustrates the synthesis of polynucleotides encoding engineered insecticidal proteins and scaffold proteins for expression in plants.

Nucleotide sequences encoding scaffold proteins and engineered insecticidal proteins for expression in plants were designed and synthesized according to methods generally described in U.S. Pat. No. 5,500,365, avoiding certain inimical problem sequences such as ATTTA and A/T rich plant polyadenylation sequences while preserving the amino acid sequence of the original scaffold or engineered insecticidal protein. The nucleotide sequences for these genes encoding engineered insecticidal proteins and scaffold proteins for expression in plants are listed below in Table 5.

TABLE 5

Polynucleotide Sequences Designed for Use in Plants Encoding Scaffold and Engineered Insecticidal Proteins.			
NUCLEOTIDE SEQ ID NO.	PROTEIN	VARIANT	
27	Cry1Da1.nno	None (scaffold protein)	
29	Cry1Da1_2.nno	Cry1Da1 + A2**	
31	Cry1Da1_3.nno	Cry1Da1 + Y316S + A2	
33	Cry1Da1_4.nno	Cry1Da1 + S374R + A2	
35	Cry1Da1_5.nno	Cry1Da1 + S374R + A2	
37	Cry1Da1_6.nno	Cry1Da1 + S282K_Y316S_I368P + A2	
39	Cry1Da1_7.nno	Cry1Da1 + S282V_Y316S_I368P + A2	
41	TIC844_9.nno	TIC844 + A2	
43	TIC844_11.nno	TIC844 + S282V_Y316S_I368P + A2	

**Variant designation "A2" indicates insertion of an alanine residue at amino acid position 2 compared to the native sequence for cloning purposes into plant expression vectors.

Example 5

Expression Cassettes for Expression of Engineered Insecticidal Proteins in Plants

This Example illustrates the construction of expression cassettes comprising polynucleotide sequences designed for use in plants which encode scaffold and engineered insecticidal proteins.

A variety of plant expression cassettes were constructed with the polynucleotide sequences encoding scaffold and engineered insecticidal proteins designed for plant expression provided in Table 5. Such expression cassettes are useful for transient expression in plant protoplasts or transformation of plant cells. Typical expression cassettes were designed with respect to the eventual placement of the protein within the cell. One set of expression cassettes was designed in a manner to allow the protein to be translated and remain in the cytosol. Another set of expression cassettes was designed to have a transit peptide contiguous with the toxin protein to allow targeting to an organelle of the cell such as the chloroplast or plastid. All expression cassettes were designed to begin at the 5' end with a promoter, which can be comprised of multiple promoter elements, enhancer elements, or other expression elements known to those of ordinary skill in the art operably linked to boost the expression of the transgene. The promoter sequence was usually followed contiguously with one or more leader sequences 3' to the promoter. An intron sequence was usually provided 3'

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to the leader sequence to improve expression of the transgene. A coding sequence for the toxin or transit peptide and coding sequence for the toxin was usually located 3' to the operably linked promoter, leader and intron configuration. A 3'UTR sequence was usually provided 3' of the coding sequence to facilitate termination of transcription and to provide sequences important for the polyadenylation of the resulting transcript. All of the elements described above were operably linked and arranged sequentially, often with additional sequences provided for the construction of the expression cassette.

Example 6

Transformation Vectors Containing a Scaffold or Engineered Insecticidal Protein Expression Cassette

This Example illustrates the incorporation of scaffold or engineered insecticidal proteins into plant tissues.

Methods for producing a transgenic plant which expresses a nucleic acid segment encoding a scaffold protein or an engineered insecticidal protein can be done utilizing variations of methods well known in the art. In general, the method comprises transforming a suitable host cell with a DNA segment which contains a promoter operatively linked to a coding region that encodes one or more of the engineered insecticidal proteins or scaffold proteins. Such a coding region is generally operatively linked to a transcription-terminating region, whereby the promoter is capable of driving the transcription of the coding region in the cell, and hence providing the cell the ability to produce the polypeptide in vivo. Vectors, plasmids, cosmids, and DNA segments for use in transforming such cells will generally comprise operons, genes, or gene-derived sequences, either native or synthetically-derived, and particularly those encoding the disclosed engineered insecticidal proteins. These DNA constructs can further include structures such as promoters, enhancers, polylinkers, or other gene sequences which can have regulating activity upon the particular genes of interest. The resultant transgenic plant, plant parts and plant cells are tested for the expression and bioactivity of the encoded protein.

Examples of methods which can be modified for obtaining transgenic plants that express Lepidopteran-active proteins include those describing, for example, Cry1A proteins (U.S. Pat. No. 5,880,275), Cry1B (U.S. patent application Ser. No. 10/525,318), Cry1C (U.S. Pat. No. 6,033,874), Cry1A/F chimeras (U.S. Pat. Nos. 7,070,982; 6,962,705, and 6,713,063), and a Cry2Ab protein (U.S. Pat. No. 7,064,249).

Example 7

Lepidopteran Activity of Engineered Insecticidal Proteins in Stably Transformed Corn

This Example illustrates the inhibitory activity exhibited by the engineered insecticidal proteins against Lepidopteran pests when expressed in corn plants and provided as a diet to the respective insect pest.

R0 transgenic corn plants expressing Cry1Da1 and Cry1Da1_7.nno proteins were produced using vectors containing the expression cassettes described in Example 6. F1 transgenic corn plants were grown from seed produced by pollinating ears of non-transformed wild-type commercial germplasm plants with pollen from R0 transformants.

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The transformed cells were induced to form plants by methods known in the art. Bioassays using plant leaf disks were performed analogous to those described in U.S. Pat. No. 8,344,207. A non-transformed plant was used to obtain tissue for a negative control. Multiple transformation events from each binary vector were assessed, and the results were tabulated.

The insecticidal activity of transgenic corn plants expressing Cry1Da1 and Cry1Da1_7.nno proteins at F1 and R0 is provided in Table 6, in addition to activity against transgenic corn plants expressing Cry1Da1 and Cry1Da1_7.nno proteins at F1 in the field. Specifically, Table 6 demonstrates the Lepidopteran activity profile for Cry1Da1_7.nno compared to the parent scaffold protein Cry1Da1 when tested against CEW, FAW, and SWC. As can be seen in Table 6, unlike Cry1Da1, Cry1Da1_7.nno demonstrates activity against both CEW and FAW in R0 and F1 bioassay and F1 field tests.

TABLE 6

Insect activity profile for Cry1Da1 and Cry1Da1_7.nno expressed in corn plants.									
Protein (SEQ ID NO.)	CEW			FAW			SWC		
	R0	F1	Field	R0	F1	Field	R0	F1	Field
Cry1Da1 (28)	-	NT	NT	+	NT	NT	-	NT	NT
Cry1Da1_7.nno (40)	+	+	+	+	+	+	-	-	-

+ Active against insect species;
- Inactive against insect species;
NT Not Tested

Example 8

Lepidopteran Activity of Engineered Insecticidal Proteins in Stably Transformed Cotton

This Example illustrates the inhibitory activity exhibited by the engineered insecticidal proteins against Lepidopteran pests when expressed in cotton plants and provided as a diet to the respective insect pest.

Cotton plants expressing Cry1Da1_7.nno and TIC844_11.nno proteins were produced using vectors containing the expression cassettes described in Example 6. The transformed cells were induced to form plants by methods known in the art. Cotton leaf tissue was used in bioassay as described in Example 7 and tested against CBW, FAW, Tobacco budworm (TBW, *Heliothis virescens*), and SBL. Table 7 shows the activity observed against these Lepidopteran species in stably transformed R₀ generation cotton.

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As can be seen in Table 7, Cry1Da1_7.nno and TIC844_11.nno demonstrated activity against two or more Lepidopteran pest species in stably transformed R₀ generation cotton.

TABLE 7

Bioassay activity profile of Cry1Da1_7.nno, and TIC844_11.nno expressed in R ₀ generation cotton.				
	Toxin			
	CBW	FAW	TBW	SBL
Cry1Da1_7.nno (SEQ ID NO: 40)	+	+	+	+
TIC844_11.nno (SEQ ID NO: 44)	+	+	-	+

+ Active against insect species;
- Inactive against insect species.

Selected transformed events were used to produce R₁ plants. R₁ plants expressing Cry1Da1_7.nno were assayed for resistance to CBW, FAW and SBL. Leaf, square and boll tissues were used in bioassay, in addition to field tests conducted in screenhouses. Table 8 shows the activity observed in these tests. As demonstrated in Table 8, Cry1Da1_7.nno demonstrated activity against CBW, FAW and SBL in bioassay and field tests.

TABLE 8

Insect activity profile of Cry1Da1_7.nno expressed in R ₁ generation cotton.									
Toxin	CBW			FAW			SBL	Screenhouse	
	Leaf	Square	Boll	Leaf	Square	Boll	Leaf	CBW	FAW
Cry1Da1_7.nno (SEQ ID NO: 40)	+	+	+	+	+	+	+	+	+

+ Active against insect species;
- Inactive against insect species.

Example 9

Lepidopteran Activity of Engineered Insecticidal Proteins in Stably Transformed Soybean

This Example illustrates the inhibitory activity exhibited by the engineered insecticidal proteins against Lepidopteran pests when expressed in soybean plants and provided as a diet to the respective insect pest.

Soybean plants expressing Cry1Da1_7.nno, TIC844_9.nno and TIC844_11.nno proteins were produced using vectors containing the expression cassettes described in Example 6. Leaf tissue was harvested and used in bioassay as described in Example 7 or, alternatively, lyophilized tissue was used in the insect diet for bioassay. Bioassay was performed against various Lepidopteran species, including SAW, SBL and Soybean Pod Worm (SPW, *Helicoverpa zea*). Table 9 shows the activity observed

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against these Lepidopteran pests in stably transformed R₀ generation soybeans. As can be seen in Table 9, Cry1Da1_7.nno and TIC844_11.nno demonstrated activity against SPW, SAW and SBL. TIC844_9.nno (TIC844 plus a bonus alanine for cloning) did not demonstrate activity against SPW.

TABLE 9

Bioassay activity profile of Cry1Da1_7.nno, TIC844_9.nno and TIC844_11.nno expressed in R ₀ generation soybean.				
	Toxin			
	SPW	SAW	SBL	
Cry1Da1_7.nno (SEQ ID NO: 40)	+	+	+	
TIC844_11.nno (SEQ ID NO: 44)	+	+	+	
TIC844_9.nno (SEQ ID NO: 42)	-	+	+	

+ Active against insect species;
- Inactive against insect species.

Selected transformed events were used to produce R₁ plants. R₁ plants expressing Cry1Da1_7.nno were assayed for resistance to SAW, SBL, SPW and Velvetbean caterpillar (VBC, *Anticarsia gemmatilis*). Leaf tissue was harvested from the R₁ generation plants and used in a feeding bioassay. Table 10 shows the activity observed in these tests. As demonstrated in Table 10, Cry1Da1_7.nno demonstrated activity against SPW, SAW and SBL.

TABLE 10

Bioassay activity profile of Cry1Da1_7.nno expressed in R ₁ generation soybean.				
	Toxin			
	SPW	SAW	SBL	VBC
Cry1Da1_7.nno (SEQ ID NO: 40)	+	+	+	-

+ Active against insect species;
- Inactive against insect species.

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Table 11 shows the results of field tests conducted in screenhouses with stably transformed R₁ generation soybean plants expressing Cry1Da1_7.nno. Species used to infest plants in the screenhouses include Black armyworm (BLAW, *Spodoptera cosmioides*), Bean shoot moth (BSM, *Crociosema aporema*), South American podworm (SAPW, *Helicoverpa gelatopoeon*), Sunflower looper (SFL, *Rachip-lusia nu*) and VBC. Table 11 shows the activity observed in these tests. As demonstrated in Table 11, Cry1Da1_7.nno demonstrated activity against BLAW, SAPW and SFL.

TABLE 11

Activity profile of Cry1Da1_7.nno expressed in R ₁ generation soybean tested in screenhouse field tests.					
	Toxin				
	BLAW	BSM	SAPW	SFL	VBC
Cry1Da1_7.nno (SEQ ID NO: 40)	+	-	+	+	-

+ Active against insect species;
- Inactive against insect species.

All of the compositions and methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of the foregoing illustrative embodiments, it will be apparent to those of skill in the art that variations, changes, modifications, and alterations may be applied to the composition, methods, and in the steps or in the sequence of steps of the methods described herein, without departing from the true concept, spirit, and scope of the invention. More specifically, it will be apparent that certain agents that are both chemically and physiologically related may be substituted for the agents described herein while the same or similar results would be achieved. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as defined by the appended claims.

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tacacaccac taccagctgg ttatgtaaca aaggaattag agtactccc agagaccgat 3420
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ctccttatgg aggaatag                                     3498

SEQ ID NO: 4      moltype = AA length = 1165
FEATURE           Location/Qualifiers
REGION            1..1165
                  note = Amino acid sequence of the engineered insecticidal
                        protein Cry1Dai_3.
source            1..1165
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 4
MEINNQNCV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLQAE QLISQRIEEF ARNQAISRLS GLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMSNAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180

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DFLNSFTIYT	DSLARSAYWG	GHLVNSFRTG	TTTNLIRSPL	YGREGNTERP	VTITASPSVP	360
IFRTLSTYITG	LDNSNPVAGI	EGVEFQNTIS	RSIYRKS GPI	DSFSELPPQD	ASVSPAIGYS	420
HRLCHATFLE	RISGPRIAGT	VFSWTHRSAS	PTNEVSPSRI	TQIPWVKAHT	LASGASVIK	480
PGFTGGDILT	RNSMGELGTL	RVTFTGRLPQ	SYIIRFRYAS	VANRSQTFRY	SQPPSYGISF	540
PKTMDAGEPL	TSRSFAHTTL	FTPITFSRAQ	EEFDLYIQSG	VYIDRIEFIP	VTATFEAEYD	600
LERAQKVUNA	LFTSTNQLGL	KTDVTDYHID	QVSNLVACLS	DEFCLDEKRE	LSEKVKHAKR	660
LSDERNLLQD	PNFRGINRQP	DRGWRGSTD	TIQGGDDVFK	ENYVTLPGTF	DECYPTYLYQ	720
KIDESKLLKAY	TRYQLRGYIE	DSQDLEIYLI	RYNAKHEIVN	VPGTGSLWPL	SVENQIGPCG	780
EPNRCAPHLE	WNPDLHSCSR	DGEKCAHHSH	HFSLDIDVGC	TDLNEDLGWV	VIFKIKTQDG	840
HARLGNLEFL	EEKPLLGEAL	ARVKRAEKKW	RDKRETLQLE	TTIVYKEAKE	SVDALFVNSQ	900
YDRLQADTNI	AMIHAADKRV	HRIREAYLPE	LSVIPGVNAA	IFEELEERIF	TAFSLYDARN	960
IIKNGDFNNG	LLCWNVKGHV	EVVEQNHNRS	VLVIPEWEAE	VSQEVVCPG	RGYILRVVAY	1020
KEGYGEGCVT	IHEIENNTDE	LKFNNCVEEE	VYPNNTVTCI	NYTATQEEYE	GTYSRNRGY	1080
DEAYGNPNPV	PADYASVYEE	KSYTDRRREN	PCESNRGYGD	YTPLPAGYVT	KELEYFPETD	1140
KVWIEIGETE	GTFIVDSVEL	LLMEE				1165

SEQ ID NO: 5 moltype = DNA length = 3498
FEATURE Location/Qualifiers
misc_feature 1..3498
 note = Nucleotide sequence used for expression in a
 bacterial cell encoding Cry1Dai_4.
source 1..3498
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 5

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cttatttaatt	ttctatattc	taatttttga	ccaggaggag	gatttatagt	aggtttacta	180
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caattgatta	gtcaaaagaat	agaagaattt	gctaggaatc	aggcaatttc	aagattggag	300
gggctaagca	atctttataa	ggctcatgtt	agagcgttta	gcgactggga	gaaagatcct	360
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tggggatatg	atacagcgac	tatcaataat	cgctatagtg	atctgactag	ccttattcat	600
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gagaattacg	ttacgctacc	gggtaccttt	gatgagtgtc	atccaacgta	tttatatcaa	2160
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gtaccaggt	cagggaagtt	atggcctctt	tctgtagaaa	atcaaattgg	acctgtgga	2340
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ctccttatgg aggaatag                                     3498

SEQ ID NO: 6      moltype = AA length = 1165
FEATURE
REGION           Location/Qualifiers
                  1..1165
                  note = Amino acid sequence of the engineered insecticidal
                        protein Cry1Dal_4.
source           1..1165
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 6
MEINNQNQCVC PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIEEF ARNQAISRL EGLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSDLTSLIH VYTNHCVDTY NQGLRRLEGR FLSDWIVYNR FRRQLTISVL 240
DIVAFPPNYD IRTYPIQTAT QLTREVYLDL PFINENLSPA ASYPTFSAAE SAIIRSPHLV 300
DFLNSFTIYT DSLARYAYWG GHLVNSFRTG TTTNLIRSPL YGREGNTERP VTITASPSVP 360
IFRTLSYITG LDNRNPVAGI EGVFQNTIS RSIYRKSGPI DSFSELPPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRIAGT VFSWTHRSAS PTNEVSPSRI TQIPWVKAHT LASGASVIK 480
PGFTGGDILT RNSMGELGTL RVTFTGRLPQ SYYIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTPITFSRAQ EEFDLYIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKVVNA LFTSTNQLGL KTDVTDYHID QVSNLVACLS DEFCLDEKRE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQP DRGWRGSDTI TIQGGDDVPK ENYVTLPGTF DECYPTYLYQ 720
KIDESKLLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHEIVN VPGTGSWLPL SVENQIGPCG 780
EPNRCAPHLE WNPDLHCSCR DGEKCAHSH HFSLDIDVGC TDLNEDLGWV VIFKIKTQDG 840
HARLGNLEFL EEKPLLGEAL ARVKRAEKKW RDKRETLQLE TTIVYKEAKE SVDALFVNSQ 900
YDRLQADTNI AMIHAADKRV HRIREAYLPE LSVIPGVNAA IFEELEERIF TAFSLYDARN 960
IKKNGDFNNG LLCWNVKGHV EVEEQNNHRS VLVIPWEWAE VSQEVVRVCPG RGYLRLVTAY 1020
KEGYGEGCVT IHEIENNTDE LKFNNCVEEE VYPNNTVTCT NYTATQEEYE GTYTSRNRGY 1080
DEAYGNPNPV PADYASVYEE KSYTDRRREN PCESNRGYGD YTPLPAGYVT KELEYFPETD 1140
KWVIEIGETE GTFIVDSVEL LLMEE                                     1165

SEQ ID NO: 7      moltype = DNA length = 3498
FEATURE
misc_feature     Location/Qualifiers
                  1..3498
                  note = Nucleotide sequence used for expression in a
                        bacterial cell encoding Cry1Dal_5.
source           1..3498
                  mol_type = other DNA
                  organism = synthetic construct

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cttattaatt ttctatatcc taattttgta ccaggaggag gatttatagt aggtttacta 180
gaattaatat ggggatttat agggccttcg caatgggata ttttttagc tcaaataggag 240
caattgatta gtcaagaagt agaagaattt gctaggaatc aggcaatttc aagattggag 300
gggctaagca actcttataa ggtctatgtt agagcgttta gcgactggga gaaagatcct 360
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gttcaagccg caaacttaca tttatctatt ttaaggatg ttctagtttt cggagaaaga 540
tggggatatg atacacgcag tatcaataat cgctatagtg atctgactag ccttattcat 600
gtttatacta accattgtgt ggatacgtat aatcagggat taaggcgttt ggaaggctcg 660
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ctccttatgg aggaatag 3498

SEQ ID NO: 8          moltype = AA  length = 1165
FEATURE              Location/Qualifiers
REGION              1..1165
                    note = Amino acid sequence of the engineered insecticidal
                    protein Cry1Da1_5.
source              1..1165
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 8
MEINNQNQCVC PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIEEF ARNQAISRLG GLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATIMN RYSDLTSLIH VYTNHCVDYV NQGLRRLEGR FLSDWIVYNR FRQLTISVL 240
DIVAFPPNYD IRTYPIQTAT QLTREVYLDL PFINENLSPA ASYPTFSAAE SAIIRSPHLV 300
DFLNGFTIYT DSLARSAYWG GHLVNSFRTG TTNLIRSLP YGREGNTERP VTITASPSVP 360
IPRTLSTYRTG LDNSNPVAGI EGVFQNTIS RSIYRKSGPI DSFSELPPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRIAGT VFSWTHRSAS PTNEVSPSRI TQIPWVKAHT LASGASVIK 480
PGFTGGDILT RNSMGELGTL RVTFTGRLPQ SYIIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTPITFSRAQ EEFDLYIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKVVNA LFTSTNQLGL KTDVTDYHID QVSNLVACLS DEFCLDEKRE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQP DRGWRGSTDI TIQGGDDVFK ENYVTLPGTF DECYPTYLYQ 720
KIDESKLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHEIVN VPGTGSWLPL SVENQIGPCG 780
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IKKNGDFPNNG LLCWNVKGHV EVBEEQNNHRS VLVIPWEAE VSQEVVCPG RGYILRVYAT 1020
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DEAYGNNPSV PADYASVYEE KSYTDRRREN PCESNRGYGD YTPLPAGYVT KELEYFPETD 1140
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SEQ ID NO: 9          moltype = DNA  length = 3498
FEATURE              Location/Qualifiers
misc_feature         1..3498
                    note = Nucleotide sequence used for expression in a
                    bacterial cell encoding Cry1Da1_6.
source              1..3498
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 9
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cttattaatt ttctatatcc taatttttga ccaggaggag gatttatagt aggtttacta 180
gaattaatat ggggatttat agggccttcg caatgggata tttttttagc tcaaatgag 240
caattgatta gtcaagaagt agaagaattt gctaggaatc agggcaattc aagattggag 300
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tggggatagc atacagcgac ttcaataat cgctatagtg atctgactag ccttattcat 600
gtttatacta accatttgtt ggatacgtat aatcagggat taaggcgttt ggaaggtcgt 660
tttcttagcg attggattgt atataatcgt ttccggagac aattgacaat ttcagtatta 720

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gatattgttg cgtttttttc aaattatgat attagaacat atccaattca aacagctact 780
cagctaacga ggggaagtcta tctggattta ccttttatta atgaaaatct ttctcctgca 840
gcaaaatato caaccttttc agctgctgaa agtgctataa ttagaagtcc tcatttagta 900
gactttttaa atagctttac catttataca gatagtctgg caggttctgc atattgggga 960
gggcacttgg taaattcttt ccgcacagga accactacta atttgataag atccccctta 1020
tatggaaggg aaggaaatcc agagcgcccc gtaactatta ccgcatcacc tagcgtacca 1080
atatttagaa cactttcata tccaacaggc cttgacaatt caaatcctgt agctggaatc 1140
gaggagtgag aattocaaaa tactataagt agaagtatct atcgtaaaag cggteccaata 1200
gattctttta gtgaattacc acctcaagat gccagcgtat ctctgcaat tgggtatagt 1260
caccgtttat gccatgcaac atttttagaa cggattagtg gaccaagaat agcaggcacc 1320
gtattttctt ggacacaccg tagtgccagc cctactaacg aagtaagtcc atctagaatt 1380
acacaaatcc catgggtaaa ggcgcatact cttgcatctg gtgcctccgt cattaaaggt 1440
ctgggattta caggtggaga tattctgact aggaatagta tgggcgagct ggggacctta 1500
cgagtaacct tcacaggaag attaccacaa agttattata tacgtttccg ttatgcttcg 1560
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aaaaacagatg tgacggatta tcatattgat caggtatcca atctagtgc gtgtttatcg 1920
gatgaatttt gtcctggatga aaagagagaa ttgtccgaga aagttaaaca tgcaaaagcg 1980
ctcagtgatg agcggaaatt acttcaagat ccaaaccttca gagggatcaa taggcaacca 2040
gaccgtggct ggagaggaag tacggatatt actatccaa gaggagatga cgtattcaaa 2100
gagaattacg ttacgctacc gggtagcttt gatgagtgct atccaacgta ttatatcaa 2160
aaaaatagatg agtcgaaatt aaaagcctat acccgttatc aattaagagg gtatatcgaa 2220
gatagtcaag acctagaaat ctatttaatt cgttacaatg caaaacacga aatagtaaat 2280
gtaccaggtg caggaagttt atggcctctt tctgtagaaa atcaaatgg acctgtgga 2340
gaaccgaatc gatcgcgccc acaccttgaa tggaaatcctg atttactg ttctctcgaga 2400
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gacgaagcct atggttaata cccttccgta ccagctgatt atgcgtcagt ctatgaagaa 3300
aaatcgtata cagatagacg aagagagaat ccttgtgaat ctaacagagg atatggagat 3360
tacacaccac taccagctgg ttatgtaaca aaggaattag agtactccc agagaccgat 3420
aaggtatgga ttgagattgg agaaacagaa ggaacattca tcgtggacag cgtggaatta 3480
ctccttatgg aggaatag 3498

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SEQ ID NO: 10      moltype = AA length = 1165
FEATURE           Location/Qualifiers
REGION            1..1165
                  note = Amino acid sequence of the engineered insecticidal
                  protein Cry1Da1_6.
source            1..1165
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 10
MEINNQNQCV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIIEF ARNQAISRLS GLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSDLTSLIH VYTNHCVDYV NQGLRRLEGR FLSDWIVYNR FRRQLTISVL 240
DIVAFFPNYD IRTYPIQTAT QLTREVYLDL PFINENLSPA AKYPTFSAAE SAIIRSPHLV 300
DPLNSFTIYT DSLARSAYWG GHLVNSFRTG TTNLIRSLPL YGREGNTERP VTITASPSVP 360
IFRTLSTYPTG LDNSNPVAGI EGVEFQNTIS RSIYRKS GPI DSELPDPPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRIAGT VFSWTHRSAS PTNEVSPSRI TQIPVWKAHT LASGASVIK 480
PGFTGGDILT RNSMGLGTL RVTFTGRLPQ SYIIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTFITFSRAQ EEFDLYIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKVVNA LFTSTNQLGL KTDVTDYHID QVSNLVACLS DEFCLDEKRE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQP DRGWRGSTDI TIQGGDDVFK ENYVTLPGTF DECYPTYLYQ 720
KIDESKLLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHEIVN VPGTGSWLPL SVENQIGPCG 780
EPNRCAPHLE WNPDLHSCSR DGEKCAHSHS HFSLDIDVGC TDNLNEDLGWV VIFKIKTQDG 840
HARLGNLEFL EEKPLLGEAL ARVKRAEKKW RDKRETLQLE TTIVYKEAKE SVDALFVNSQ 900
YDRLQADTNI AMIHAAKRV HRIREAYLPE LSVIPGVNAA IFEELEERIF TAFSLYDARN 960
IKNKGDFFNG LLCWNVKGHV EVEEQNNHRS VLVIPWEWAE VSQEVVRVCPG RGYILRVYAT 1020
KEGYGEGCVT IHEIENNTDE LKFNNCVEEE VYPNNTVTCI NYTATQEEYE GTYTSRNRGY 1080
DEAYGNPNSV PADYASVYEE KSYTDRRREN PCBSNRGYGD YTPLPAGYVT KELEYFPETD 1140
KWWIEIGETE GTFIVDSVEL LLMEE 1165

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SEQ ID NO: 11      moltype = DNA length = 3498

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FEATURE                               Location/Qualifiers
misc_feature                           1..3498
                                         note = Nucleotide sequence used for expression in a
                                         bacterial cell encoding Cry1Da1_7.
source                                1..3498
                                         mol_type = other DNA
                                         organism = synthetic construct

SEQUENCE: 11
atggaaataa ataatacaaaa ccaatgtgtg ccttacaatt gttaaagtaa tcctaaggag 60
ataatattag gcgaggaagag gctagaaca ggaataactg tagcagacat ttcattaggg 120
cttattaat tttatatatt taattttgtg ccaggaggag gatttatagt aggtttacta 180
gaattaatat ggggatttat agggccttcg caatgggata tttttttagc tcaaatgtag 240
caattgatta gtcaagaat agagaattt gctaggaatc aggcatttc aagattggag 300
gggctaagca atctttataa ggtctatgtt agagcgttta gcgactggga gaaagatcct 360
actaatcctg ctttaaggga agaaatgcgt atacaattta atgacatgaa tagtgctctc 420
ataacggcta ttccactttt tagagttaa aattatgaag ttgctctttt atctgtatat 480
gttcaagccg caaaccttaca tttatctatt ttaagggatg tttcagtttt cggagaaaga 540
tggggatatg atacagcgac tatcaataat cgctatagtg atctgactag ccttattcat 600
gtttatacta accattgtgt ggatacgtat aatcagggat taaggcgttt ggaaggtcgt 660
tttcttagcg attgattgt atataatcgt ttccggagac aattgacaat ttcagtatta 720
gatattgttg cgttttttcc aaattatgat attagaacat atccaattca aacagctact 780
cagctaacga ggggaagtcta tctggattta ccttttatta atgaaaatct ttctcctgca 840
gcagtataac caaccttttc agctgctgaa agtgctataa ttagaagtcc tcatttagta 900
gactttttta atagctttac catttataca gatagtctgg cactgtctgc atattgggga 960
gggcaacttg taaattcttt ccgcacagga accactacta atttgataag atccccctta 1020
tatggaaggg aagaaaatac agagcgcccc gtaactatta ccgcatcacc tagcgtacca 1080
atatttagaa cactttcata tccaacaggg cttgacaatt caaatcctgt agctggaatc 1140
gagggagtg aattccaaaa tactataagt agaagtatct atcgtaaaag cggtccaata 1200
gattctttta ttgaattacc acctcaagat gccagcgtat ctctgcaat tgggtatagt 1260
caccgtttat gccatgcaac atttttagaa cggatttagtg gaccaagaat agcaggcacc 1320
gtattttctt ggacacaccc tagtgccagc cctactaacg aagtaagtcc atctagaatt 1380
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cctggattta caggtggaga tattctgact aggaatagta tgggcgagct ggggacctta 1500
cgagtaacct tcacaggaaa agttaccaca agttattata tacgttttcc ttatgcttcg 1560
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ccaaaaataa tggacgcagg tgaaccacta acatctcgtt cggtcgtcca tacaacactc 1680
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ctcagtgatg agcgggaatt acttcaagat ccaaaactca gagggatcaa taggcaacca 2040
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gagaattacg ttacgctacc gggtaacctt gatgagtgtc atccaacgta tttatatcaa 2160
aaaatagatg agtcgaaatt aaaagcctat acccgttacc aattaagagg gtatatcgaa 2220
gatagtcgag cttagaaatt ctatttaatt cgttacatg caaaacacga aatagtaaat 2280
gtaccaggta caggaagttt atggcctctt tctgtagaaa atcaaatgg accctgtgga 2340
gaaccgaatc gatgcgcgcc acaccttgaa tggaaatcctg atttacactg ttctgcgaga 2400
gacggggaaa aatgtgcaca tcattctcat cattctctt tggacattga tgttgatgt 2460
acagacttaa atgaggactt aggtgtatgg gtgatattca agattaagac gcaagatggc 2520
cacgcacgac tagggaatct agagtttctc gaagagaac cattattagg agaagcacta 2580
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gaggtagaag aacaaaacaa tcaccgttca gtccctggtta tcccagaatg ggaggcagaa 3000
gtgtcacaa ggttcctgtt ctgtccaggt cgtggctata tccttcgtgt tacagcgtac 3060
aaagagggat atggagaagg ttgcgtaacg atccatgaga tcgagaacaa tacagacgaa 3120
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aattatactg cgactcaaga agaatatgag ggtacgtaca cttctcgtaa tcgaggatat 3240
gacgaagcct atggtaataa cccttccgta ccagctgatt atgcgtcagt ctatgaagaa 3300
aaatcgata cagatagacg aagagagaat ccttgtgaat ctaacagagg atatggagat 3360
tacacaccac taccagctgg ttatgtaaac aaggaattag agtacttccc agagaccgat 3420
aaggtatgga ttgagattgg agaaaacgaa ggaacattca tcgtggacag cgtggaatta 3480
ctccttatgg aggaatag                                     3498

SEQ ID NO: 12                        moltype = AA length = 1165
FEATURE                               Location/Qualifiers
REGION                               1..1165
                                         note = Amino acid sequence of the engineered insecticidal
                                         protein Cry1Da1_7.
source                                1..1165
                                         mol_type = protein
                                         organism = synthetic construct

SEQUENCE: 12
MEINNQNCV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIEF ARNQAISRLS GLSNLYKVYV RAFSDWEKDP 120

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TNPALREEMR	IQFNDMNSAL	ITAIPLFRVQ	NYEVALLSVY	VQAANLHLSI	LRDVSVFGER	180
WGYDTATINN	RYSDLTSLIH	VYTNHCVDY	NQGLRRLEGR	FLSDWIVYNR	FRQLTISVL	240
DIVAFPPNYD	IRTYPIQTAT	QLTREVYLDL	PFINENLSPA	AVYPTFSAAE	SAIIRSPHLV	300
DPLNSFTIYT	DSLARSAYWG	GHLVNSPRTG	TTNLIRSL	YREGNTERP	VTITASPSVP	360
IFRTLSYPTG	LDNSNPVAGI	EGVEFQNTIS	RSIYRKSGPI	DSFSELPPQD	ASVSPAIGYS	420
HLRCHATFLE	RISGPRIAGT	VFSWTHRSAS	PTNEVSPSRI	TQIPWVKAHT	LASGASVIKG	480
PGFTGGDILT	RNSMGELGTL	RVTFTGRLPQ	SYIIRFRYAS	VANRSGTFRY	SQPPSYGISF	540
PKTMDAGEPL	TSRSFAHTTL	FTPITFSRAQ	EEFDLYIQSG	VYIDRIEFIP	VTATFEAEYD	600
LERAQKVUNA	LFTSTNQLGL	KTDVTDYHID	QVSNLVACLS	DEFCLDEKRE	LSEKVKHAKR	660
LSDERNLLQD	PNFRGINRQP	DRGWRGSTDI	TIQGGDDVFK	ENYVTLPGTF	DECYPTYLYQ	720
KIDESKLKAY	TRYQLRGYIE	DSQDLEIYLI	RYNAKHEIVN	VPGTGSLWPL	SVENQIGPCG	780
EPNRCAPHLE	WNPDLHCSCR	DGEKCAHSHS	HFSLDIDVGC	TDLNEDLGWV	VIFKIKTQDG	840
HARLGNLEFL	EEKPLLGEAL	ARVKRAEKKW	RDKRETLQLE	TTIVYKEAKE	SVDALFVNSQ	900
YDRLQADTNI	AMIHAADKRV	HRIREAYLPE	LSVIPGVNAA	IFEELEERIF	TAFSLYDARN	960
IKNGDFNNG	LLCWNVKGHV	EVVEQNNHRS	VLVIPEWEAE	VSQEVVRCPG	RGYILRVYAT	1020
KEGYGEGCVT	IHEIENNTDE	LKPNNCVEEE	VYPNNTVTCT	NYTATQEEYE	GTYSRNRGY	1080
DEAYGNPNVS	PADYASVYEE	KSYTDRRREN	PCESNRGYGD	YTPLPAGYVT	KELEYFPETD	1140
KVWIEIGETE	GTFIVDSVEL	LLMEE				1165

SEQ ID NO: 13 moltype = DNA length = 3420
 FEATURE Location/Qualifiers
 misc_feature 1..3420
 note = Nucleotide sequence used for expression in a
 bacterial cell encoding TIC844.
 source 1..3420
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 13

atggagatca	acaaccagaa	ccagtgcgtc	ccgtacaact	gcctgagcaa	ccctaaggag	60
atcatccttg	gtgaggaaac	cctggagacc	ggcaacaccg	tagccgacat	tagcctgggc	120
ctcatcaact	tcctctacag	caacttcgtg	cccgccgggtg	gcttcatcgt	gggcctctcg	180
gagcttatct	ggggcttcac	cgcccggtcc	cagtgggaca	tcctcctcgc	ccagatcgag	240
caactgatca	gccagcggat	cgaggagttc	gctaggaacc	aggccatctc	ccgcctggag	300
ggactctcca	acctctacaa	ggtgtacgtg	cgcgcgttca	gcgactggga	gaaggaccgc	360
accaaaccgg	ccctcccgga	ggaaatgcgt	atccagtcca	acgatatgaa	ctcggccctc	420
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gtctgccca	gcgcgggcta	cattctcagg	gtcaccgctt	acaaggaggg	ctacggtgag	3000
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aacagcagcg	ttcctgctga	ctacgcttcc	gcctacgagg	agaaggccta	cacggatggc	3240
cgcagggaca	acccttgcca	gagcaaccgc	ggctacggcg	actaacctcc	cctgcccgcc	3300
ggctacgtta	ccaaggagct	ggagtacttc	cgggagactg	acaaggtgtg	gatcgagatc	3360
ggcgagaccg	agggcacctt	catcgtagac	agcgtggagc	tgctcctgat	ggaggagtag	3420

SEQ ID NO: 14 moltype = AA length = 1139
 FEATURE Location/Qualifiers
 REGION 1..1139
 note = Amino acid sequence of the chimeric protein TIC844.
 source 1..1139
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 14

MEINQNQCVC	PYNCLSNPKE	IILGEERLET	GNTVADISLG	LINFLYSNFV	PGGGFIVGLL	60
ELIWFIGIPS	QWDIFLAQIE	QLISQRIEEF	ARNQAISRLE	GLSNLYKVYV	RAFSDWKDP	120
TNPALREEMR	IQFNDMNSAL	ITAIPLFRVQ	NYEVALLSVY	VQAANLHLSI	LRDVSVFGER	180
WGYDTATINN	RYSDLTSLIH	VYTNHCVDY	NQGLRRLRLEGR	FLSDWIVYNR	FRRLTISVL	240
DIVAFPPNDY	IRTYPIQTAT	QLTREVYLDL	PFINENLSPA	ASYPTFSAAE	SAIIRSPHLV	300
DPLNSFTIYT	DSLARYAYWG	GHLVNSFRTG	TTTNLIRSPL	YGREGNTERP	VTITASPSVP	360
IFRTLSTIYT	LDNSNPVAGI	EGVEFQNTIS	RSIYRKSGPI	DSFSELPPQD	ASVSPAIGYS	420
HRLCHATFLE	RISGPRIAGT	VFSWTHRSAS	PTNEVSPSRI	TQIPVWKAHT	LASGASVIK	480
PGFTGGDILT	RNSMGEGLTL	RVTFTGRLPQ	SYIIRFRYAS	VANRSGTFRY	SQPPSYGISF	540
PKTMDAGEPL	TSRSFAHTLL	FTPTFSRAQ	EEFDLYIQSG	VYIDRIEFIP	VTATFEAEYD	600
LERAQKAVNE	LFTSSNQIGL	KTDVTDYHID	QVSNLVECLS	DEFCLDEKKE	LSEKVKHAKR	660
LSDERNLLQD	PNFRGINRQL	DRGWRGSTD	TIQGGDDVFK	ENYVTLTGTF	DECYPTYLYQ	720
KIDESKLKAY	TRYQLRGYIE	DSQDLEIYLI	RYNAKHETVN	VPGTGSLWPL	SAPSPIGKCA	780
HSHHFLSLDI	DVGCTDLNED	LGWVWIFKIK	TQDGHARLGN	LEFLEEKPLV	GEALARVKRA	840
EKKWRDKREK	LEWETNIVYK	EAKESVDALF	VNSQYDRLQA	DTNIAMIHAA	DKRVHSIREA	900
YLPFLSVIPG	VNAIFEELE	GRIPTAFSLY	DARNVIKNGD	FNNGLSCWNV	KGHVDVEEQN	960
NHRSVLVPE	WEAEVSQEV	VCPGRGYILR	VTAYKEGYGE	GCVTIHEIEN	NTDELKFSNC	1020
VEEEVYPNNT	VTCDNYTATQ	EEYEGTYTSR	NRGYDGAYES	NSSVPADYAS	AYEEKAYTDG	1080
RDRNPCESEN	GYGDTPLPA	GYVTKLEYF	PETDKVWIEI	GETEGTFIVD	SVELLMLMEE	1139

SEQ ID NO: 15 moltype = DNA length = 3420
 FEATURE Location/Qualifiers
 misc_feature 1..3420
 note = Nucleotide sequence used for expression in a
 bacterial cell encoding TIC844_2.
 source 1..3420
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 15

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gaattaatat	ggggatttat	agggccttcg	caatgggata	tttttttagc	tcaaattgag	240
caattgatta	gtcaagaagt	agaagaattt	gctaggaatc	aggcaatttc	aagattggag	300
gggctaagca	atctttataa	gggtctatgt	agagcgttta	gcgactggga	gaaagatcct	360
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gttcaagccg	caaaacttaca	tttatctatt	ttaagggatg	tttcagtttt	cggagaaaga	540
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gtttatacta	accattgtgt	ggatacgtat	aatcagggat	taaggcgttt	ggaaggtcgt	660
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SEQ ID NO: 16      moltype = AA length = 1139
FEATURE           Location/Qualifiers
REGION            1..1139
                  note = Amino acid sequence of the engineered insecticidal
                  chimeric protein TIC844_2.
source            1..1139
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 16
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TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSDLTSLIH VYTNHCVDY NQGLRLRLEGR FLSDWIVYNR FRRQLTISVL 240
DIVAFPPNDV IRTYPIQTAT QLTREVYLDL PFINENLSPA ASYPTFSAAE SAIIRSPHLV 300
DFLNSFTIYT DSLARSAYWG GHLVNSFRTG TTTNLIRSPL YGREGNTERP VTITASPSVP 360
IFRTLSTYTG LDNSHPVAGI EGEVFQNTIS RSIYRKSGPI DSFSELPPQD ASVSPAIGYS 420
HRLCHATFLE RLSGPRFAGT FFSWTHRSAS PTNEVSPSRI TQIPWVKAHT LASGASVIK 480
PGFTGGDILT RNSMGEGLTL RVTFTGRLPQ SYYIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTFITFSRAQ EEFDLYIQSG VYIDRIEFIP VTATFEAEYD 600
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LSDERNLLQD PNFRGINRQL DRGWRGSDTI TIQGGDDVFK ENYVTLTGTF DECYPTYLYQ 720
KIDESKLLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHETVN VPGTGSWLPL SAPSPIGKCA 780
HSHHFLSLDI DVGCTDLNED LGWVVFVKIK TQDGHARLGN LEFLEEKPLV GEALARVKRA 840
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YLPESLVIPI VNAALFEELE GRIFTAFSLY DARNVIKNGD FNNGLSCWNV KGHVDVEEQN 960
NHRSLVWVPE WEAEVVSQEV R VCPGRGYILR VTAYKEGYGE GCVTIHEIEN NTDELKFSNC 1020
VEEEVYPNNT VTCNDYTATQ EEEGYTTSR NRGYDGAYES NSSVPADYAS AYEKAYTDG 1080
RRDNPCESNR GYGDYTPLPA GYVTKLEYF PETDKVWIEI GETEGTFIVD SVELLLMEE 1139

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SEQ ID NO: 17      moltype = DNA length = 3420
FEATURE           Location/Qualifiers
misc_feature       1..3420
                  note = Nucleotide sequence used for expression in a
                  bacterial cell encoding TIC844_4.
source            1..3420
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 17
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SEQ ID NO: 18      moltype = AA  length = 1139
FEATURE            Location/Qualifiers
REGION             1..1139
                   note = Amino acid sequence of the engineered insecticidal
                   chimeric protein TIC844_4.
source             1..1139
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 18
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TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQANLHLSI LRDVSVFGER 180
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SEQ ID NO: 19      moltype = DNA  length = 3420
FEATURE            Location/Qualifiers
misc_feature        1..3420
                   note = Nucleotide sequence used for expression in a
                   bacterial cell encoding TIC844_5.
source             1..3420
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 19
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cttagtgatg agcgggaattt acttcaagat ccaaaactta gagggatcaa tagacaacta 2040
gaccgtggct ggagaggaag tacggatatt accatccaag gaggcgatga cgtattcaaa 2100
gagaattacg ttacgctatt gggtaacctt gatgagtgct atccaacgta ttatatcaa 2160
aaaatagatg agtcgaattt aaaagcctat acccggttacc aattaagagg gtatatcgaa 2220
gatagtcgaag ctttagaatt ctatttaatt cgctacaatg ccaaacacga aacagtaaat 2280
gtgccaggtg cgggttctct atggccgctt tcagcccaaa gtccaatcgg aaaatgtgcc 2340
catcattccc atcattttct cttggacatt gatgttggtg gtacagactt aaatgaggac 2400
ttagtgatg ggggtatgaa caagattaag acgcaagatg gccatgcaag actaggaaat 2460
ctagaatttc tcgaagagaa accattagta ggagaagcac tagctcgtgt gaaaagagcg 2520
gagaaaaaat ggagagacaa acgtgaaaaa ttggaatggg aaacaaatat tgtttataaa 2580
gaggcaaaag aatctgtaga gcgtttatgt gtaaaactct aatatgtag attacaagcg 2640
gataccaaca tcgcgatgat tcctgcggca gataaacgcg ttcatagcat tcgagaagct 2700
tatctgcctg agctgtctgt gattccgggt gtcaatgcgg ctatttttga agaattagaa 2760
gggcgtatgt tcactgcaat ctccctatat gatgcgagaa atgtcaattaa aaatgggtgat 2820
tttaataatg gcttatcctg ctggaacgtg aaagggcatg tagatgtaga agaacaaaac 2880
aaccacgctt cggtccttgt tgttccggaa tgggaagcag aagtgtcaca agaagttcgt 2940
gtctgtccgg gtcgtggcta tatcctctgt gtcacagcgt acaaggaggg atatggagaa 3000
ggttgcgtaa ccattcatga gatcgagaac aatacagacg aactgaagtt tagcaactgt 3060
tagaagaggg aagtatatcc aaacaacacg gtaacgtgta atgattatc tgcgactcaa 3120
gaagaatatg agggtagcga cacttctcgt aatcgaggat atgacggagc ctatgaaagc 3180
aattcttctg taccagctga ttatgcatca gccatgaag aaaaagcata tacagatgga 3240
cgaagagaca atccttgtga atctaacaga ggatatgggg attcacacc actaccagct 3300
ggctatgtga caaagaatt agagtacttc ccagaaaccg ataaggtatg gattgagatc 3360
ggagaaacgg aaggaacatt catcgtggac agcgtggaat tacttcttat ggaggaatag 3420

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SEQ ID NO: 20      moltype = AA length = 1139
FEATURE           Location/Qualifiers
REGION            1..1139
                  note = Amino acid sequence of the engineered insecticidal
                  chimeric protein TIC844_5.
source            1..1139
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 20
MEINNQNCV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIEEF ARNQAISRLG GLSNLYKVVY RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATIMN RYSDLTSLIH VYTNHCVDY NQGLRRLEGR FLSDWIVYNR FRQLTISVL 240
DIVAFPPNDV IRTYPIQTAT QLTREVLYDL PFINENLSPA AKYPTFSAAE SAIIRSPHLV 300
DFLNSFTIYT DSLARSAYWG GHLVNSFRTG TTNLIRSPL YGREGNTERP VTITASPSVP 360
IFRTLSTPTG LDNSNPVAGI EGVEFQNTIS RSIYRKS GPI DSFSELPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRTAGT FSWTHRSAS PTNEVSPSRI TQIPWVKAHT LASGASVIK 480
PGFTGGDILT RNSMGELGTL RVTFTGRLPQ SYIIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTPITFSRAQ EEFDLIYQSG VYIDRIEFIP VTATFEEAYD 600
LERAQKAVNE LFTSSNQIGL KTDVTDYHID QVSNLVECLS DEFCLDEKKE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQL DRGWRGSTD I TQGGDDVFK ENYVTLGTF DECYPYLYQ 720
KIDESKLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHETVN VPGTGSWLPL SAPSPIGKCA 780

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SEQ ID NO: 21      multype = DNA length = 3420
FEATURE            Location/Qualifiers
misc_feature       1..3420
                   note = Nucleotide sequence used for expression in a
                   bacterial cell encoding TIC844_6.
source             1..3420
                   mol_type = other DNA
                   organism = synthetic construct

```

```
SEQ ID NO: 22      moltype = AA  length = 1139
FEATURE            Location/Qualifiers
REGION            1..1139
                  note = Amino acid sequence of the engineered insecticidal
                        chimeric protein TIC844 6.
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-continued

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source          1..1139
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 22
MEINNQNQCVC PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIEEF ARNQAISRLG GLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSDLTSLIH VYTNHCVDYV NQGLRRLLEGR FLSDWIVYNR FRRLTISVL 240
DIVAFPPNYD IRTYPIQTAT QLTREVYLDL PFINENLSPA ASYPTFSAAE SAIIRSPHLV 300
DFLNSFTIYT DSLARYAYWG GHLVNSFRTG TTNLIRSLP YGREGNTERP VTITASPSVP 360
IFRTLSTYITG LDNRNPVAGI EGVEFQNTIS RSIYRKSGPI DSFSELPPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRIAGT VFSWTHRSAS PTNEVSPSRI TQIPWVKAHT LASGASVIK 480
PGFTGGDILT RNSMGELGTL RVTFTGRLPQ SYIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTPITFSRAQ EEFDLVIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKAVNE LFTSSNQIGL KTDVTDYHID QVSNLVECLS DEFCLDEKKE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQL DRGWRGSTD I QGGGDDVFK ENYVTLGTF DECYPTYLYQ 720
KIDESKLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHETVN VPGTGLWPL SAPSPIGKCA 780
HHSHHFLSDI DVGCTDLNED LGVWVIFKIK TQDGHARLGN LEFLEEKPLV GEALARVKRA 840
EKKWRDKREK LEWETNIVYK EAKESVDALF VNSQYDRLQA DTNIAMIHAA DKRVHSIREA 900
YLPESLVIPG VNAAIFEELE GRIFTAFSLY DARNVIKNGD FNNGLSCWNV KGHVDVEEQN 960
NHRSLVLPVE WEAEVSVQVR VCPGRGYILR VTAYKEGYGE GCVTIHEIEN NTDELKFSNC 1020
VEEEVYPNMT VTCNDYTATQ EEEGTYSYTSR NRGYDGAYES NSSVPADYAS AYEKAYTDG 1080
RRDNPCESNR GYGDYTPLPA GYVTKLEYF PETDKVWIEI GETEGTFIVD SVELLLMEE 1139

SEQ ID NO: 23      moltype = DNA length = 3420
FEATURE            Location/Qualifiers
misc_feature        1..3420
                    note = Nucleotide sequence used for expression in a
                        bacterial cell encoding TIC844_7.
source              1..3420
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 23
atggaaataa ataatacaaaa ccaatgtgtg ccttacaatt gttaaagtaa tcctaaggag 60
ataatattag cgcaggaaag gctagaaaca ggaataactg tagcagacat ttcattaggg 120
cttatttaatt ttcatatttc taatttttga ccaggaggag gatttatagt aggtttacta 180
gaattaatat ggggacttcg agggccttcg caatgggata tttttttagc tcaaattgag 240
caattgatta gtcaaaagaat agaagaattt gctaggaatc aggcaatttc aagattggag 300
gggctaagca atctttataa ggtctatggt agagcgttta gcgactggga gaaagatcct 360
actaatcctg cttaagggga agaaatgcgt atacaattta atgacatgaa tagtgctctc 420
ataacggcta ttccactttt tagagttcaa aattatgaag ttgctctttt atctgtatat 480
gttcaagcgc caacttaca tttatctatt ttaagggatg tttcagtttt cggagaaaga 540
tggggatagc atacagcgac tatcaataat cgctatagtg atctgactag ccttattcat 600
gtttatacta accatttgtt ggatacgtat aatcagggat taagggcgtt ggaaggtcgt 660
tttcttagcg attggattgt atataatcgt ttccggagac aattgacaat ttcagtatta 720
gatattgttg cgttttttcc aaattatgat attagaacat atccaattca aacagctact 780
cagctaacga ggggaagtcta tctggattta ccttttatta atgaaaatct ttctcctgca 840
gcaagctatc caaccttttc agctgctgaa agtgctataa ttagaagtc tcatattaga 900
gactttttta atagcttttc catttataca gatagtcctg cagcttctgc atattgggga 960
gggcacttgg taaattcttt ccgcacagga accactacta atttgataag atccccttta 1020
tatggaaggg aaggaaatac agagcgcccc gtaactatta ccgcatcacc tagcgtacca 1080
atatttagaa cactttcata tagaacaggc cttgacaatt caaatcctgt agctggaatc 1140
gagggagtg aattccaaaa tactataagt agaagtatct atcgtaaaag cgggtccaata 1200
gattctttta gtgaattacc acctcaagat gccagcgtat ctctcgcaat tgggtatagt 1260
caccgtttat gccatgcaac atttttagaa cggattagtg gaccaagaat agcagggcacc 1320
gtattttctt ggacacaccc tagtgccagc cctactaacg aagtaagtc atctagaatt 1380
acacaaattc catgggtaaa ggcgcatact cttgcatctg gtgcctcgt cattaaagg 1440
cctggattta caggtggaag tattctgact aggaatagta tgggcagact ggggacctta 1500
cgagtaacct tcacaggaag attaccacaa agttattata tacgtttccg ttatgcttcg 1560
gtagcaataa ggagtggtac atttagatat tcacagccac cttcgtagg aatttcatt 1620
ccaaaaacta tggacgcagg tgaaccacta acatctcgtt cgttcgctca tacaacactc 1680
ttcactccaa taaccttttc acgagctcaa gaagaatttg atctatacat ccaatcggt 1740
gtttatatag atcgaattga atttataacc gttactgcaa catttgaggc agaatatgat 1800
tagaaaagag caaaaaaggc gttgaatgag ctgtttactt cttccaatca aatcggtgta 1860
aaaacagatg tgacggatta tcatattgat caagtatcca atttagtgta gtgtttatct 1920
gatgaatttt gtctggatga aaaaaaagaa ttgtccgaga aagtcacaac tgcgaagcga 1980
cttagtgatg agcgggaattt acttcaagat ccaacttata gagggatcaa tagacaacta 2040
gaccgtggct ggagaggaag tacggatatt accatccaag gaggcgatga cgtattccaa 2100
gagaattacg ttacgctatt ggttaccttt gatgagtgct atccaacgta tttatatcaa 2160
aaaatagatg agtcgaattt aaaaagcctat acccggtacc aattaagagg gtatatcgaa 2220
gatagtcgaag acttagaaat ctatttaatt cgctacaatg ccaaacacga aacagtaaat 2280
gtgccaggta cgggttcctt atggccgctt tcagccccc aatccaatcg aaaaatgtgc 2340
catcattccc atcattcttc cttggacatt gatgttgat gtacagactt aaatgaggac 2400
ttaggtgtat ggggtgatatt caagattaag acgcaagatg gccatgcaag actaggaaat 2460
ctagaatttc tcgaagagaa accattagta ggagaagcac tagctcgtgt gaaaaagagc 2520
gagaaaaaat ggagagacaa acgtgaaaaa ttggaatggg aaacaaatat tgtttataa 2580
gagggcaaaa aatctgtaga tgcctttatt gtaaactctc aatatgatag attacaagcg 2640
gataccaaca tcgcgatgat tcatcgcgca gataaacgcg ttcatagcac tcgagaagct 2700
tatctgcctg agctgtctgt gattccgggt gtcaatgcgg ctatttttga agaattagaa 2760

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gggcgatatt tcaactgcatt ctccttatat gatgcgagaa atgtcattaa aaatggtgat 2820
ttaaataatg gcttatccctg ctggaacgtg aaagggcatg tagatgtaga agaacaaaac 2880
aaccaccgtt cggtccttgt tgttccggaa tgggaagcag aagtgtcaca agaagttcgt 2940
gtctgtccgg gtcgtggcta tatecttcgt gtcacacgtg acaaggaggg atatggagaa 3000
ggttgcgtaa ccattcatga gatcgagaa aatacacagc aactgaagtt tagcaactgt 3060
gtagaagagg aagtatatcc aaacaacacg gtaacgtgta atgattatac tgcgactcaa 3120
gaagaatatg agggtagcta cacttctcgt aatcgaggat atgacggagc ctatgaaagc 3180
aattcttctg taccagctga ttatgcatca gcctatgaag aaaaagcata tacagatgga 3240
cgaagagaca atccttctga atctaacaga ggatatgggg attcacaccc actaccagct 3300
ggctatgtga caaaagaatt agagtacttc ccagaaaccg ataaggtatg gattgagatc 3360
ggagaaacgg aaggaacatt catcgtggac agcgtggaat tacttcttat ggaggaatag 3420

SEQ ID NO: 24      moltype = AA length = 1139
FEATURE           Location/Qualifiers
REGION           1..1139
                  note = Amino acid sequence of the engineered insecticidal
                  chimeric protein TIC844_7.
source           1..1139
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 24
MEINNQNCV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWFIGIPS QWDIFLAQIE QLISQRIEEF ARNQAISRL ELSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSLDLSLIH VYTNHCVDYI NQGLRRLEGR FLSDWIVYNR FRRQLTISVL 240
DIVAFPPNYD IRTYPIQTAT QLTREVYLDL PFINENLSPA ASYPTFSAAE SAIIRSPHLV 300
DFLNSFTIYT DSLARSAYWG GHLVNSFRTG TTNLIRLSP YGREGNTERP VTITASPSVP 360
IFRTLSTYRTG LDNSNPVAGI EGVEFQNTIS RSIYRKS GPI DSFSELPPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRIAGT VFSWTHRSAS PTNEVSPSRI TQIPVWKAHT LASGASVIK 480
PGFTGGDILT RNSMGLGTL RVTFTGRLPQ SYIIRFRYAS VANRSCTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTL FTFITFSRAQ EEFDLIYIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKAVNE LFTSSNQIGL KTDVTDYHID QVSNLVECLS DEFCLDEKKE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQL DRGWRGSDTI TIQGGDDVFK ENYVTLTGTF DECYPTYLYQ 720
KIDSKLLKAY TRYQLRGYIE DSQDLIYILI RYNAKHETVN VPGTGSWLPL SAPSPIGKCA 780
HHSHHFLSLDI DVGCTDLNED LGVWVIFKIK TQDGHARLGN LEFLEEKPLV GEALARVKRA 840
EKKWRDRKREK LEWETNIVYK EAKESVDALF VNSQYDRLQA DTNIAMIHAA DKRVHSIREA 900
YLPFLSVIPG VNAIFEELE GRIFTAFSLY DARNVIKNGD FNNGLSCWNV KGHVDVEEQN 960
NHRSLVLVPE WEAESVQEV R VCPGRGYILR VTAYKEGYGE GCVTIHEIEN NTDELKFSNC 1020
VEEEVYPNNT VTCNDYATQ EEEYEGTYSR NRGYDGAYES NSSVPADYAS AYEKAYTDG 1080
RRDNPCESNR GYGDYTPLPA GYVTKLEYF PETDKVWIEI GETEGTIVD SVELLMLME 1139

SEQ ID NO: 25      moltype = DNA length = 3420
FEATURE           Location/Qualifiers
misc_feature      1..3420
                  note = Nucleotide sequence used for expression in a
                  bacterial cell encoding TIC844_8.
source           1..3420
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 25
atggaaataa ataatacaaaa ccaatgtgtg ccttacaatt gttaaagtaa tcctaaggag 60
ataatattag gcgaggaaga gctagaaaca ggaataactg tagcagacat ttcattaggg 120
cttattaatt ttctatatcc taatttttga ccaggaggag gatttatagt aggtttacta 180
gaattaatat ggggatttat agggccttcg caatgggata tttttttagc tcaaatgtag 240
caattgatta gtcacaagata agaagaattt gctaggaatc aggcgaattt aagattggag 300
gggctaagca atctttataa ggtctatgtt agagcgttta gcgactggga gaaagatcct 360
actaatcctg cttaaaggga agaaatgcgt atacaattta atgacatgaa tagtgctctc 420
ataacggcta ttcactttt tagagtccaa aattatgaag ttgctctttt atctgtatat 480
gttcaagccg caaacttaca tttatctatt ttaaggatg tttcagtttt cggagaaaga 540
tggggatatg atacagcgag tatcaataat cgctatagtg atctgactag ccttattcat 600
gtttatacta accattgtgt ggatacgtat aatcagggat taaggcgttt ggaaggctcg 660
tttcttagcg attggattgt atataatcgt ttccggagac aattgacaat ttcagtatta 720
gatattgttg cgttttttcc aaattatgat attagaacat atccaattca aacagctact 780
cagctaacga ggggaagtcta tctggattta ccttttatta atgaaaatct ttctcctgca 840
gcagtataat caaccttttc agctgctgaa agtgcataaa ttagaagtc tcatattagta 900
gactttttaa atagctttac catttatata gatagtctgg cagcttctgc atattgggga 960
gggcacttgg taaattcttt ccgcacagga accactacta atttgataag atccccttta 1020
tatggaaggg aaggaaatag gtaactatta ccgcacaccc gtaactatc tagcgtacca 1080
atatttagaa cactttcata tccaacaggc cttgacaatt caaatcctgt agctggaatc 1140
gagggagtg aattocaaaa tactataagt agaagtatct atcgtaaaa cggtccaata 1200
gattctttta gtgaattacc acctcaagat gccagcgtat ctccgtcaat tgggtatagt 1260
caccgtttat gccatgcaac atttttagaa cggattagtg gaccaagaat agcaggcacc 1320
gtattttctt ggacacacgg tagtgccagc cctactaacg aagtaagtcc atctagaatt 1380
acacaaatc acagggtaaa ggcgcatact cttgcatctg gtgcctccgt cattaaagg 1440
cctggattta caggtggaga tattctgact aggaatagta tgggcgagct ggggacctta 1500
cgagtaacct tcacaggaa attaccacaa agttattata tacgtttccg ttatgcttcg 1560
gtagcaataa ggaagtgtac atttagatat tcacagccac cttcgtatgg aatttcattt 1620
ccaaaaacta tggacgcagg tgaaccacta acatctcgtt cgttcgctca tacaacactc 1680
ttcactccaa taaccttttc acgagctcaa gaagaatttg atctatacat ccaatcggtt 1740

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gtttatatag atcgaattga atttataccg gttactgcaa catttgagcg agaatatgat 1800
ttagaaagag cacaaaaggg ggtgaatgag ctgtttactt cttccaatca aatcgggtta 1860
aaaacagatg tgacgggatta tcatattgat caagtatcca atttagttga gtgtttatct 1920
gatgaatttt gtctggatga aaaaaaagaa ttgtccgaga aagtcaaaaca tgcgaagcga 1980
cttagtgatg agcgggaattt acttcaagat ccaaacttta gagggatcaa tagacaacta 2040
gaccgtggct ggagagggaag tacggatatt accatccaag gaggcgatga cgtattcaaa 2100
gagaattacg ttacgctatt gggtagcctt gatgagtgc atccaacgta tttatatcaa 2160
aaaatagatg agtcgaattt aaaagcctat acccgttacc aattaagagg gtatatcgaa 2220
gatagtcaag acttagaatt ctatttaatt cgctacaatg ccaaacacga aacagtaaat 2280
gtgccaggta cgggttcctt atggccgctt tcagcccaa gtccaatcgg aaaatgtgcc 2340
catcattccc atcattttct cttggacatt gatgttgat gtacagactt aaatgaggac 2400
ttaggtgtat ggggtgatatt caagattaag acgcaagatg gccatgcaag actaggaaat 2460
ctagaatttc tcgaagagaa accattagta ggagaagcac tagctcgtgt gaaaagagcg 2520
gagaaaaaat ggagagacaa acgtgaaaaa ttggaatggg aaacaaatat tgtttataaa 2580
gaggcaaaag aatctgtaga tgctttattt gtaaaacttc aatatgatag attacaagcg 2640
gataccaaca tgcgcatgat tcatgcggca gataaacgcg ttcatagcac tcgagaagct 2700
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ggcgctattt tcaactgatt ctccctatat gatgcgagaa atgtcattaa aaatggtgat 2820
ttaataatg cgttatcctg ctggaacgtg aaaggcgatg tagatgtaga agaacaaaac 2880
aaccaccgtt ggttccttgt tgttccggaa tgggaagcag aagtgtcaca agaagttcgt 2940
gtctgtccgg gtcgtggcta tatccttcgt gtcacagcgt acaaggaggg atatggagaa 3000
ggttgcgtaa ccattcatga gatcgagaa aatacagacg aactgaagtt tagcaactgt 3060
gtagaagagg aagtataacc aaacaacacg gtaacgtgta atgattatac tgcgactcaa 3120
gaagaatatg agggtagcga cacttctcgt aatcgaggat atgacggagc ctatgaaagc 3180
aattcttctg taccagctga ttatgcata gcctatgaag aaaaagcata tacagatgga 3240
cgaagagaca atccttctga atctaacaga ggatatgggg attacacacc actaccagct 3300
ggctatgtga caaagaatt agagtacttc ccagaaaccg ataaggtatg gattgagatc 3360
ggagaacagg aaggaaacatt catcgtggac agcgtggaat tacttcttat ggaggaatag 3420

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SEQ ID NO: 26      moltype = AA length = 1139
FEATURE           Location/Qualifiers
REGION            1..1139
                  note = Amino acid sequence of the engineered insecticidal
                  chimeric protein TIC844_8.
source            1..1139
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 26
MEINNQNCVQV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIEEF ARNQAISRL EGLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSLTSLIH VYTNHCVDY NQGLRRLEGR FLSDWIVYNR FRRQLTISVL 240
DIVAFPPNDV IRTYPIQTAT QLTREYVLDL PFINENLSPA AVYPTFSAE SAIIRSPHLV 300
DFLNSFTIYT DSLARSAYWG GHLVNSFRTG TTTNLIRSPL YGREGNTERP VTITASPSVP 360
IFRTLSYPTG LDNSNPVAGI EGEVFQNTIS RSIYRKS GPI DSFSELPQD ASVSPAIGYS 420
HRLCHATFLE RISGPRTAGT VFSWTHRSAS PTNEVSPSRI TQIPWVKAHT LASGASVIK 480
PGFTGGDILT RNSMGELGTL RVTFTGRLPQ SYYIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTTL FTPITFSRAQ EEFDLYIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKAVNE LFTSSNQIGL KTDVTDYHID QVSNLVECLS DEFCLDEKKE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQL DRGWRGSDTI TIQGGDDVFK ENYVTLTGTF DECYPTYLYQ 720
KIDESKLLKAY TRYQLRGYIE LGSQDLEIYLI RYNAKHETVN VPGTGSWLPL SAPSPIGKCA 780
HSHHFSLDI DVGCTDLNED LGVWVIFKIK TQDGHARLGN LEFLEEKPLV GEALARVKRA 840
EKKWRDKREK LEWETNIVYK EAKESVDALF VNSQYDRLQA DTNIAMIHAA DKRVHSIREA 900
YLPESLVI PG VNAIFEELE GRIFTAFSLY DARNVIKNGD FNNGLSWNV KGHVDVEEQN 960
NHRSLVVPE WEAEVSGEVR VCPGRGYILR VTAYKEGYGE GCVTIHEIEN NTDELKFSNC 1020
VEEEVYPNNT VTCNDYTATQ EEYEGTYSR NRGYDGAYES NSSVPADYAS AYEKAYTDG 1080
RRDNPCESNR GYGDYTPLPA GYVTKLEYF PETDKVWIEI GETEGTFIVD SVELLMLME 1139

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SEQ ID NO: 27      moltype = DNA length = 3498
FEATURE           Location/Qualifiers
misc_feature       1..3498
                  note = Synthetic DNA sequence designed for plant expression
                  encoding Cry1Da1.
source            1..3498
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 27
atggagatca acaaccagaa ccagtgcgtc ccgtacaact gcctgagcaa ccctaaggag 60
atcatcctgg gtgaggaacg cctggagacc ggcaacaccg tagccgacat tagcctgggc 120
ctcatcaact tccctctacg caactctcgt cccggcgggt gcttcatcgt gggcctcctg 180
gagcttatct tgggcttcat cggcccggtc cagtgggaca tcttctcgc ccagatcgag 240
caactgatca gccagcggat cgaggagttc gctaggaacc aggccatctc ccgcctggag 300
ggactctcca accctacaaa ggtgtacgtg cgcgcgttca gcgactggga gaaggaccg 360
accaaccggg cctcccgaga ggaatgcgt atccagttca acgatatgaa ctccggcctc 420
atcacccgca tcccgctctt ccgcgtgcag aactacgagg tggccctcct gtcctgtac 480
gttcaagccg ccaacctcca cctctccatc ctccgcgacg tgagcgtgtt cggcgagcgc 540
tggggctacg acaccgccac catcaacaac cgctactccg acctcacctc cctcatccac 600
gtttacacca accactgcgt ggacacgtac aaccagggcc tccgcgcgct ggagggcgcg 660
tctctctcgc actggatcgt gtacaaccgc ttccgcgcgc agctccacct ctccgctcgt 720

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-continued

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gacatcgtcg ccttcttttc caactacgac atccgcaccc accctatcca gaccgccacc 780
cagctcaccg gcgaggtcta cctcgacctc cegttcatca acgagaacct cagcccggcc 840
gccagctacc cgaccttttc cgcgcgtgag tccgccatca ttcgcagccc gcacctcgtg 900
gaattctcca actccttcac catctacacc gaetccctcg cccgctacgc ctactggggc 960
ggtcacctcg tgaactcctt ccgcacccgc accactacca acctcatccg cagcccgtc 1020
tacggccgcg agggcaacac cgagcgcccc gtgaccatca ccgccagccc gagcgtgccc 1080
atcttcgcga ccttcagcta catcacccgc ctggacaaca gcaacctgtg ggcggggc 1140
gagggcggtg agttccagaa caccatctcc agggagcatc accgcaagag cggccctatc 1200
gacagcttca gcgagctgcc tcttcaggac gccagcgtga gccctgccat cggctacagc 1260
cacaggtgtg gccacggcac ctctctggag cgcctcagcg gccctcgcat cgcggggc 1320
gtgttctcgt ggacccacgc cagcgccctc cctacgaacg aggtgtctcc tagtcgcatc 1380
acccagatcc cttgggtcaa ggcacacacc ctggcctagt gcgctagtgt catcaagggc 1440
cctggcttca ccggtggtag catcctgacc aggaactcta tgggcgagct gggcactctg 1500
agggctcact tcaactggcg cctgcctcag tcttactaca tccgctccg ctacgctagt 1560
gtcgctaacc gctctgttac tttccgctac tctcagcctc cgtcttacgg tatctctttc 1620
cctaagacta tggacgctgg tgagcctctg accagtagga gcttcgctca cactactctg 1680
ttcactccta tacttttctc tagggctcag gaggagtctg acctgtacat ccagtctggt 1740
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cttgagcgcg ccagaagggt ggtgaacgcc ctcttacta gcaactaaca gctaggcctg 1860
aagactgacg tgcacgacta ccacatcgac caagtgaagc acctagtggc ctgctctccc 1920
gacgagttct gcctcgacga gaagcgcgag ctgtccgaga aggtgaagca cgcgaagcgc 1980
ctctccgacg agcgcaacct gctccaggac cccaacttca ggggcaccaa caggcagccc 2040
gaccgcggtc ggcgcggctc caccgacatc accatccagg gcggtgacga cgtattcaag 2100
gagaactacg ttaccctccc cggcaccttc gacgagtggt accccacctc cctctaccag 2160
aagatcgacg agtccaagct gaaggcctac acccgctacc agctccgagg ctacatcgag 2220
gaactccagg actcggaaat ctacctcacc cgctacaacg ccaagcacga gatcgtgaac 2280
gtgctctgga ccgcgagctc ctggcctctc agcgtggaga accagatcgg cctctgcggc 2340
gagcctaacc gctcgccccc tcacctcgag tggaaacctg acctcactg ctctgacagg 2400
gacggcgaga agtgcgcccc ccatagccac cacttctctc tggacatcga cgtgggctgc 2460
accgacctga acgaggacct gggcgtgtgg gttatcttca agatcaagac ccaggacggt 2520
cacgcccagg tgggtaacct ggagttcctt gaggaaaagc ctctgctggg tgaggccctg 2580
gccagggtca agagggtgga gaagaaatgg agggataaga gggagacctc gcagctggag 2640
accactatcg tctacaagga ggctaaggag tctgtcgatg ctctgttctg caactctcag 2700
tacgatagac tgcaagctga taccacatc gctatgatcc acgctgcgga taagcgggtc 2760
caccggatcc gggaggctta ccttcgggag ctttctgtca tcccggtgtg caacgctgcg 2820
atcttcgagg aacttgagga acggatcttc actgcgttta gtctttacga tgcgcggaac 2880
atcatcaaga acggggactt caacaatggt ctgctgtgct ggaacgtcaa gggctcatgt 2940
gaggtcgagg aacaaaacaa tcatcgtagt gtccttgta ttcctgagtg ggaggcggag 3000
gtctctcaag aggtccgtgt ttccccgggg cgtgggtaca ttcttctgtg tactgcgtac 3060
aaggaggggg acggggaggg gtgcgttact attcatgaga ttgagaacaa tactgatgag 3120
cttaagttca acaattgtgt tgaggaggag gtttaccga acaatactgt tacgtgcac 3180
aactacacgg caacgcaaga ggaatacag gggacgtaca cctcgcgtaa tagaggggat 3240
gatgaggcgt accgaaccaa cccgtcggtt ccagcagatt atgcctcggt ttatgaggag 3300
aagtctgaca cggatagacg acgcgagaat ccatgtgagt caaatcgag atacggagat 3360
tacacaccat taccagcagg atacgttaca aaggagttag aatactccc ggaacagat 3420
aaagtttgga ttgaaatcgg agaaacagaa ggaacattca tcgtcgactc agtagaattg 3480
ttgttgatgg aagaatga 3498

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SEQ ID NO: 28      moltype = AA length = 1165
FEATURE           Location/Qualifiers
REGION            1..1165
                  note = Amino acid sequence of Cry1Da1 encoded by a
                  synthetic DNA sequence.
source            1..1165
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 28
MEINQNCQCV PYNCLSNPKE IILGEERLET GNTVADISLG LINFLYSNFV PGGGFIVGLL 60
ELIWGFIGPS QWDIFLAQIE QLISQRIIEF ARNQAISRLS GLSNLYKVYV RAFSDWEKDP 120
TNPALREEMR IQFNDMNSAL ITAIPLFRVQ NYEVALLSVY VQAANLHLSI LRDVSVFGER 180
WGYDTATINN RYSDLTSLIH VYTNHCVDYTY NQGLRRLEGR FLSDWIVYNR FRRQLTISVL 240
DIVAFFPNYD IRTYPIQTAT QLTREYVLDL PFINENLSPA ASYPTFSAAE SAIIRSPHLV 300
DFLNSFTIYT DSLARYAYWG GHLVNSFRGT TTNLIRSPL YGREGNTERP VTITASPSVP 360
IFRTLSTIYG LDNSNPVAGI EGVEFQNTIS RSIYRKSGPI DSFSELPPOD ASVSPAIGYS 420
HRLCHATFLE RISGPRIAGT VFSWTHRSAS PTNEVSPSRI TQIPVWKAHT LASGASVIK 480
PGFTGGDILT RNSMGLGTL RVTFTGRLPQ SYIIRFRYAS VANRSGTFRY SQPPSYGISF 540
PKTMDAGEPL TSRSFAHTLL FTPITFSRAQ EEFDLYIQSG VYIDRIEFIP VTATFEAEYD 600
LERAQKVVNA LFTSTNQLGL KTDVTDYHID QVSNLVACLS DEFCLDEKRE LSEKVKHAKR 660
LSDERNLLQD PNFRGINRQP DRGWRGSTDI TIQGGDDVFK ENYVTLPGTF DECYPTYLYQ 720
KIDESKLLKAY TRYQLRGYIE DSQDLEIYLI RYNAKHEIVN VPGTGSWLPL SVENQIGPCG 780
EPNRCAPHLE WNPDLHCSR DGEKCAHSHS HFSLDIDVGC TDLNEDLGWV VIFKIKTQDG 840
HARLGNLEFL EEKPLLGEAL ARVKRAEKKW RDKRETQLQE TTIVYKEAKE SVDALFVNSQ 900
YDRLQADTNI AMIHAAKRV HRIREAYLPE LSVIPGVNAA IFEELEERIF TAFSLYDARN 960
I1KNGDFNNG LLCWNVKGHV EVEEQNNHRS VLVIPWEWAE VSQEVVRCPG RGYILRVYAT 1020
KEGYGEGCVT IHEIENNTDE LKFNNCVEEE VYPNNTVTCI NYTATQEEYE GTYTSRNRGY 1080
DEAYGNPNPV PADYASVYEE KSYTDRRREN PCBSNRGYGD YTPLPAGYVT KELEYFPETD 1140
KWWIEIGETE GTFIVDSVEL LLME 1165

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SEQ ID NO: 29      moltype = DNA length = 3501

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FEATURE                      Location/Qualifiers
misc_feature                  1..3501
                              note = Synthetic DNA sequence designed for plant expression
                              encoding Cry1Da1 with an additional Alanine residue
                              inserted at position 2.
source                       1..3501
                              mol_type = other DNA
                              organism = synthetic construct

```

```

SEQUENCE: 29
atggctgaga tcaacaacca gaaccagtgc gtcccgtaca actgctgag caaccctaag 60
gagatcatcc tgggtgagga acgcctggag accggcaaca ccgtagccga cattagcctg 120
ggcctcatca acttctctta cagcaacttc gtgcccgcg gtggttccat cgtgggctc 180
ctggagctta tctggggctt catcgcccg tcccagtggg acatcttctt cgcccagatc 240
gagcaactga tcagccagcg gatcgaggag ttcgctagga accaggccat ctcccgcctg 300
gagggactct ccaacctcta caaggtgtac gtgcgcgctg tcagcgactg ggagaaggac 360
ccgaccaacc cggccctccg cgaggaatat cgtatccagt tcaacgatat gaactcggcc 420
ctcatcaccg ccatcccgct ctcccgctg cagaactacg aggtggccct cctgtccgtg 480
tacgttcaag ccgccaacct ccacctctcc atcctccgct acgtgagcgt gttcggcgag 540
cgctggggct acgacaccgc caccatcaac aaccgctact ccgacctcac ctccctcacc 600
cacgtttaca ccaaccactg cgtggacacg tacaaccagg gcctccgccc cctggagggc 660
cgcttctctg ccgactggat cgtgtacaac cgcttccgccc gccagctcac catctccgtc 720
ctggacatcg tcgccttctt tcccaactac gacatccgca cctaccctat ccagaccgcc 780
accagctca ccccgagagt ctacctcgac ctcccgctta tcaacgagaa cctcagcccg 840
gccgcagact acccgacctt ctcccgctg gagtccgcca tcattcgagc ccgcgacctc 900
gtggacttcc tcaactcctt caccatctac accgaactcc tcgcccgtta cgctactggt 960
ggcgggtcac tcgtgaactc ctcccgacc gccaccacta ccaacctcat ccgcagcccg 1020
ctctacggcc gcgaggcgaa caccgagcgc ccggtgacca tcaccgccag ccgcagcgtg 1080
cccatcttcc gcacctcag ctacatcacc ggcttgaca acagcaacc tgtggcgggc 1140
atcgaggggc tggagtcca gaacaccatc tccaggagca tctaccgcaa gagcgccct 1200
atcgacagct tcagcgagct gccctctcag gacgccagcg tgagccctgc catcggtac 1260
agccacagcg tcgtgccagc cacttctctg gagcgcatca gcggccctcg catcgcgggc 1320
accgtgttct cgtggaccga ccgcagcgcc tctctacga acgaggtgtc tctagtctgc 1380
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gggtgtgata tcgacaggat cgaattctac ccctgaccgc ccagcttcca ggccgagtag 1800
gaccttgagc gcgccagaaa ggtggtgaac gccctcttca ctgacactaa ccagctaggc 1860
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cgctctccg acgagcgcaa cctgctccag gaccccaact tcaggggcat caacaggcag 2040
cccgaccgcg gctggcgagg ctccaccgac atcaccatcc agggcggtga cgacgtattc 2100
aaggagactc acgttacctt ccccgccacc ttcgacaggt gttaccctac ctacctctac 2160
cagaagactg acgagtccea gctgaaggcc tacaccgctc accagctccg cggctacatc 2220
gaggactccc aggaacctga aatctacctc atccgctaca acgccaagca cgagatcgtg 2280
aacgtgctcg gcaccggcag cctctggcct ctccagctgg agaaccagat cggcccttgc 2340
ggcgagccta accgctgcgc cctcactc gaagtggaaac ctgacctcca ctgctcgtgc 2400
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cagtacgata gactgcaagc tgataccaac atcgctatga tccacgctgc ggataagcgg 2760
gtccaccgga tccgggagcg ttaccttccg gagctttctg tcatcccggg tgtcaacgct 2820
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aacatcatca agaaccggga cttcaacaat ggtctgctgt gctggaaact caagggtcat 2940
gtcgaggtcg aggaacaaaa caatcatcgt agtgtcctg tcatctctga gtgggaggcg 3000
gaggtctctc aagaggtccg tgtttgcccg gggcggtggg acattcttcc tgttactcgc 3060
tacaaggagg ggtacgggga ggggtgcgtt actattcatg agattgagaa caatactgat 3120
gagcttaagt tcaacaattg tgttgaggag gaggtttacc cgaacaatac tgttactgtc 3180
atcaactaca cggcaacgca agaggaatac gaggggagct acacctcgcg taatagaggg 3240
tatgatgagg cgtacgaaaa caaccgctcg gttccagcag attatgcctc ggtttatgag 3300
gagaagtcgt acacggatag acgacgcgag aatccatgtg agtcaaatcg aggatacggg 3360
gattacacac cattaccagc aggatcgtt acaaaggagt tggaatactt cccggaacaa 3420
gataaagttt ggattgaat cggagaacaa gaaggaacat tcatcgctga ctcagtagaa 3480
ttgtgttgta tggagaagt a

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SEQ ID NO: 30               moltype = AA length = 1166
FEATURE                      Location/Qualifiers
REGION                      1..1166
                              note = Amino acid sequence of Cry1Da1 encoded by a
                              synthetic DNA sequence wherein an additional Alanine
                              residue has been inserted at position 2.
source                       1..1166
                              mol_type = protein
                              organism = synthetic construct
SEQUENCE: 30

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MAEINNQNQC	VPYNCLSNPK	EIILGEERLE	TGNTVADISL	GLINFLYSNF	VPGGGFIVGL	60
LELIWFIGP	SQWDIFLAQI	EQLISQRIE	FARNQAISRL	EGLSNLYKVY	VRAFSDWEKD	120
PTNPALREEM	RIQFNDMNSA	LITAIPLFRV	QNYEVALLSV	YVQAANLHLS	ILRDVSVFGE	180
RWGYDTATIN	NRYSDLTSLI	HVYTNHCVDI	YNQGLRRLEG	RFLSDWIVYN	RFRRLTISV	240
LDIVAFFPNY	DIRTYIQTAT	TQLTREYVLD	LPFINENLSP	AASYPTFSAA	ESAIIRSPHL	300
VDFLNSFTIY	TDSLARYAYW	GGHLVNSFRT	GTTTNLIRSP	LYGREGNTER	PVTITASPSV	360
PIFRTLSYIT	GLDNSNPVAG	IEGVEFQNTI	SRSIYRKSGP	IDSFSELPPQ	DASVSPAIGY	420
SHRLCHATFL	ERISGPRIAG	TVFSWTHRSA	SPTNEVSPSR	ITQIPWVKAH	TLASGASVIK	480
GGPGFTGGDIL	TRNSMGELGT	LRVTFTGRLP	QSYIIRFRYA	SVANRSQTFR	YSQPPSYGIS	540
FPKTMADAGEP	LTSRSFAHTT	LFTPIITFSRA	QEEFDLYIQS	GVYIDRIEFI	PVTATFEAEY	600
DLERAQKVVN	ALFTSTNQLG	LKTDVTDYHI	DQVSNLVACL	SDEFCLDEKR	ELSEKVKHAK	660
RLSDERNLLQ	DPNFRGINRQ	PDRGWRGSTD	ITIQQGDDVF	KENYVTLPGT	FDECYPTYLY	720
QKIDESKLKA	YTRYQLRGYI	EDSQDLEIYL	IRYNAKHEIV	NVPGTGLWP	LSVENQIGPC	780
GEPNRCAPHL	EWNPDLCSC	RDGEKCAHHS	HHFSLDIDVG	CTDLNEDLGV	WVIFKIKTQD	840
GHARLGNLEF	LEEKPLIGEA	LARVKRAEKK	WRDKRETLLQ	ETTIVYKEAK	ESVDALFVNS	900
QYDRLQADTN	IAMIHADDKR	VHRIREAYLP	ELSVIPGVNA	AIFEELEERI	FTAFSLYDAR	960
NIKNGDFMN	GLLCWNVKGH	VEVEEQNNHR	SVLVIPEWEA	EVSQEVVCP	GRGYILRVTA	1020
YKEGYGEGCV	TIHIEIENNTD	ELKFNNCCEE	EVPNNNTVTC	INYTATQEEY	EGTYTSRNRG	1080
YDEAYGNFNS	VPADYASVYE	EKSYTDRRRE	NPCESNRGY	DYTPLPAGYV	TKELEYFPET	1140
DKVWIEIGET	EGTFIVDSVE	LLLMEE				1166

SEQ ID NO: 31
 FEATURE
 misc_feature
 source

moltype = DNA length = 3501
 Location/Qualifiers
 1..3501
 note = Synthetic DNA sequence designed for plant expression
 encoding Cry1Da1_3 with an additional Alanine residue
 inserted at position 2.
 1..3501
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 31

atggctgaga	tcaacaacca	gaaccagtgc	gtcccgtaga	actgctctgag	caaccctaag	60
gagatcatcc	tgggtgagga	acgcctggag	accggcaaca	ccgtagccga	cattagccctg	120
ggcctcatca	acttctctca	cagcaacttc	gtgcccgccg	gtggcttcat	cgtgggcctc	180
ctggagctta	tctggggctt	categgcccg	tccagtgagg	acatcttctc	cgcccagatc	240
gagcaactga	tcagccagcg	gacgcaggag	tgccttagga	accaggccat	ctcccgcctg	300
gaggagactc	ccaacctcta	caaggtgtac	gtgcgcgcgt	tcagcgactg	ggagaaggac	360
cagaccaacc	cggccctccg	cgaggaaatg	cgtatccagt	tcaacgatat	gaactcggcc	420
ctcatcaccg	ctatcccgct	cttcgcgcgt	cagaactacg	agggtggccct	cctgtccgctg	480
tacgttcaag	ccgccaaact	ccacctctcc	atcctccgcg	acgtgagcgt	gttcggcgag	540
cgctggggct	acgacaccgc	caccatcaac	aaccgctact	ccgacctcac	ctccctcact	600
cacgtttaca	ccaaccactg	cgtggacacg	tacaaccagg	gcctccgcgc	cctggagggc	660
cgcttctctc	cgactggatg	cgtgtacaac	cgcttccgcc	gccagctcac	catctccgtc	720
ctggacatcg	tcgcctctct	tcccaactac	gacatccgca	cctaccctat	ccagaccgcc	780
acccagctca	cccgcgaggt	ctacctcgac	ctcccggtca	tcaacgagaa	cctcagcccg	840
gccgccagct	accgcacact	ctccgcgcgt	gagtcgcgca	tcatttcgag	cccgcacctc	900
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ggcgggtcac	tcgtgaaact	cttcgcgacc	ggcaccacta	ccaacctcat	ccgcagcccg	1020
ctctacggcc	gcagggccaa	caccgagcgc	ccggtagacca	tcaccgccag	cccgaagcgt	1080
cccatcttcc	gcaccctcag	ctacatcacc	ggcctggaca	acagcaaccc	tgtggcgggc	1140
atcgaggggc	tggagtccca	gaacaccatc	tccaggagca	tctaccgcaa	gagcggccct	1200
atcgacagct	tcagcgagct	gcctctctag	gacgcccagc	tgagccctgc	catcgccctac	1260
agccacagcg	tgtgccacgc	acacctctct	gagcgcacat	gcggccctcg	catcgccggc	1320
accgtgttct	cgtggaccga	ccgcagcgcc	tctctacga	acgaggtgct	tcctagtcgc	1380
atcacccaga	tcctctgggt	caaggccccc	accctggcta	gtggcgctag	tgtcatcaag	1440
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ctgaggggtca	cttctactgg	ccgcctgcct	cagctctact	acatccgctt	ccgctacgct	1560
agtgtcgcta	accgctctgg	tactttccgc	tactctcagc	ctccgtctta	cggtatctct	1620
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cgctctctcg	acgagcgcaa	cctgctccag	gaccccaact	tcaggggcat	caacaggcag	2040
cccgaaccgc	gctggcgccg	ctccaccgac	atcaccatcc	agggcggtga	cgacgtattc	2100
aaggagaact	acgttaccct	ccccggcacc	ttcgacaggt	gttaccgccac	ctacctctac	2160
acgaagatcg	acgagtcctc	gctgaaggcc	tacacccgct	accagctccg	cggtacatc	2220
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agggacggcg	agaagtgcgc	ccaccatagc	caccacttct	ctctggacat	cgacgtgggc	2460
tgcaccgacc	tgaacgagga	cctgggcgtg	tgggttatct	tcaagatcaa	gacccaggac	2520
ggtcacgccca	ggctgggttaa	cttggagttc	cttgaggaaa	agcctctgct	gggtgaggcc	2580
ctggccaggg	tcagaggggc	tgagaagaaa	tggaggagata	agagggagac	cctgcagctg	2640
gagaccacta	tcgtctacaa	ggaggctaag	gagctctgtc	atgctctgtt	cgtcaactct	2700
cagtacgata	gactgcaagc	tgataccaac	atcgctatga	tccacgtgc	ggataagcgg	2760
gtccaccgga	tcggggagcg	ttaccttcgc	gagctttctg	tcacccgggg	tgtcaacgct	2820
gcgatcttcc	aggaacttga	ggaacggatc	ttactgcgtg	ttagtcttta	cgatgcgcgg	2880

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aacatcatca agaacgggga cttcaacaat ggtctgctgt gctggaacgt caagggtcat 2940
gtcgaggtcg aggaacaaaa caatcatcgt agtgtocttg tcattcctga gtgggaggcg 3000
gaggtctctc aagaggtccg tgtttgcccg gggcggtggg acattctctg tgttactgcg 3060
tacaaggagg ggtacgggga ggggtgcgtt actattcatg agattgagaa caatactgat 3120
gagcttaagt tcaacaattg tgttgaggag gaggtttacc cgaacaatac tgttacgtgc 3180
atcaactaca cggcaacgca agaggaatac gaggggacgt acacctcgcg taatagaggg 3240
tatgatgagg cgtacggaaa caaccgcgtc gtccagcgag attatgcctc ggtttatgag 3300
gagaagtcgt acacggatag acgacgcgag aatccatgtg agtcaaatcg aggatacgga 3360
gattacacac cattaccagc aggatacgtt acaaaggagt tggaatactt cccggaacaa 3420
gataaagttt ggattgaatc cggagaacaa gaaggaacat tcatcgctga ctcagtagaa 3480
ttgtgtgtga tggagaatg a                                     3501

SEQ ID NO: 32      moltype = AA length = 1166
FEATURE           Location/Qualifiers
REGION            1..1166
                  note = Amino acid sequence of the engineered insecticidal
                  protein Cry1Da1_3 encoded by a synthetic DNA sequence
                  wherein an additional Alanine residue has been inserted at
                  position 2.
source            1..1166
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 32
MAEINNQNQC VPYNCLSNPK EIILGEERLE TGNTVADISL GLINFLYSNF VPGGGFIVGL 60
LELIWGFIPG SQWDIFLAQI EQLISQRIE FARNQAISRL EGLSNLYKVY VRAFSDWEKD 120
PTNPALREEM RIQFNDMNSA LITAIPLFRV QNYEVALLSV YVQAANLHLS ILRDVSVFGE 180
RWGYDTATIN NRYSDLTSLI HVYTNHCVDI YNQLRLRLEG RFLSDWIVYN RFRRLTISV 240
LDIVAFFPNY DIRTYPIQTA TQLTREYVLD LPFINENLSP AASYPTFSAA ESAIIRSPHL 300
VDFLNSFTIY TDSLARSAYW GGHVNSFRT GTTTLIRSP LYGREGNTER PVTITASPSV 360
PIFRTLSEYIT GLDNSNPVAG IEGVEFQNTI SRSIYRKSGP IDSFSELPPO DASVSPAIGY 420
SHRLCHATFL ERISGPRVAG TVFSWTHRSA SPTNEVSPSR ITQIPWVKAH TLASGASVIK 480
GPGFTGGDIL TRNSMGELGT LRVFTTGRLP QSYIIRFRYA SVANRSGTFR YSQPPSYGIS 540
FPKTMIDAGEP LTRSFAHTT LFTPTIFSRA QEEFDLYIQS GVIYIDRIEF PVTATFEAEY 600
DLERAQKVVN ALFTSTNQLG LKTDVTDYHI DQVSNLVACL SDEFCLDEKR ELSEKVKHAK 660
RLSDERNLLQ DPNFRGINRQ PDRGWRGSTD ITIQGGDDVF KENYVTLPGT FDECYPTYLY 720
QKIDESKLKA YTRYQLRGIY EDSQDLEIYL IRYNAKHEIV NVPGTGSLWP LSVENQIGPC 780
GEPNRCAPHL EWNPDLCSC RDGEKCAHHS HHFSLDIDVG CTDLNLDELGV WVIFKIKTQD 840
GHARLGNLEF LEEKPLLGEA LARVKRAEKK WRDKRETLLQ ETTIVYKEAK ESVDALFVNS 900
QYDRLOADTN IAMIHADKR VHRIREAYLP ELSVIPGVNA AIFEELEERI FTAFSLYDAR 960
NIIKNGDFNN ELWCWNVKGH VEVEEQNNHR SVLVIPEWEA EVSQEVRVCP GRGYILRVTA 1020
YKEGYGEGCV TIHEIENNTD ELKFNNNCVEE EVYPNNTVTC INYTATQEEY EGTYSRNRG 1080
YDEAYGNNSP VPADYASVYE EKSYYDRRRE NPCESNRGYG DYTPLPAGYV TKELEYFPET 1140
DKVWIBIGET BGTFFIVDSVE LLLMEE                                     1166

SEQ ID NO: 33      moltype = DNA length = 3501
FEATURE           Location/Qualifiers
misc_feature       1..3501
                  note = Synthetic DNA sequence designed for plant expression
                  encoding Cry1Da1_4 with an additional Alanine residue
                  inserted at position 2.
source            1..3501
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 33
atggctgaga tcaacaacca gaaccagtgc gtcccgtaga actgcctgag caaccctaag 60
gagatcatcc tgggtgagga acgcctggag accggcaaca ccgtagccga cattagcctg 120
ggcctcatca acttccctta cagcaacttc gtgcccgccg gtggcttcat cgtgggcctc 180
ctggagctta tctggggttt catcggcccg tccagtgagg acatcttctc cgccagatc 240
gagcaactga tcagccagcg gatcgaggag ttcgctagga accaggccat ctcccgcctg 300
gagggactct ccaacctcta caaggtgtac gtgcgcgcgt tcagcgactg ggagaaggac 360
ccgaccaacc cggccctccg cgaggaaatg cgtatccagt tcaacgatav gaactcggcc 420
ctcctaccgg ccatcccgtc cttccgcgtg cagaactacg aggtggccct cctgtccgtg 480
tacgttcaag ccgccaacct ccacctctcc atcctccgcg acgtgagcgt gttcggcgag 540
cgctggggct acgacacgcg caccatcaac aaccgctact ccgacctcac ctccctcatc 600
cacgtttaca ccaaccactg cgtggacacg tacaaccagg gcctccgcgg cctggaggggc 660
cgcttctctc ccgactggat cgtgtacaac cgcttccgcc gccagctcac catctccgtc 720
ctggacatcg tgcctctctt tcccaactac gacatccgca cctacctatc ccagaccgcc 780
accagctca cccgcgaggt ctacctcgac ctcccggtta tcaacgagaa cctcagcccg 840
gccgccagct acccgacctt ctccgcgcgt gagtccgcca tcattcgtag cccgcacctc 900
gtggacttcc tcaactcctt caccatctac accgactccc tcgcccgtta cgctactagg 960
ggcggtcacc tcgtgaactc cttccgcacc gccaccacta ccaacctcat ccgcagcccg 1020
ctctacggcc gcgagggcaa caccgagcgc ccggtgacca tcaccgccag cccgagcgtg 1080
cccatcttcc gcacctcacc ctacatcacc ggcctggaca acaggaaccc tgtggcgggc 1140
atcgagggcg tggagttcca gaacaccatc tccaggagca tctaccgcaa gagcggccct 1200
atcgacagct tcagcgagct gcctcctcag gacgccagcg tgagccctgc catcggtac 1260
agccacaggc tgtgccagcg caccttctct gagcgcatca cggccctcgc catcgcgggc 1320
accgtgttct cgtggaccca ccgcagcgcc tctcctacga acgaggtgtc tctagtctgc 1380
atcaccaga tcccttgggt caaggcccac accctggcta gtggcgtag tgtcatcaag 1440
ggcctggct tcaccggtg tgacatctg accaggaact ctatggcgga gctgggcact 1500

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ctgaggggtca ctttactggt cgcctgcct cagtcttact acatccgctt ccgctacgct 1560
agtgtcgcta accgctctgg tactttccgc tactctcagc ctcggtctta cggtatctct 1620
ttccctaaga ctatggagcg tggtagcct ctgaccagta ggagcttcgc tcacactact 1680
ctgttcactc ctatcacttt ctctagggct caggaggagt tcgacctgta catccagtct 1740
gggtgtgata tcgacaggat cgagttcctc cccgtgaccg ccacgttcga ggccgagtac 1800
gaccttgagc ggcgccagaa ggtggtgaac gccctcttca ctgacctaa ccagctaggc 1860
ctgaagactg acgtgaccga ctaccacatc gaccaagtga gcaacctagt ggctgctc 1920
tcgacagagt tctgctcgca cgagaagcgc gagctgtccg agaaggtgaa gcacgccaag 1980
cgctctccgc acgagcgcaa cctgctccag gaccccaact tcaggggcat caacaggcag 2040
cccgaccgcg gctggcgcggt cccaccgac atccacctcc agggcggtga cgacgtattc 2100
aaggagaact acgttacctt ccccgccacc ttcgacgagt gttacccac ctacctctac 2160
cagaagactg acgagtcgaa gctgaaggcc tacacccgct accagctccg cggtacatc 2220
gaggactccc aggaactgga aatctacctc atccgctaca acgccaagca cgagatcgtg 2280
aacgtgcctg gcacggcgag cctctggcct ctcagcgtgg agaaccagat cgcccttgc 2340
ggcgagccta accgctgcgc cctcaccctc gagtggaaac ctgacctcca ctgctcgtgc 2400
agggacggcg agaagtgcgc caccatagc caccacttct ctctggacat cgacgtgggc 2460
tgcaccgacc tgaacgagga cctgggctgt tgggttatct tcaagatcaa gaccaggac 2520
ggtcacgcca ggctgggtta cctggagttc cttgaggaag agcctctgct gggtagggcc 2580
ctggccaggg tcaagagggt tgagaagaaa tggagggata agaggagagc cctgcagctg 2640
gagaccacta tcgtctacaa ggaggctaag gagtctgtcg atgctctgtt cgtcaactct 2700
cagtacgata gactgcaagc tgataccaac atcgctatga tccacgctgc ggataagcgg 2760
gtccaccgga tccgggaggg ttaccttccg gagctttctg tcatcccggtg tgtcaacgct 2820
gcgatcttcg aggaacttga ggaacggatc ttcactgcgt ttagtcttta cgatgcgctg 2880
aacatcatca agaacgggga cttcaacaat ggtctgctgt gctggaacgt caagggtcat 2940
gtcgaggtcg aggaacaaaa caatcatcgt agtgcctgt tcatctctga gtgggagggc 3000
gaggtctctc aagaggtcgc tgtttgcccg gggcggtgggt acattctctg tgttactgcy 3060
tacaaggagg ggtacgggga ggggtgcgtt actattcatg agattgagaa caatactgat 3120
gagcttaagt tcaacaattg tgttgaggag gaggtttacc cgaacaatac tgttactgcy 3180
atcaactaca cgcgaacgca agaggaatac gaggggacgt acacctcgcg taatagaggg 3240
tatgatgagg cgtacggaaa caaccgctcg gttccagcag attatgcctc ggtttatgag 3300
gagaagtcgt acacggatag acgacgcgag aatccatgtg agtcaaatcg aggatacgga 3360
gattacacac cattaccagc aggatcgtt acaaaggagt tggataactt cccggaaaca 3420
gataaagttt ggattgaaat cggagaaaca gaaggaaact tcatcgctga ctcagtagaa 3480
ttgttgttga tggaagaatg a 3501

```

SEQ ID NO: 34 moltype = AA length = 1166
 FEATURE Location/Qualifiers
 REGION 1..1166
 note = Amino acid sequence of the engineered insecticidal
 protein Cry1Da1_4 encoded by a synthetic DNA sequence
 wherein an additional Alanine residue has been inserted at
 position 2.
 source 1..1166
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 34

```

MAEINNQNQC VPYNCLSNPK EIILGEERLE TGNTVADISL GLINFLYSNF VPGGGFIVGL 60
LELIWFIGP SQWDIFLAQI EQLISQRIE FARNQAI SRL EGLSNLYKVY VRAFSDWEKD 120
PTNPALREEM RIQFNDMNSA LITAIPLFRV QNYEVALLSV YVQAANLHLS ILRDVSFVGE 180
RWGYDTATIN NRYSDLTSLI HVYTNHCVDT YNQLRRLEG RFLSDWIVYN RFRRLTISV 240
LDIVAFFPNY DIRTYPIQTA TQLTREYVLD LPFINENLSP AASYPTFSAA ESAIIRSPHL 300
VDFLNSFTIY TDSLARYAYW GHLVNSFRT GTTNLIRSP LYGREGNTER PVTITASPSV 360
PIFRTLSTYIT GLDNRNPVAG IEGVEFQNTI SRSIYRKSGP IDSFSELPPQ DASVSPAIGY 420
SHRLCHATFL ERISGPRIAG TVFSWTHRSA SPTNEVSPSR ITQIPWVKAH TLASGASVIK 480
GPGFTGDDL L TRNSMGLGT LRVFTTGRLP QSYIYRFRYA SVANRSCTFR YSQPPSYGIS 540
FPKTMADAGEP LTRSFAHTT LFTPTIFSRA QEEFDLYIQS GVIYDRIEFI PVTATFEAEY 600
DLERAQKVVN ALFTSTNQLG LKTDVTDYHI DQVSNLVACL SDEFCLDEKR ELSEKVKHAK 660
RLSDERNLLQ DPNFRGINRQ PDRGWRGSTD ITIQGGDDVF KENYVTLPGT FDECYPTYLY 720
QKIDESKLKA YTRYQLRGIY EDSQDLEIYL IRYNAKHEIV NVPGTGSLWP LSVENQIGPC 780
GEPNRCAPHL EWNPDLCSC RDGEKCAHHS HHFSLDIDVG CTDLNEDLGV WVIFKIKTQD 840
GHARLGNLEF LEEKPLLEGE LARVKRAEKK WRDKRETLQL ETTIVYKEAK ESDALFVNS 900
QYDRLQADTN IAMIHAADKR VHRIREAYLP ELSVIPGVNA AIFEELEERI FTAFSLYDAR 960
NIIKNGDFNN GLLCWNVKGH VEVEEQNNHR SVLVIPEWEA EVSQEVRVCP GRGYILRVTA 1020
YKEGYGEGCV TIHLEINNTD ELKFNNCVEE EYYPNNTVTC INYTATQEEY EGTYSRNRG 1080
YDEAYGNPNP VPADYASVYE EKSYPDTRRE NPCESNRGYG DYTPLPAGYV TKELEYFPET 1140
DKVWIEIGET EGTFTIVDSVE LLLMEE 1166

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SEQ ID NO: 35 moltype = DNA length = 3501
 FEATURE Location/Qualifiers
 misc_feature 1..3501
 note = Synthetic DNA sequence designed for plant expression
 encoding Cry1Da1_5 with an additional Alanine residue
 inserted at position 2.
 source 1..3501
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 35

```

atggctgaga tcaacaacca gaaccagtgc gtcccgtaga actgcctgag caaccctaag 60
gagatcatcc tgggtgagga acgctgggag accggcaaca ccgtagccga cattagctgc 120

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ggcctcatca acttctctta cagcaacttc gtgcccggcg gtggttcat cgtgggctc 180
ctggagctta tctggggctt catcgcccg tcccagtggt acatcttct cggccagatc 240
gagcaactga tcagccagcg gatcgaggag ttcgctagga accaggccat ctcccgcctg 300
gagggactct ccaacctcta caaggtgtac gtgcgcgcgt tcagcgactg ggagaaggac 360
cggaccaacc cggccctccg cgaggaaatg cgtatccagt tcaacgatat gaactcggcc 420
ctcatcaccg ccattcccgt ctcccgctg cagaactacg aggtggccct cctgtccgtg 480
tacgttcaag ccgccaacct ccacctctcc atcctcccg acgtgagcgt gttcggcgag 540
cgctggggct acgacacgcg caccatcaac aaccgctact ccgacctcac ctccctcatc 600
cacgtttaca ccaaccactg cgtggacacg tacaaccagg gcctccgcgg cctggagggc 660
cgcttctctc ccgactggat cgtgtacaac cgcttccg ccagctcac catctccgtc 720
ctggacatcg tcgcttctt tcccaactac gacatccgca cctacctat ccagacggcc 780
acccagctca cccgcgaggt ctacctcgac ctcccgttca tcaacgagaa cctcagcccg 840
gcccgcagct acccgacctt ctccgcgct gagtccgcca tcattcgag cccgcacctc 900
gtggacttcc tcaactcctt caccatctac accgactccc tcgcccgcag cgctactgg 960
ggcggtcacc tcgtgaactc ctccgcacc ggccaccata ccaacctcat ccgcagcccg 1020
ctctacggcc gcgagggcaa ccgcgagcgc ccggtgacca tcaccggcag cccgagcgtg 1080
cccatcttcc gcaccctcag ctaccgcacc ggctgggaca acagcaaccc tgtggcgggc 1140
atcgaggggc tggagtcca gaacaccatc tccaggagca tctaccgcaa gagcggccct 1200
atcgacagct tcagcgagct gcctcctcag gacgcccagc tgagccctgc catcggtac 1260
agccacagcg tgtgccacgc cacttctctg gacgcacatc gcggccctcg catcgcgggc 1320
accgtgttct cgtggaccca ccgcagcgcc tctctacga acgaggtgtc tcttagtgcg 1380
atcacccaga tcccttgggt caaggcccac accctggcta gtggcgctag tgtcatcaag 1440
ggccctggct tcacgggttg tgacatctg accaggaact ctatgggcca gctgggact 1500
ctgaggggtc ctctcactgg ccgcctgcct cagtcttact acatccgctt ccgctacgct 1560
agtgtcgcta acccttctgg tacttctcgc ctccgtctta cggtatctct 1620
ttccctaaga ctatggagcg tggtagcct ctgaccagta ggagcttgcg tcacactact 1680
ctgttctact ctatcacttt ctctagggct caggaggagt tcgacctgta catccagctc 1740
ggtgtgtaca tcgacaggtt cgagttcatc ccgtgaccg ccacgttcga ggccgagtag 1800
gaccttgagc gcgcaccagaa ggtgggtgaac gccctcttca ctagcactaa ccagctaggc 1860
ctgaagactg acgtgaccga ctaccacatc gaccaagtga gcaacctagt ggctgcctc 1920
tccgacaggt tctgctcga cgagaagcgc gagctgtccg agaaggtgaa gcacgccaag 1980
cgctctctcg acgagcgcaa cctgctccag gaccccaact tcaggggcat caacaggcag 2040
cccgaaccgg gctggcgcgg ctccaccgac atcaccatcc agggcggtga cgacgtattc 2100
aaggagaact acgttacctt ccccgccacc ttcgacgagt gttaccccac ctacctctac 2160
cagaagatcg acgagtccea gctgaaggcc tacaccgct accagctccg cggctacatc 2220
gaggactccc aggcacctga aatctacctc atccgctaca acgccaagca cgagatcgtg 2280
aacgtgcctg gcaccggcag cctctggcct ctacgcgtgg agaaccagat cggcccttgc 2340
ggcgagccta accgctgcgc cctcaccctc gagtggaaac ctgacctcca ctgctcgtg 2400
agggacggcg agaagtgcgc ccaccatagc caccacttct ctctggacat cgacgtgggc 2460
tgacccgacc tgaacgagga cctggcgctg tgggttatct tcaagatcaa gaccaggac 2520
ggtcacgcca ggtcgggtaa cctggagttc cttgaggaag agcctctgct gggtagggcc 2580
ctggccaggg tcaagagggc tgagaagaaa tggagggata agaggggagac cctgcagctg 2640
gagaccacta tcgtctacaa ggaggctaag gactctgtcg atgctctgtt cgtcaactct 2700
cagtacgata gactgcaagc tgataccaac atcgctatga tccagctgc ggataagcgg 2760
gtccaccgga tcggggaggc ttaccttccg gagcttctcg tcatcccggg tgtcaacgct 2820
gcgatcttct aggaacttga ggaacggatc ttcactgcgt ttactcttta cgatgcgagg 2880
aacatcatca agaacgggga ctccaacaat ggtctgctgt gctggaacgt caagggtcat 2940
gtcgaggtcg aggaacaaaa caatcatcgt agtgccttgc tcattctcga gtgggaggcg 3000
gaggtctctc aagaggtccg tgtttgccg gggcggtggg acattcttcg tgttactgcg 3060
tacaaggagg ggtacgggga ggggtgcgtt actattcatg agattgagaa caatactgat 3120
gagcttaagt tcaacaattg tgttaggag gaggtttacc cgaacaatac tgttactgtc 3180
atcaactaca cgtcaacgga agagggaatc acacctcgcg taatagaggg 3240
tatgatgagg cgtacggaaa caaccgctcg gttccagcag attatgcctc ggtttatgag 3300
gagaagtcgt acacggatag acgacgcgag aatccatgtg agtcaaatcg aggatacgg 3360
gattacacac cattaccagc aggatacgtt acaaaggagt tggaaatact cccggaacaa 3420
gataaagttt ggattgaatc cggagaaaca gaaggaaatc tcatcgctga ctcagtagaa 3480
ttgtgttgta tggaagaatg a 3501

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SEQ ID NO: 36      moltype = AA length = 1166
FEATURE            Location/Qualifiers
REGION             1..1166
                   note = Amino acid sequence of the engineered insecticidal
                   protein Cry1Da1_5 encoded by a synthetic DNA sequence
                   wherein an additional Alanine residue has been inserted at
                   position 2.
source             1..1166
                   mol_type = protein
                   organism = synthetic construct

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```

SEQUENCE: 36
MAEINNQNQC VPYNCLSNPK EILGEERLE TGN TVADISL GLINFLYSNF VPGGGFIVGL 60
LELIWFIGP SQWDFLAQI EQLISQRIE FARNQAISRL EGLSNLYKVY VRAFSDWEKD 120
PTNPALREEM RIQFNDMMSA LITAIPLFRV QNYEVALLSV YVQAANLHLS ILRDVSVFGE 180
RWGYDTATIN NRYSDLSLI HVYTNHCVDI YNQLRLRLEG RFLSDWIVYN RFRRLTISV 240
LDIVAFFPNY DIRTYPIQTA TQLTREYVLD LPFINENLSP AASYPTFSAA ESAIIRSPHL 300
VDFLNSFTIY TDSLARSAYW GGHVNSFRT GTTTLIRSP LYGREGNTER PVTITASPSV 360
PIFRTLSTYR GLDNSNPVAG IEGVEFQNTI SRSIYRKSGP IDSFSLEPPQ DASVSPAIGY 420
SHRLCHATPL ERISGPRIAG TVFSWTHRSA SPTNEVSPSR ITQIPWVKAH TLASGASVIK 480
GPGFTGGDIL TRNSMGBELG LRVFTTGRLP QSYIIRFRIA SVANRSGTFR YSQPPSYGIS 540
FKPTMDAGEP LTRSFAHTT LFTPITFSRA QEEFDLYIQS GVIIDRIEPI PVTATFEAEY 600

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DLERAQKVVN	ALFTSTNQLG	LKTDVTDYHI	DQVSNLVACL	SDEFCLDEKR	ELSEKVKHAK	660
RLSDERNLLQ	DPNFRGINRQ	PDRGWRGSTD	ITIQQGDDVF	KENYVTLPGT	FDECYPTYLY	720
QKIDESKLKA	YTRYQLRGYI	EDSQDLEIYL	IRYNAKHEIV	NVPGTGSLWP	LSVENQIGPC	780
GEPNRCAPHL	EWNPDLCSC	RDGEKCAHHS	HHFSLDIDVG	CTDLNEDLGV	WVIFKIKTQD	840
GCHARLGNLEF	LEEKPLLGEA	LARVKRAEKK	WRDKRETLQL	ETTIVYKEAK	ESVDALFVNS	900
QYDRLQADTN	IAMIHAADKR	VHRIREAYLP	ELSVIPGVNA	AIFEELEERI	FTAFSLYDAR	960
NIKNGDFNN	GLLCWNVKGH	VEVEEQNNHR	SVLVIPEWEA	EVSQEVVCP	GRGYILRVTA	1020
YKEGYGEGCV	TIHEIENNTD	ELKFNNCCEE	EVYPNNTVTC	INYTATQEEY	EGTYTSRNRG	1080
YDEAYGNNS	VPADYASVYE	EKSYTDRRRE	NPCESTRGYG	DYTPLPAGYV	TKELEYFPET	1140
DKVWIBIGET	EGTFIVDSVE	LLLMEE				1166

SEQ ID NO: 37 moltype = DNA length = 3501

FEATURE Location/Qualifiers

misc_feature 1..3501

 note = Synthetic DNA sequence designed for plant expression

 encoding Cys1Dal 6 with an additional Alanine residue

 inserted at position 2.

source 1..3501

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 37

atggctgaga	tcaacaacca	gaaccagtgc	gtcccgtaga	actgcctgag	caaccctaag	60
gagatcatcc	tgggtgagga	acgcctggag	accggcaaca	ccgtagccga	cattagcctg	120
ggcctcatca	actcctctta	cagcaacttc	gtgcccgccg	gtggcctcat	cgtgggcctc	180
ctggagctta	tctggggctt	catcgcccg	tcccagtggg	acatcttctt	cgccagatc	240
gagcaactga	tcagccagcg	gacgagggag	tgcgtagga	accaggccat	ctcccgcctg	300
gagggactct	ccaacctcta	caagggtgtac	gtgcgcgcgt	tcagcgactg	ggagaaggac	360
ccgaccaacc	cgccctcccg	cgaggaatat	cgtatccagt	tcaacgatat	gaactcggcc	420
ctcatcacgc	ccatcccgct	cttcccgctg	cagaactacg	aggtggccct	cctgtccgtg	480
tacgttcaag	ccgccaacct	ccacctctcc	atcctcccg	acgtgagcgt	gttcggcgag	540
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gattacacac	cattaccagc	aggataacgt	acaaaggagt	tggaaatact	cccggaaaca	3420
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ttgtgtgtga tggaagaatg a 3501

SEQ ID NO: 38      moltype = AA length = 1166
FEATURE           Location/Qualifiers
REGION            1..1166
                  note = Amino acid sequence of the engineered insecticidal
                  protein Cry1Da1_6 encoded by a synthetic DNA sequence
                  wherein an additional Alanine residue has been inserted at
                  position 2.
source            1..1166
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 38
MAEINNQNQC VPYNCLSNPK EIILGEERLE TGNTVADISL GLINFLYSNF VPGGGFIVGL 60
LELIWGFIFG SQWDIFLAQI EQLISQRIE FARNQAISRL EGLSNLYKVV VRAFSDWEKD 120
PTNPALREEM RIQFNDMNSA LITAIPLFRV QNYEVALLSV YVQAANLHLS ILRDVSVFGE 180
RWGYDTATIN NRYSDLTSLI HVYTNHCVDI YNQGRLRLEG RFLSDWIVYN RFRRLTISV 240
LDIVAFFPNY DIRTYPIQTA TQLTREVYLD LPFINENLSP AAKYPTFSAA ESAIIRSPHL 300
VDFLNSFTIY TDSLARSAYW GGLVNSFRT GTTNLIRSP LYGREGNTER PVTITASPSV 360
PIFRTLSTYPT GLDNSNPVAG IEGVEFQNTI SRSIYRKSGP IDSFSELPPQ DASVSPAIGY 420
SHRLCHATFL ERISGPRIAG TVFSWTHRSA SPTNEVSPSR ITQIPWVKAH TLASGASVIK 480
GPGFTGGDIL TRNSMGELGT LRVFTTGRLP QSYIIRFRYA SVANRSGTFR YSQPPSYGIS 540
FPKTMADAGEP LTRSFAHTT LFTPTIFSRA QEEFDLYIQS GVIYDRIEFI PVTATFEAEY 600
DLERAQKVVN ALFTSTNQLG LKTDVTDYHI DQVSNLVACL SDEFCLDEKR ELSEKVKHAK 660
RLSDERNLLQ DPNFRGINRQ PDRGWRGSTD ITIQGGDDVF KENVYTLPGT FDECYPTYLY 720
QKIDESKLKA YTRYQLRGIY EDSQDLEIYL IRYNAKHEIV NVPGTGSLWP LSVENQIGPC 780
GEPNRCAPHL EWNPDLCSC RDGEKCAHHS HHFSLDIDVG CTDLNELDGV WVIFKIKTD 840
GHARLGNLEF LEEKPLLGEA LARVKRAEKK WRDKRETLLQ ETTIVYKEAK ESDALFVNS 900
QYDRLQADTN IAMIHAADKR VHRIREAYLP ELSVIPGVNA AIFEELEERI FTAFSLYDAR 960
NIIKNGDFNN GLLCWNVKGH VEVEEQNNHR SVLVIPEWEA EVSQEVVRCP GRGYILRVTA 1020
YKEGYGEGCV TIHIEINNTD ELKFNNCREE EVYPNNTVTC INYTATQEEY EGTYSRNRG 1080
YDEAYGNPNS VPADYASVYE EKSYPDTRRE NPCESNRGYD DYTPLPAGYV TKELEYFPET 1140
DKVWIEIGET EGTFTIVDSVE LLLMEE 1166

SEQ ID NO: 39      moltype = DNA length = 3501
FEATURE           Location/Qualifiers
misc_feature       1..3501
                  note = Synthetic DNA sequence designed for plant expression
                  encoding Cry1Da1_7 with an additional Alanine residue
                  inserted at position 2.
source            1..3501
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 39
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ggcctcatca acttctctta cagcaacttc gtgcccgagg gtggcttcat cgtggggcctc 180
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gagggactct ccaacctcta caaggtgtac gtgctgcgct tcagcgactg ggagaaggac 360
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gattacacac cattaccagc aggatacgtt acaaaggagt tggataactt cccggaaca 3420
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SEQ ID NO: 40      moltype = AA length = 1166
FEATURE           Location/Qualifiers
REGION            1..1166
                  note = Amino acid sequence of the engineered insecticidal
                  protein Cry1Da1_7 encoded by a synthetic DNA sequence
                  wherein an additional Alanine residue has been inserted at
                  position 2.
source            1..1166
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 40
MAEINNQNQC VPYNCLSNPK EIILGEERLE TGNTVADISL GLINFLYSNF VPGGGFIVGL 60
LELIWGFIPG SQWDIFLAQI EQLISQRIE FARNQAISRL EGLSNLYKVY VRAFSWDEKD 120
PTNPALREEM RIQFNDMNSA LITAIPLFRV QNYEVALLSV YVQAANLHLS ILRDVSVFGE 180
RWGYDTATIN NRYSDLTSLI HVYTNHCVDI YNQLRRLLEG RFLSDWIVYN RFRRQLTISV 240
LDIVAFFPNY DIRTYPIQTA TQLTREYVLD LPFINENLSP AAVYPTFSAA ESAIIRSPHL 300
VDFLNSFTIY TDSLARSAYW GGHLVNSFRT GTTTNLIRSP LYGREGNTER PVTITASPSV 360
PIFRTLSTYPT GLDNSNPVAG IEGVEFQNTI SRSIYRKSGP IDSFSELPPQ DASVSPAIGY 420
SHRLCHATFL ERISGPRIAG TVFSWTHRSA SPTNEVSPSR ITQIPWVKAH TLASGASVIK 480
PGPFTGGDIL TRNSMGELGT LRVFTFGRLP QSYIIRFRIA SVANRSGTFR YSQPPSYGIS 540
FKPTMDAGEP LTRSFAHTT LFTPTIFSRA QEEFDLYIQS GVIYDRIEFI PVTATFEAEY 600
DLERAQKVVN ALFTSTNQLG LKTDVTDYHI DQVSNLVACL SDEFCLDEKR ELSEKVKHAK 660
RLSDEBNLLQ DPNFRGINRQ PDRGWRGSTD ITIQGGDDVF KENYVTLPGT FDECYPTYLY 720
QKIDESKLKA YTRYQLRGYI EDSQDLEIYL IRYNAKHEIV NVPGTGSLWP LSVENQIGPC 780
GEPNRCAPHL EWNPDLCSC RDGEKCAHHS HHFSLDIDVG CTDLNEGLV WVIFKIKTQD 840
GHARLGNLEF LEEKPLDGEA LARVKRAEKK WRDKRETLQL ETTIVYKEAK ESVDALFVNS 900
QYDRLLQADTN IAMIHADKR VHIRREAYLP ELSVIPGVNA AIFEELEERI FTAFSLYDAR 960
NIIKNGDFMN GLLCWNVKGH VEVEEQNNHR SVLVIPEWEA EVSQEVRVCP GRGYILRVTA 1020
YKEGYGEGCV TIHIEIENND ELKFNNCCEE EYYPNNTVTC INYTATQEEY EGYTTSRNRG 1080
YDEAYGNPNS VPADYASVYE EKSYPDTRRE NPCESNRGYV DYTPLPAGYV TKELEYFPET 1140
DKVWIEIGET EGTFIIVDSVE LLLMEE 1166

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SEQ ID NO: 41      moltype = DNA length = 3423
FEATURE           Location/Qualifiers
misc_feature       1..3423
                  note = Synthetic DNA sequence designed for plant expression
                  encoding TIC844 with an additional Alanine residue
                  inserted at position 2.
source            1..3423
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 41
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cacgtttaca ccaaccactg cgtggacacg tacaaccagg gcctccgccc cctggagggc 660
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gcccaccaca gccaccactt ctcttggac atcgatgtgg gctgcaccga cctgaacgag 2400
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SEQ ID NO: 42      moltype = AA length = 1140
FEATURE           Location/Qualifiers
REGION            1..1140
                  note = Amino acid sequence of the chimeric protein TIC844
                  encoded by a synthetic DNA sequence wherein an additional
                  Alanine residue has been inserted at position 2.
source            1..1140
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 42
MAEINNQNQC VPYNCLSNPK EIILGEERLE TGNTVADISL GLINFLYSNF VPGGGFIVGL 60
LELIWFIGP SQWDFILAQI EQLISQRIE FARNQAISRL EGLSNLYKVY VRAFSDWKED 120
PTNPALREEM RIQFNDMNSA LITAIPLFRV QNYEVALLSV YVQAANLHLS ILRDVSVFGE 180
RWGYDTATIN NRYSDLTSLI HVTNHCVDI YNQLRRLEG RFLSDWIVYN RFRRQLTISV 240
LDIVAFFPNY DIRTYPIQTA TQLTREYVLD LPFINENLSP AASYPTFSAA ESAIIRSPHL 300
VDFLNSFTIY TDSLARYAYW GGHLVNSFRT GTTTNLIRSP LYGREGNTER PVTITASPSV 360
PIFRTLSTYIT GLDMSNPVAG IEGVEFQNTI SRSIYRKS GP IDSFSLEPPQ DASVSPAIGY 420
SHRLCHATFL ERISGPRIAG TVFSWTHRSA SPTNEVSPSR ITQIPWVKAH TLASGASVIK 480
GPGFTGGDIL TRNSMGELGT LRVTFTRGLP QSYIIRFRYA SVANRSGTFR YSQPPSYGIS 540
FPKTM DAGEP LTRSFAHTT LFTPIFTRSA QEEFDLYIQS GVIIDRIEF PVTATFEAEY 600
DLERAQKAVN ELFTSSNQIG LKTDVTDYHI DQVSNLVECL SDEFCLDEKK ELSEKVKHAK 660
RLSDERNLLQ DPNFRGINRQ LDRGWRGSTD ITIQGGDDVF KENYVTLTGT FDECYPTYLY 720
QKIDESKLKA YTRYQLRGYI EDSQDLEIYL IRYNAKHETV NVPGTGSLWP LSAPSPIGKC 780
AHHSHHFSLD IDVGCTDLNE DLGVVIFKI KTQDGHARLG NLEFLEEKPL VGEALARVKR 840
AEKKWRDKRE KLEWETNIVY KEAKESVDAL FVNSQYDRLQ ADTNIAIHA ADKRVSIRE 900
AYLPESLVIP GVNAAIPEEL EGRIFTAFSL YDARNVINKG DFNNGSLCWN VKGHVDVEEQ 960
NNHRSVLVVP EWEAEVSEQV RVCPRGYIL RVTAYKEGYG EGCVTIHEIE NNTDELKFSN 1020
CVEEEVYPNN TVTCDNYTAT QEEYEGTYTS RNRGYDGAYE SNSSVPADYA SAYEEKAYTD 1080
GRRDNPCESN RGYGDYTLPL AGVVTKELEY FPETDKVWIE IGETEGTFIV DSVELLMLME 1140

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SEQ ID NO: 43      moltype = DNA length = 3423
FEATURE           Location/Qualifiers

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misc_feature      1..3423
                  note = Synthetic DNA sequence designed for plant expression
                  encoding TIC844_8 with an additional Alanine residue
                  inserted at position 2.
source            1..3423
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 43
atggctgaga tcaacaacca gaaccagtgc gtcccgtaga actgcctgag caaccctaag 60
gagatcatcc tgggtgagga acgcctggag accggcaaca ccgtagccga cattagcctg 120
ggcctcatca acttctctta cagcaacttc gtgcccgccg gtggcttcat cgtgggcctc 180
ctggagctta tctggggctt catcgccccc tcccagtgag acatcttctc cgcacagatc 240
gagcaactga tcagccagcg gatcgaggag ttcgctagga accaggccat ctcccgcctg 300
gagggactct ccaacctcta caaggtgtac gtgcgcgcgt tcagcgactg ggagaaggac 360
cgcaccaacc cggccctccg cgaggaaatg cgtatccagt tcaacgatat gaactcggcc 420
ctcatcaccg ccatcccgct ctcccgctg cagaactacg aggtggccct cctgtccgtg 480
tacgttcaag cgcaccaact ccacctctcc atcctccgcg acgtgagcgt gttcggcgag 540
cgctggggct acgacaccgc caccatcaac aaccgctact ccgacctcac ctccctcatc 600
cacgtttaca ccaaccatg cgtggacacg tacaaccagg gcctccgcgc cctggagggc 660
cgcttctctc cgactggat cgtgtacaac cgcttccgcc gccagctcac catctccgtc 720
ctggacatcg tcgccttctt tcccaactac gacatccgca cctaccttat ccagaccgcc 780
accagctca cccgcgaggt ctacctgac ctcccggtta tcaacgagaa cctcagcccg 840
gccgcgctct acccgacctt ctccgcgcgt gactccgcca tcatttcgag cccgcacctc 900
gtggacttcc tcaactctt caccatctac accgactccc tcgcccgcag cgctactggt 960
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cccatcttcc gcacctctag ctaccccacc ggctcggaac acagcaaccc tgtggcgggc 1140
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atcacccaga tctctgggtt caaggccacc accctggcta gtggcgctag tgtcatcaag 1440
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tccgatgagt tctgctcga cgagaagaag gagctgtccg agaaggtgaa gcatgccaa 1980
cgtctcagcg acgagaggaa tctctctcag gaccccaatt tcgcgcgcac caacaggcag 2040
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aaggagaact acgtgactct cctgggcact ttcgacgagt gctacctac ctactgttac 2160
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aacgtgcccc gtactgtgtt cctctggccg ctgagcgccc ccagcccgat cggcaagtgt 2340
gccaccacca gccaccactt ctctctggac atcgatgtgg gctgcaccga cctgaacgag 2400
gacctcggag tctgggtcat ctcaagatc aagaccaggg acggccacgc gcgcctgggc 2460
aacctggagt tctctgagga gaagccctg gtccgtgagg ctctggccag ggtcaagagg 2520
gctgagaaga agtgaggagg caagcgcgag aagctcgagt gggagaccaa catcgtttac 2580
aaggaggcca aggagagcgt cgacccctg ttctggaact ccagtaaga ccgcctgcag 2640
gccgacacca acatcgccat gatccacgct gccgacaaga ggggtgcacag cattcgcgag 2700
gcctacctgc ctgagctgct cgtgatccct ggtgtgaacg ctgccatctt tgaggagctg 2760
gaaggccgca tctttaccgc attctccctg tacgacgccc gcaacgtgat caagaacggg 2820
gacttcaaca atggcctcag ctgctggaac gtcaagggcc acgtggacgt cgaggaaacg 2880
aacaaccacc gctccgtcct ggtcgtccca gagtgggagg ctgaggtctc ccaagaggtc 2940
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ggccgcaggg acaacccttg cgagagcaac cgcggctacg gcgactacac tcccctgccc 3300
gccgctacg ttaccaagga gctggagtag ttcccgagga ctgacaaggt gtggatcgag 3360
atcgccgaga ccgagggcac cttcatcggt gacagcgtag agctgctcct gatggaggag 3420
tag 3423

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SEQ ID NO: 44      moltype = AA length = 1140
FEATURE            Location/Qualifiers
REGION             1..1140
                  note = Amino acid sequence of the engineered insecticidal
                  chimeric protein TIC844_8 encoded by a synthetic DNA
                  sequence wherein an additional Alanine residue has been
                  inserted at position 2.
source             1..1140
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 44
MAEINNQNQC VPYNCLSNPK EIILGEERLE TGNTVADISL GLINFLYSNF VPGGGFIVGL 60

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LELIWGFIPG	SQWDIFLAQI	EQLISQRIE	FARNQAISRL	EGLSNLYKVY	VRAFSDWEKD	120
PTNPALREEM	RIQNDMMNSA	LITAIPLFRV	QNYEVALLSV	YVQAANLHLS	ILRDVSVFGE	180
RWGYDTATIN	NRYSDLTSLI	HVYTNHCVD	YNQGLRRLEG	RFLSDWIVYN	RFRRLTISV	240
LDIVAFFPNY	DIRTYPIQTA	TQLTREVYLD	LPFINENLSP	AAVYPTFSAA	ESAIIRSPHL	300
VDFLNSFTIY	TDSLARSAYW	GGHLVNSFRT	GTTTNLIRSP	LYGREGNTER	PVTITASPSV	360
PIFRTLSTYPT	GLDNSNPVAG	IEGVFQNTI	SRSIYRKSGP	IDSFSELPPQ	DASVSPAIGY	420
SHRLCHATFL	ERISGPRIAG	TVFSWTHRSA	SPTNEVSPSR	ITQIPWVKAH	TLASGASVIK	480
GPGFTGGDIL	TRNSMGELGT	LRTVFTGRLP	QSYIIRFRYA	SVANRSGTFR	YSQPPSYGIS	540
FPKTMADAGEP	LTSRSFAHTT	LFTPITFSRA	QEEFDLYIQS	GVYIDRIEFI	PVTATFEAEY	600
DLERAQKAVN	ELFTSSNQIG	LKTDVTDYHI	DQVSNLVECL	SDEFCLDEKK	ELSEKVKHAK	660
RLSDERNLLQ	DPNFRGINRQ	LDRGWRGSTD	ITIQQGDDVF	KENYVTLTGT	FDECYPTYLY	720
QKIDESKLKA	YTRYQLRGIY	EDSQDLEIYL	IRYNAKHETV	NVPGTGSLWP	LSAPSPIGKC	780
AHSHHFLSLD	IDVGCTDLNE	DLGVWVIFKI	KTQDGHARLG	NLEFLEEKPL	VGEALARVKR	840
AEKKWRDKRE	KLEWETNIVY	KEAKESVDAL	FVNSQYDRLQ	ADTNAMIHA	ADKRVHSIRE	900
AYLPESLVIP	GVNAAIPEEL	EGRIPTAFSL	YDARNVIKNG	DFNNGSLCWN	VKGHVVDVEEQ	960
NNHRSVLVVP	EWAEVVSQEV	RVCPCRGYIL	RVTAYKEGYG	EGCVTIHEIE	NNTDELKFSN	1020
CVVEEVYPNM	TVTCDNDYAT	QEEYEGTYTS	RNRGYDGAYE	SNSSVPADYA	SAYEEKAYTD	1080
GRRDNPCESN	RGYGDYTPLP	AGYVTKELEY	FPETDKVWIE	IGETEGTFIV	DSVELLLMEE	1140

What is claimed is:

1. An engineered insecticidal protein comprising an amino acid sequence as set forth in any of SEQ ID NO: 12, SEQ ID NO: 26, SEQ ID NO:4, SEQ ID NO:6, SEQ ID NO:8, SEQ ID NO:10, SEQ ID NO:18, SEQ ID NO:20, SEQ ID NO:22, SEQ ID NO:24, SEQ ID NO:32, SEQ ID NO: 34, SEQ ID NO:36, or SEQ ID NO:38.

2. The engineered insecticidal protein of claim 1, wherein the engineered insecticidal protein exhibits inhibitory activity against an insect species of the order Lepidoptera.

3. The engineered insecticidal protein of claim 2, wherein the Lepidoptera is selected from the group consisting of a *Spodoptera* and a *Helicoverpa*.

4. The engineered insecticidal protein of claim 2, wherein the Lepidoptera is selected from the group consisting of *Helicoverpa zea* and *Spodoptera frugiperda*.

5. A polynucleotide encoding the engineered insecticidal protein of claim 1, wherein the polynucleotide is operably linked to a heterologous promoter.

6. A polynucleotide encoding an engineered insecticidal protein, wherein the polynucleotide comprises a nucleotide sequence that

a. comprises the polynucleotide sequence as set forth in any of SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO:31, SEQ ID NO: 33, SEQ ID NO: 35, or SEQ ID NO: 37; or

b. encodes the engineered insecticidal protein of claim 1.

7. A host cell comprising the polynucleotide of claim 6, wherein the host cell is selected from the group consisting of a bacterial host cell or a plant host cell.

8. The host cell of claim 7, wherein the bacterial host cell is selected from the group consisting of *Agrobacterium*, *Rhizobium*, *Bacillus*, *Brevibacillus*, *Escherichia*, *Pseudomonas*, *Klebsiella*, and *Erwinia*; and wherein the *Bacillus* species is a *Bacillus cereus* or a *Bacillus thuringiensis*, said *Brevibacillus* is a *Brevibacillus laterosporus*, and said *Escherichia* is an *Escherichia coli*.

9. The host cell of claim 7, wherein the plant host cell is selected from the group of plants consisting of monocots and dicots.

10. An insect inhibitory composition comprising the engineered insecticidal protein of claim 1.

11. The insect inhibitory composition of claim 10, further comprising at least one insect inhibitory agent different from the engineered insecticidal protein.

12. The insect inhibitory composition of claim 11, wherein the at least one insect inhibitory agent is selected from the group consisting of an insect inhibitory protein, an insect inhibitory dsRNA molecule, and an insect inhibitory chemistry.

13. The insect inhibitory composition of claim 11, wherein the at least one other pesticidal agent exhibits activity against one or more pest species of the orders Lepidoptera, Coleoptera, Hemiptera, Homoptera, or Thysanoptera.

14. A seed comprising:

a. an insect inhibitory effective amount of the engineered insecticidal protein of claim 1; or

b. a polynucleotide set forth in any of SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO:31, SEQ ID NO: 33, SEQ ID NO: 35, or SEQ ID NO: 37.

15. A method of controlling a Lepidopteran pest, the method comprising contacting the Lepidopteran pest with an inhibitory amount of the engineered insecticidal protein of claim 1.

16. A transgenic plant cell, plant or plant part comprising the engineered insecticidal protein of claim 1.

17. A method of controlling a Lepidopteran pest, comprising exposing the pest to the transgenic plant cell, plant or plant part of claim 16, wherein said plant cell, plant or plant part expresses a Lepidopteran inhibitory amount of the engineered insecticidal protein.

18. A commodity product derived from the plant cell, plant, or plant part of claim 16, wherein the product comprises a detectable amount of the engineered insecticidal protein.

19. The commodity product of claim 18, wherein the product is selected from the group consisting of plant biomass, oil, meal, animal feed, flour, flakes, bran, lint, hulls, and processed seed.

20. A method of producing a seed comprising the engineered insecticidal protein of claim 1, the method comprising:

a. planting at least one seed comprising the engineered insecticidal protein of claim 1;

b. growing plants from said seed; and

c. harvesting seed from said plants, wherein said harvested seed comprises the engineered insecticidal protein of claim 1.

21. A recombinant polynucleotide molecule encoding the engineered insecticidal protein of claim 1, comprising a nucleotide sequence selected from the group consisting of SEQ ID NO: 11, SEQ ID NO: 25, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 17, SEQ ID NO: 19, SEQ ID NO: 21, SEQ ID NO: 23, SEQ ID NO: 31, SEQ ID NO: 33, SEQ ID NO: 35, or SEQ ID NO: 37; and further comprising a polynucleotide sequence encoding an insect inhibitory agent different from the engineered insecticidal protein.

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